

Neurocritical Care Board Review: 250 Practice Questions (Test Mode)

250 single-best-answer questions written to the content of *The Practice of Neurocritical Care*, 2nd Edition (Neurocritical Care Society, 2021), organized by chapter. Answers with explanations are in the companion answer key.

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Part I: General Critical Care

Airway, Mechanical Ventilation, and Respiratory Disease

Question 1

Airway, Mechanical Ventilation, and Respiratory Disease · Control of breathing / abnormal respiratory patterns · easy

A 68-year-old man with atrial fibrillation presents with vertigo, dysphagia, and hiccups. Blood pressure is 148/82 mmHg, heart rate 78, SpO₂ 94% on room air. On examination he has right-sided Horner syndrome, right palatal weakness, and crossed sensory loss. Over the next several hours the nurse notes that his breathing has become completely irregular, with breaths of varying depth separated by unpredictable pauses, without any crescendo-decrescendo pattern. Which lesion location best accounts for this breathing pattern?

- A. Midbrain tectum
- B. Bilateral cerebral hemispheres
- C. Dorsolateral pons at the parabrachial nucleus
- D. Ventral medulla involving the pre-Botzinger complex
- E. Cervical spinal cord above C3

Question 2

Airway, Mechanical Ventilation, and Respiratory Disease · Neuromuscular respiratory failure: FVC and NIF thresholds · moderate

A 34-year-old woman weighing 60 kg is admitted with ascending weakness and areflexia five days after a diarrheal illness. She is now unable to lift her head off the pillow, has a nasal voice, and is pooling oral secretions. Blood pressure is 132/78 mmHg, heart rate 104, respiratory rate 26, SpO₂ 96% on 2 L nasal cannula. Bedside testing shows a forced vital capacity of 0.9 L and a negative inspiratory force of -15 cmH₂O. Arterial blood gas shows pH 7.34, PaCO₂ 48 mmHg, PaO₂ 82 mmHg. Which of the following is the most appropriate next step?

- A. Continue observation with forced vital capacity measurements every 4 hours
- B. Perform endotracheal intubation and initiate mechanical ventilation
- C. Start bilevel noninvasive positive pressure ventilation
- D. Start high-flow nasal cannula oxygen at 40 L/min
- E. Begin incentive spirometry and chest physiotherapy

Question 3

Airway, Mechanical Ventilation, and Respiratory Disease · Induction agents for intubation · moderate

A 58-year-old man with a Hunt and Hess grade IV subarachnoid hemorrhage requires urgent intubation for airway protection. Blood pressure is 88/50 mmHg on low-dose norepinephrine, heart rate 96, SpO₂ 93% on a nonrebreather. His history includes a myocardial infarction with drug-eluting stent placement two months ago and a documented anaphylactic reaction to eggs and soy. Which induction agent is most appropriate?

- A. Propofol 1.5 mg/kg intravenously
- B. Midazolam 0.2 mg/kg intravenously
- C. Etomidate 0.3 mg/kg intravenously
- D. Ketamine 2 mg/kg intravenously
- E. Fentanyl 3 mcg/kg intravenously as the sole induction agent

Question 4

Airway, Mechanical Ventilation, and Respiratory Disease · Neuromuscular blocking agents · moderate

A 71-year-old woman with acetylcholine receptor antibody positive myasthenia gravis presents in crisis after a urinary tract infection. She has ptosis, dysarthria, and paradoxical abdominal breathing. Blood pressure is 138/80 mmHg, heart rate 110, respiratory rate 32, SpO₂ 91% on 6 L nasal cannula. Forced vital capacity is 0.7 L and PaCO₂ is 56 mmHg. She last ate two hours ago, and rapid sequence intubation is planned. Which neuromuscular blocking agent is most appropriate?

- A. Succinylcholine 1.5 mg/kg intravenously
- B. Succinylcholine 4 mg/kg intramuscularly
- C. Cisatracurium 0.15 mg/kg intravenously
- D. No paralytic; induce with propofol alone
- E. Rocuronium 1 mg/kg intravenously

Question 5

Airway, Mechanical Ventilation, and Respiratory Disease · Hyperventilation, cerebral blood flow, and ICP · moderate

A 28-year-old man with severe traumatic brain injury has an intracranial pressure of 32 mmHg despite sedation, hyperosmolar therapy, and cerebrospinal fluid drainage. The respiratory rate is increased on the ventilator, and over the next several hours his PaCO₂ falls to 25 mmHg. His intracranial pressure decreases to 18 mmHg and his cerebral perfusion pressure improves from 58 to 72 mmHg, and the team plans to maintain these settings overnight. Which of the following best describes the principal risk of this strategy?

- A.** Hypocarbica causes cerebral vasoconstriction, increasing the volume of hypoperfused brain and decreasing total cerebral blood flow despite the improved intracranial pressure and cerebral perfusion pressure numbers
- B.** Hypocarbica causes cerebral vasodilation, increasing cerebral blood volume and producing rebound intracranial hypertension
- C.** The resulting respiratory acidosis directly worsens cytotoxic cerebral edema
- D.** The increased respiratory rate raises intrathoracic pressure, impairing jugular venous drainage and raising intracranial pressure
- E.** Sustained hypocarbica causes barotrauma and pneumothorax before it affects cerebral perfusion

Question 6

Airway, Mechanical Ventilation, and Respiratory Disease · Readiness for a spontaneous breathing trial · hard

It is morning rounds in the neurocritical care unit and the team is selecting patients for a spontaneous breathing trial. Which of the following intubated patients meets criteria for a trial?

- A.** A 24-year-old with diffuse axonal injury whose intracranial pressure is 26 mmHg on hypertonic saline, on PEEP of 5 cmH₂O and FiO₂ of 0.4
- B.** A 61-year-old three days after coiling of a ruptured anterior communicating artery aneurysm, intracranial pressure 12 mmHg, RASS -1, SpO₂ 96% and PaO₂ 88 mmHg on FiO₂ of 0.4, PEEP of 5 cmH₂O, PaCO₂ 40 mmHg, heart rate 88, mean arterial pressure 78 mmHg off vasopressors, who opens his eyes but does not follow commands
- C.** A 45-year-old in myasthenic crisis with a negative inspiratory force of -12 cmH₂O on PEEP of 5 cmH₂O and FiO₂ of 0.4
- D.** A 58-year-old with acute respiratory distress syndrome on PEEP of 12 cmH₂O and FiO₂ of 0.6
- E.** A 70-year-old with septic shock and a mean arterial pressure of 55 mmHg on escalating norepinephrine

Question 7

Airway, Mechanical Ventilation, and Respiratory Disease · Extubation decision in brain-injured patients · moderate

A 66-year-old man is on hospital day 6 after a left basal ganglia intracerebral hemorrhage. He has passed a 30-minute spontaneous breathing trial on pressure support with a rapid shallow breathing index of 62 breaths per minute per liter. He is on PEEP of 5 cmH₂O and FiO₂ of 0.3, he has a strong cough with suctioning, minimal secretions, and an audible cuff leak. He opens his eyes spontaneously and tracks the examiner across the room, but he is aphasic and does not follow commands. Which of the following is the most appropriate next step?

- A.** Perform percutaneous dilatational tracheostomy within 24 hours
- B.** Continue mechanical ventilation and repeat the spontaneous breathing trial daily until he follows commands
- C.** Give intravenous methylprednisolone for 24 hours and then reassess for extubation
- D.** Extubate to supplemental oxygen with close monitoring
- E.** Extubate directly to bilevel noninvasive positive pressure ventilation

Question 8

Airway, Mechanical Ventilation, and Respiratory Disease · Tracheostomy timing and technique · moderate

A 55-year-old woman with a large right middle cerebral artery infarct has been intubated for airway protection for three days and remains unable to protect her airway. The family asks whether performing a tracheostomy now rather than in the second week would improve her chances of recovery. Which statement best reflects the evidence described for tracheostomy in this population?

- A.** Early tracheostomy reduced mortality and improved functional outcome compared with late tracheostomy in ventilated stroke patients
- B.** Early tracheostomy in ventilated stroke patients shortened intensive care unit stay but significantly increased bleeding complications
- C.** A randomized trial of early tracheostomy up to day 3 versus late tracheostomy on days 7 to 14 in ventilated stroke patients did not improve outcomes, but early tracheostomy was safe and feasible and may decrease sedation needs
- D.** Surgical tracheostomy has a lower overall complication rate, about 3 percent, than percutaneous dilatational tracheostomy, about 12 percent
- E.** Neurocritically ill patients wean from the ventilator more slowly after tracheostomy than general intensive care unit patients

Question 9

Airway, Mechanical Ventilation, and Respiratory Disease · Post-extubation laryngeal edema · easy

A 63-year-old woman is extubated on day 8 after a traumatic subdural hematoma evacuation. No cuff leak had been documented before extubation. Ten minutes after extubation she develops a high-pitched inspiratory sound audible across the room, suprasternal retractions, and rising work of breathing. Blood pressure is 156/88 mmHg, heart rate 118, respiratory rate 30, SpO₂ 91 percent on a nonrebreather mask. Lung fields are clear without wheezing. Which of the following is the most appropriate next step?

- A. Administer nebulized epinephrine and intravenous methylprednisolone
- B. Administer nebulized albuterol 2.5 mg every 20 minutes
- C. Proceed directly to percutaneous dilatational tracheostomy
- D. Continue supplemental oxygen by nonrebreather mask alone and observe
- E. Administer magnesium 2 g intravenously over 20 minutes

Question 10

Airway, Mechanical Ventilation, and Respiratory Disease · COPD exacerbation management · easy

A 70-year-old man with severe chronic obstructive pulmonary disease is admitted to the neurocritical care unit after a small thalamic intracerebral hemorrhage. On day 2 he becomes dyspneic and wheezy with increased purulent sputum. He is alert and protecting his airway. Blood pressure is 142/84 mmHg, heart rate 104, respiratory rate 26, SpO₂ 84 percent on room air. He is started on nebulized albuterol and ipratropium, prednisone 40 mg orally daily, and antibiotics. To what oxygen saturation range should supplemental oxygen be titrated?

- A. 80 to 84 percent
- B. 84 to 88 percent
- C. 88 to 92 percent
- D. 92 to 96 percent
- E. 96 to 100 percent

Question 11

Airway, Mechanical Ventilation, and Respiratory Disease · ARDS in the brain-injured patient · hard

A 44-year-old man, 70 kg ideal body weight, with severe traumatic brain injury and diffuse cerebral edema has a parenchymal intracranial pressure monitor reading 16 mmHg. Three days after admission he develops fever and hypoxemia. Chest radiograph shows new bilateral opacities, echocardiography shows normal biventricular function without volume overload, and on FiO₂ of 0.8 with PEEP of 10 cmH₂O his PaO₂ is 68 mmHg with a PaCO₂ of 41 mmHg. He is currently ventilated with tidal volumes of 550 mL and a plateau pressure of 32 cmH₂O. Which of the following is the most appropriate ventilator strategy?

- A. Increase the tidal volume to 10 mL/kg of ideal body weight to prevent any rise in PaCO₂
- B. Start high-frequency oscillatory ventilation, which has been shown to reduce mortality in this setting
- C. Initiate venovenous extracorporeal membrane oxygenation as first-line rescue therapy
- D. Reduce the tidal volume to 6 mL/kg of ideal body weight and accept permissive hypercapnia with a PaCO₂ of 60 to 70 mmHg
- E. Reduce the tidal volume to 6 mL/kg of ideal body weight with a plateau pressure of 30 cmH₂O or less, modifying the protocol by increasing the respiratory rate to avoid hypercarbia, while monitoring intracranial pressure

Acid-Base and Electrolyte Disorders

Question 12

Acid-Base and Electrolyte Disorders · Hyperglycemic crises: HHS versus DKA · easy

A 58-year-old man with type 2 diabetes is brought to the emergency department after several days of increasing confusion and polyuria. He is obtunded and does not follow commands. Temperature 38.1 C, heart rate 118/min, blood pressure 96/58 mmHg. Fingerstick glucose is 950 mg/dL. Laboratory studies show sodium 130 mEq/L, potassium 5.3 mEq/L, chloride 89 mEq/L, bicarbonate 17 mmol/L, serum osmolality 355 mOsm/kg, and beta-hydroxybutyrate 50.7 mg/dL. Arterial blood gas shows pH 7.41, PCO₂ 27 mmHg, base excess -6.6 mmol/L. Noncontrast head CT is normal. Which of the following is the most likely diagnosis?

- A. Diabetic ketoacidosis
- B. Alcoholic ketoacidosis
- C. Hyperosmolar hyperglycemic syndrome with minor ketoacidotic features
- D. Lactic acidosis from septic shock
- E. Salicylate intoxication

Question 13

Acid-Base and Electrolyte Disorders · Anion gap and delta/delta interpretation · hard

A 42-year-old man is on hospital day 3 after severe traumatic brain injury from a motor vehicle collision. He remains on mechanical ventilation, a continuous hypertonic saline infusion, pentobarbital, and two vasopressors. Today's arterial blood gas shows pH 7.20 and PCO₂ 37 mmHg with bicarbonate 14 mEq/L. The basic metabolic profile shows sodium 152 mEq/L, chloride 122 mEq/L, and total CO₂ 14 mEq/L. Which of the following best characterizes his acid-base disorder?

- A. Isolated hyperchloremic (normal anion gap) metabolic acidosis
- B. Anion gap metabolic acidosis with a coexisting hyperchloremic metabolic acidosis
- C. Anion gap metabolic acidosis with a superimposed metabolic alkalosis
- D. Respiratory acidosis with an appropriate metabolic compensation
- E. Primary respiratory alkalosis with a compensatory metabolic acidosis

Question 14

Acid-Base and Electrolyte Disorders · Sodium bicarbonate in metabolic and lactic acidosis · moderate

A 66-year-old woman with pneumococcal meningitis and septic shock is intubated in the neurocritical care unit. She is on norepinephrine at an escalating dose. Arterial blood gas shows pH 7.15, PCO₂ 28 mmHg, bicarbonate 10 mEq/L, and serum lactate is 7.2 mmol/L. Creatinine is 1.2 mg/dL and there is no hyperkalemia. The bedside nurse asks whether bicarbonate should be given to improve her pressor responsiveness. Which of the following is the most appropriate next step?

- A. Continue targeted antimicrobial and hemodynamic therapy while optimizing brain oxygenation and tissue perfusion
- B. Give sodium bicarbonate 50 mEq intravenous push
- C. Start a sodium bicarbonate infusion titrated to a pH above 7.30
- D. Arrange emergent hemodialysis for acid clearance
- E. Increase the ventilator rate to drive the PCO₂ below 25 mmHg

Question 15

Acid-Base and Electrolyte Disorders · Drug-induced normal anion gap metabolic acidosis · easy

Which of the following antiseizure medications is most likely to produce a normal anion gap (hyperchloremic) metabolic acidosis through inhibition of carbonic anhydrase?

- A. Lamotrigine
- B. Carbamazepine
- C. Phenytoin
- D. Topiramate
- E. Levetiracetam

Question 16

Acid-Base and Electrolyte Disorders · Severe metabolic alkalemia management · moderate

A 71-year-old woman with a cerebellar hemorrhage has had high-output nasogastric suction and recurrent emesis for 4 days. She is euvolemic after 48 hours of volume repletion with normal saline plus aggressive potassium chloride and has no heart or liver failure. She becomes lethargic with circumoral paresthesias and carpopedal spasms. Arterial blood gas shows pH 7.64, PCO₂ 52 mmHg, bicarbonate 54 mEq/L; potassium is 3.1 mEq/L and ionized calcium is low. Telemetry shows frequent premature ventricular complexes. Which of the following is the most appropriate next step?

- A. Continue normal saline with additional potassium chloride repletion alone
- B. Start acetazolamide to induce bicarbonate diuresis
- C. Deepen sedation to reduce minute ventilation and allow the PCO₂ to rise further
- D. Administer sodium bicarbonate and begin enteral free water
- E. Infuse dilute hydrochloric acid 0.10 to 0.15 N through a central venous catheter until the pH falls below 7.6

Question 17

Acid-Base and Electrolyte Disorders · Therapeutic hyperventilation and cerebral CO₂ reactivity · moderate

A 29-year-old man is 8 hours out from a severe traumatic brain injury with diffuse cerebral edema. An external ventricular drain shows an ICP of 24 mmHg. He is normotensive with a mean arterial pressure of 85 mmHg. Over the past hour the resident has increased the respiratory rate, and the arterial blood gas now shows pH 7.55 and PCO₂ 22 mmHg. There is no brain tissue oxygen or jugular venous oximetry monitoring in place. Which of the following is the most appropriate next step?

- A. Maintain the PCO₂ at 20 to 25 mmHg until the ICP is consistently below 20 mmHg
- B. Lower the PCO₂ below 20 mmHg to obtain maximal cerebral vasoconstriction
- C. Adjust ventilation back to a PCO₂ of 35 to 40 mmHg, reserving mild brief hypocapnia of 30 to 34 mmHg as a bridge to other therapies with brain tissue or jugular venous oxygen monitoring
- D. Maintain the PCO₂ at 25 to 30 mmHg for the first 24 hours after injury
- E. Allow the PCO₂ to rise above 45 mmHg to augment cerebral blood flow

Question 18

Acid-Base and Electrolyte Disorders · Hyponatremia in acute intracranial pathology: SIADH versus CSWS · moderate

A 54-year-old woman is on day 5 after an aneurysmal subarachnoid hemorrhage that has been secured by coiling. She is arousable but has a new mild left pronator drift, and transcranial Doppler velocities are rising. Serum sodium has fallen from 138 to 126 mEq/L over 36 hours. Plasma osmolality is 262 mOsm/kg, urine osmolality is 520 mOsm/kg, and urine sodium is 65 mEq/L. She is on no diuretics and thyroid, adrenal, pituitary, and renal function are normal. Clinical volume status is difficult to assess at the bedside. Which of the following is the most appropriate treatment?

- A. Begin a hypertonic sodium chloride solution
- B. Restrict free water intake to 1 liter per day
- C. Start a vasopressin receptor antagonist
- D. Start demeclocycline
- E. Give a 1 liter bolus of 5% dextrose in water

Question 19

Acid-Base and Electrolyte Disorders · Central diabetes insipidus: criteria for ADH analog therapy · moderate

A 34-year-old man with a severe traumatic brain injury and basilar skull fracture is intubated and sedated on hospital day 2. Over the past 3 consecutive hours his urine output has been 380, 420, and 400 mL/hr. Serum sodium has risen from 142 to 149 mEq/L, urine osmolality is 180 mOsm/kg, and urine specific gravity is 1.003. He is not receiving mannitol, hypertonic saline, or diuretics, and his glucose is 130 mg/dL. Which of the following is the most appropriate next step?

- A. Continue free water via the feeding tube alone and recheck the sodium in 12 hours
- B. Start a 0.45% sodium chloride infusion
- C. Start fludrocortisone
- D. Initiate antidiuretic hormone analog therapy along with free water replacement
- E. Start a 3% sodium chloride infusion

Question 20

Acid-Base and Electrolyte Disorders · Correcting hypernatremia in a patient with cerebral edema · hard

A 65-year-old man weighing 100 kg is on hospital day 6 after a 45 mL left basal ganglia hemorrhage with perilesional edema. His hypertonic saline infusion was weaned and stopped over the past 24 hours, and his sodium goal is 145 to 155 mEq/L. Overnight he developed pulmonary edema and received two doses of a loop diuretic. This morning the sodium has risen from 157 to 160 mEq/L with bicarbonate 20 mEq/L and a critical chloride of 126 mEq/L. He has a functioning feeding tube, and follow-up imaging still shows perilesional edema. Which of the following is the safest way to bring his sodium back into the goal range?

- A. 5% dextrose in water at 145 mL/hr
- B. Enteral free water flushes of approximately 285 mL every 4 hours
- C. Enteral free water flushes of approximately 585 mL every 4 hours
- D. 0.45% sodium chloride at 70 mL/hr
- E. Replace half of the calculated free water deficit over the first 24 hours at a rate of up to 2 mEq/L per hour

Question 21

Acid-Base and Electrolyte Disorders · Magnesium: evidence base and therapeutic uses · easy

Adjunctive magnesium therapy has demonstrated a clear outcome benefit in which of the following conditions?

- A. Acute ischemic stroke when magnesium is given in the prehospital setting
- B. Aneurysmal subarachnoid hemorrhage
- C. Severe traumatic brain injury
- D. Spontaneous intracerebral hemorrhage
- E. Eclamptic seizures

Cardiovascular Monitoring and Complications

Question 22

Cardiovascular Monitoring and Complications · Invasive arterial pressure monitoring: damping artifact · moderate

A 62-year-old woman with a Hunt-Hess 3 subarachnoid hemorrhage is in the neurocritical care unit on a nicardipine infusion. Her radial arterial line displays a systolic blood pressure of 92 mmHg and a mean arterial pressure of 78 mmHg, while a simultaneous oscillometric cuff on the same arm reads 126/70 mmHg. She is awake, warm, and well perfused with a urine output of 60 mL/h. The arterial waveform appears blunted and rounded with loss of the dicrotic notch. The transducer is leveled at the fourth intercostal space in the midaxillary line and has been zeroed. Which of the following is the most appropriate next step?

- A. Start a norepinephrine infusion to raise the systolic blood pressure above 120 mmHg
- B. Replace the fluid-filled tubing with stiffer, longer pressure tubing to improve signal fidelity
- C. Flush the catheter and tubing, check for air bubbles, clots, or kinks, and confirm the flush bag is adequately pressurized
- D. Remove the arterial catheter and manage blood pressure with oscillometric cuff readings every 15 minutes
- E. Re-zero the transducer at the level of the external auditory meatus to reflect cerebral perfusion pressure

Question 23

Cardiovascular Monitoring and Complications · Pulse pressure variation and fluid responsiveness · easy

A 58-year-old man with a large left middle cerebral artery infarct is sedated and mechanically ventilated in volume-control mode with a tidal volume of 8 mL/kg. He is in sinus rhythm at 96 beats per minute. His blood pressure is 88/52 mmHg and his lactate is rising. The arterial waveform is well damped, and the monitor reports a pulse pressure variation of 18% over the respiratory cycle. Which of the following is the most appropriate interpretation and next step?

- A. The stroke volume is sensitive to preload, and a fluid bolus is likely to increase cardiac output
- B. The stroke volume is insensitive to preload, and further fluid is unlikely to increase cardiac output
- C. The value indicates volume overload, and intravenous furosemide should be given
- D. The value indicates vasoplegia, and norepinephrine should be started before any fluid is given
- E. The value is uninterpretable, and a pulmonary artery catheter should be placed to measure occlusion pressure

Question 24

Cardiovascular Monitoring and Complications · Bedside ultrasound of the inferior vena cava · moderate

A 71-year-old woman admitted with a cerebellar hemorrhage is spontaneously breathing on nasal cannula and becomes hypotensive with a blood pressure of 84/48 mmHg after a period of poor oral intake and vomiting. Bedside ultrasound in the sagittal plane just proximal to the hepatic vein shows an inferior vena cava diameter of 1.6 cm that collapses by approximately 70% with inspiration. Which of the following best describes the estimated right atrial pressure and the expected response to volume?

- A. Right atrial pressure of 10 to 20 mmHg, and the patient is unlikely to respond to fluid
- B. Right atrial pressure of 10 to 20 mmHg, and the patient is likely to respond to fluid
- C. Right atrial pressure of 6 to 12 mmHg, and the patient is unlikely to respond to fluid
- D. Right atrial pressure of 0 to 5 mmHg, and the patient is unlikely to respond to fluid
- E. Right atrial pressure of 0 to 5 mmHg, and the patient is likely to respond to fluid

Question 25

Cardiovascular Monitoring and Complications · Pulmonary artery catheter: evidence and preload estimation · hard

A 66-year-old man with a poor-grade subarachnoid hemorrhage develops shock of unclear cause on hospital day 5. He has a central venous catheter in place reading a central venous pressure of 14 mmHg. A consultant proposes floating a pulmonary artery catheter, arguing that a wedge pressure will settle whether the patient needs volume. Which of the following responses is best supported by the evidence cited in the chapter?

- A.** Place the catheter, because the pulmonary artery occlusion pressure reliably reflects left ventricular end-diastolic volume in critically ill patients
- B.** Defer the catheter, because randomized trials and a meta-analysis showed no benefit in mortality or hospital days and neither central venous pressure nor occlusion pressure reliably predicts preload
- C.** Defer the catheter, because a central venous pressure of 14 mmHg establishes that the left ventricle is adequately filled and excludes fluid responsiveness
- D.** Place the catheter, because thermodilution cardiac output is directly proportional to the magnitude of the blood temperature decrease it measures
- E.** Place the catheter, because ventricular arrhythmia during insertion is the only recognized complication and it is self-limited

Question 26

Cardiovascular Monitoring and Complications · QT interval monitoring · easy

A neurocritical care unit is establishing a protocol for monitoring the corrected QT interval in patients receiving drugs that may cause torsades de pointes. Which of the following correctly pairs the upper limit of a normal corrected QT interval with the electrocardiographic lead most commonly used to measure it?

- A.** 0.40 seconds in men and 0.42 seconds in women, measured in lead V1
- B.** 0.42 seconds in men and 0.44 seconds in women, measured in lead I
- C.** 0.45 seconds in men and 0.46 seconds in women, measured in lead aVF
- D.** 0.45 seconds in men and 0.46 seconds in women, measured in lead II
- E.** 0.50 seconds in men and 0.50 seconds in women, measured in lead V5

Question 27

Cardiovascular Monitoring and Complications · End-tidal carbon dioxide during cardiopulmonary resuscitation · easy

A 70-year-old intubated man in the neurocritical care unit suffers a pulseless arrest. Continuous waveform capnography during chest compressions shows an end-tidal carbon dioxide of 8 mmHg. After a change of compressor and two more minutes of high-quality compressions, the end-tidal carbon dioxide abruptly rises to 40 mmHg. Which of the following is the most appropriate interpretation and next step?

- A.** Return of spontaneous circulation has likely occurred; pause compressions to check the rhythm and pulse
- B.** The endotracheal tube has migrated into the esophagus; remove it and reintubate
- C.** Compressions have become inadequate; deepen compressions and reassess in two minutes
- D.** A tension pneumothorax has developed; perform immediate needle decompression
- E.** Iatrogenic hypercarbia has developed; increase the ventilator respiratory rate

Question 28

Cardiovascular Monitoring and Complications · Carbon dioxide targets in traumatic brain injury · moderate

A 34-year-old man with severe traumatic brain injury is sedated and mechanically ventilated with an intracranial pressure monitor reading 14 mmHg and a cerebral perfusion pressure of 68 mmHg. His pupils are equal and reactive and there are no signs of herniation. Capnography shows an end-tidal carbon dioxide of 24 mmHg, and an arterial blood gas confirms an arterial partial pressure of carbon dioxide of 27 mmHg. Which of the following is the most appropriate ventilator adjustment?

- A.** Continue current settings, since an arterial carbon dioxide of 25 to 30 mmHg is the standard target in traumatic brain injury
- B.** Increase the respiratory rate to bring the arterial carbon dioxide to 20 to 25 mmHg
- C.** Reduce minute ventilation to bring the arterial carbon dioxide to 35 to 45 mmHg
- D.** Reduce minute ventilation to allow the arterial carbon dioxide to rise above 45 mmHg to augment cerebral blood flow
- E.** Leave minute ventilation unchanged and increase positive end-expiratory pressure to improve carbon dioxide clearance

Question 29

Cardiovascular Monitoring and Complications · Neurogenic stress cardiomyopathy after subarachnoid hemorrhage · moderate

A 24-year-old woman presents with a Hunt-Hess 4, Fisher 4 subarachnoid hemorrhage. On hospital day 1 she develops chest discomfort and dyspnea while remaining hemodynamically stable, with a blood pressure of 126/74 mmHg. Her electrocardiogram shows a prolonged QT interval and inverted T waves in the precordial leads, and her troponin is elevated. Bedside echocardiography demonstrates apical ballooning with hyperdynamic basal segments. Urgent coronary angiography shows coronary vessels free of obstruction or spasm. Which of the following is the most likely diagnosis and an appropriate therapy?

- A.** Non-ST-elevation myocardial infarction, treated with aspirin, statin, and urgent percutaneous coronary intervention
- B.** Cardiac tamponade, treated with emergent ultrasound-guided pericardiocentesis
- C.** Stanford type A aortic dissection, treated with emergent surgical consultation
- D.** Saddle pulmonary embolism, treated with systemic alteplase
- E.** Takotsubo cardiomyopathy, treated with beta blockade

Question 30

Cardiovascular Monitoring and Complications · ST-elevation myocardial infarction with intracranial injury · hard

A 55-year-old man is brought in after an all-terrain vehicle crash. His Glasgow Coma Scale score is 13, and head computed tomography shows a basilar skull fracture and a mild left frontal contusion. On telemetry he develops ST-segment elevations, and a 12-lead electrocardiogram confirms 2 mm of ST elevation in leads I, aVL, and V5 through V6. He is hemodynamically stable. The nearest hospital capable of percutaneous coronary intervention is 100 minutes away by air. Which of the following is the most appropriate management?

- A. Administer streptokinase and then transport by air to the interventional center
- B. Give oxygen and metoprolol and transport by air to the percutaneous coronary intervention center
- C. Administer tenecteplase and then transport by air to the interventional center
- D. Give aspirin with a 2,000 unit intravenous heparin bolus and observe on telemetry locally
- E. Give sublingual nitroglycerin, digoxin, and clopidogrel and observe on telemetry locally

Endocrine Issues in Neurocritical Care

Question 31

Endocrine Issues in Neurocritical Care · Glycemic target in acute ischemic stroke · easy

A 68-year-old man with type 2 diabetes mellitus presents 2 hours after sudden onset of right hemiparesis and global aphasia. NIHSS is 14. Blood pressure is 168/88 mm Hg, heart rate 82/min. Non-contrast head CT shows no hemorrhage and an early left MCA hypodensity; CTA shows a left M2 occlusion. Fingerstick glucose is 268 mg/dL. He receives intravenous thrombolysis and is admitted to the neuro ICU on an insulin protocol. Which of the following blood glucose targets is recommended for this patient by current American Heart Association/American Stroke Association guidelines?

- A. 80-110 mg/dL
- B. 80-130 mg/dL
- C. 140-180 mg/dL
- D. Greater than 80 mg/dL and less than 200 mg/dL
- E. Less than 250 mg/dL

Question 32

Endocrine Issues in Neurocritical Care · Diabetic ketoacidosis - potassium before insulin · moderate

A 34-year-old woman with type 1 diabetes mellitus is admitted to the neuro ICU after a generalized convulsive seizure. She has vomited for 2 days and stopped taking her insulin. Blood pressure is 96/58 mm Hg, heart rate 122/min, respirations 28/min and deep. Laboratory studies: glucose 480 mg/dL, arterial pH 7.12, bicarbonate 9 mEq/L, anion gap 24, serum and urine ketones positive, sodium 132 mEq/L, potassium 3.0 mEq/L, creatinine 1.4 mg/dL. She has received 1 L of 0.9% normal saline over the past hour. Which of the following is the most appropriate next step?

- A. Hold insulin and infuse potassium chloride 20 mEq per hour until the potassium exceeds 3.3 mEq/L
- B. Give regular insulin 0.1 U/kg as a bolus followed by an infusion at 0.1 U/kg/hr
- C. Begin a regular insulin infusion at 0.14 U/kg/hr without a bolus
- D. Administer sodium bicarbonate 100 mEq in 400 mL infused over 2 hours
- E. Replace phosphate and begin a regular insulin infusion at 0.1 U/kg/hr

Question 33

Endocrine Issues in Neurocritical Care · Emergency treatment of hypoglycemia · easy

A 57-year-old man with alcohol use disorder, malnutrition, and type 2 diabetes mellitus is brought to the emergency department unresponsive. Temperature is 35.8 C, blood pressure 110/70 mm Hg, heart rate 96/min. He is comatose with a GCS of 6, has no focal deficits, and is diaphoretic. Fingerstick glucose is 28 mg/dL. A peripheral intravenous line is in place. Which of the following is the most appropriate next step?

- A. Administer 25 grams of 50% dextrose intravenously, then intravenous thiamine
- B. Administer oral glucose gel delivering 15-20 grams of simple carbohydrate
- C. Administer glucagon 1 mg intramuscularly
- D. Administer 25 grams of 50% dextrose intravenously alone
- E. Administer intravenous thiamine, then 25 grams of 50% dextrose intravenously

Question 34

Endocrine Issues in Neurocritical Care · Cerebral salt wasting versus SIADH after SAH · hard

A 52-year-old woman is on postoperative day 6 after clipping of a ruptured anterior communicating artery aneurysm (Hunt-Hess grade 3). Over 24 hours she becomes lethargic. Serum sodium has fallen from 138 to 126 mEq/L. Serum osmolality is 268 mOsm/kg, urine sodium 68 mmol/L, urine osmolality 480 mOsm/kg. Urine output has been 300-350 mL/hr, net fluid balance is negative 2.6 L, weight is down 2 kg, and BUN and hematocrit have risen. Heart rate is 108/min with orthostatic hypotension and low central venous pressure. Morning cortisol and TSH are normal. Which of the following is the most appropriate management?

- A. Restrict free water intake to 1 L per day
- B. Give 0.9% normal saline plus 3% hypertonic saline with enteral sodium chloride, and add fludrocortisone
- C. Start a vasopressin receptor antagonist
- D. Start furosemide together with sodium chloride tablets
- E. Administer desmopressin 1 mcg intravenously

Question 35

Endocrine Issues in Neurocritical Care · Central diabetes insipidus - vasopressin versus desmopressin · moderate

A 24-year-old man is in the neuro ICU on hospital day 2 after a motorcycle crash causing severe traumatic brain injury (admission GCS 5) with diffuse cerebral edema and an external ventricular drain. Urine output rises to 400 mL/hr for 3 consecutive hours. Urine specific gravity is 1.002, urine osmolality 120 mOsm/kg, serum sodium has risen from 146 to 156 mEq/L, and serum osmolality is 312 mOsm/kg. He remains hypotensive on norepinephrine 0.3 mcg/kg/min, and his intracranial pressure is labile. Which of the following is the most appropriate therapy?

- A. Subcutaneous desmopressin 1-2 mcg every 6 to 8 hours
- B. Intranasal desmopressin 10-40 mcg daily
- C. Oral desmopressin 0.1-0.6 mcg daily
- D. Continuous intravenous vasopressin titrated to urine output, adjusted as often as every 30 minutes
- E. Enteral fludrocortisone 100 mcg daily

Question 36

Endocrine Issues in Neurocritical Care · Myxedema coma · moderate

A 74-year-old woman is admitted in winter after being found unresponsive at home. Temperature is 33.4 C, heart rate 44/min, blood pressure 92/60 mm Hg, respirations 8/min. Examination shows dry coarse skin, non-pitting periorbital edema, macroglossia, and no focal deficits; head CT is unremarkable. Laboratory studies: sodium 126 mEq/L, glucose 54 mg/dL, creatinine 1.6 mg/dL, with leukocytosis and anemia. TSH is markedly elevated and free T4 is undetectable. Her family reports she stopped taking levothyroxine several months ago. Which of the following is the most appropriate next step in management?

- A. Administer intravenous levothyroxine alone
- B. Perform active external rewarming and defer hormone replacement pending repeat thyroid studies
- C. Administer intravenous hydrocortisone 50-100 mg every 8 hours before starting intravenous levothyroxine
- D. Administer oral liothyronine alone
- E. Administer intravenous levothyroxine together with fludrocortisone

Question 37

Endocrine Issues in Neurocritical Care · Pituitary apoplexy with acute adrenal insufficiency · moderate

A 58-year-old man taking warfarin for atrial fibrillation presents with the abrupt onset of severe headache, nausea, and double vision that began 4 hours ago. Examination shows a partial left third nerve palsy, a bitemporal field cut, and reduced visual acuity in the left eye; he is drowsy but arousable. Blood pressure is 82/48 mm Hg and heart rate 118/min despite 2 L of crystalloid. Sodium is 128 mEq/L and glucose is 52 mg/dL. MRI demonstrates hemorrhage into a pituitary macroadenoma with suprasellar extension. Which of the following is the most appropriate immediate treatment?

- A. Intravenous hydrocortisone 100-500 mg as a bolus, followed by a 2-4 mg per hour infusion
- B. Immediate transsphenoidal decompression before any hormonal therapy
- C. Intravenous levothyroxine
- D. Dexamethasone 4 mg every 6 hours started only after cosyntropin stimulation testing returns
- E. Oral fludrocortisone 50-200 mcg daily

Non-Neurological Trauma, Burns, and Thermal Injury

Question 38

Non-Neurological Trauma, Burns, and Thermal Injury · Primary survey - tension pneumothorax decompression · easy

A 22-year-old man is brought to the trauma bay after a high-speed motor vehicle collision. He is agitated and dyspneic. Blood pressure is 84/48 mmHg, heart rate 132 beats/min, respiratory rate 40 breaths/min, and SpO₂ 85% on a non-rebreather mask. Examination shows subcutaneous emphysema over the left chest, absent breath sounds on the left, hyperresonance to percussion on the left, distended neck veins, and tracheal deviation to the right. Immediate needle decompression is planned. At which site should the needle be placed?

- A. Second intercostal space at the midclavicular line on the left
- B. Fifth intercostal space at the midaxillary line on the left
- C. Second intercostal space at the midclavicular line on the right
- D. Fifth intercostal space at the midaxillary line on the right
- E. Seventh intercostal space at the posterior axillary line on the left

Question 39

Non-Neurological Trauma, Burns, and Thermal Injury · Damage control resuscitation and tranexamic acid · moderate

A 34-year-old man arrives 50 minutes after a motorcycle crash. Blood pressure is 86/50 mmHg, heart rate 128 beats/min, and respiratory rate 30 breaths/min. He is pale and diaphoretic. FAST examination shows hemoperitoneum and the pelvic radiograph is normal. Two large-bore intravenous lines are in place, the massive transfusion protocol has been activated, and packed red blood cells, plasma, and platelets are being given in equal ratios while the operating room is prepared. Which of the following is the most appropriate additional pharmacologic intervention?

- A. Infuse 2 L of normal saline before further blood products
- B. Start a norepinephrine infusion titrated to a systolic blood pressure above 120 mmHg
- C. Administer hydroxyethyl starch to expand intravascular volume
- D. Administer tranexamic acid now
- E. Withhold tranexamic acid until at least 6 hours after injury

Question 40

Non-Neurological Trauma, Burns, and Thermal Injury · Massive hemothorax · easy

A 41-year-old man sustains a stab wound to the left lateral chest. Blood pressure is 88/54 mmHg and heart rate is 124 beats/min. Breath sounds are decreased on the left with a dull percussion note. A 36 French thoracostomy tube is placed in the left sixth intercostal space at the midaxillary line, with immediate return of 1,700 mL of blood. Which of the following is the most appropriate next step?

- A. Proceed to operative thoracotomy
- B. Place a second thoracostomy tube in the left chest
- C. Obtain computed tomographic angiography of the chest
- D. Observe and repeat a chest radiograph in 4 hours
- E. Perform pericardiocentesis

Question 41

Non-Neurological Trauma, Burns, and Thermal Injury · Burn resuscitation - rule of nines and Parkland formula · hard

An 80-kg man is rescued from a house fire. He has deep partial-thickness and full-thickness burns involving the entire anterior torso and circumferentially involving the right upper extremity. He is intubated for deep facial burns and a hoarse voice. Using the rule of nines and the Parkland formula, which of the following is the most appropriate initial fluid regimen for the first 24 hours?

- A. Normal saline at approximately 270 mL/hr for the first 8 hours, then approximately 540 mL/hr for the next 16 hours
- B. 5% albumin bolus followed by lactated Ringer's at approximately 300 mL/hr for 24 hours
- C. Lactated Ringer's at approximately 360 mL/hr for the full 24 hours
- D. Lactated Ringer's at approximately 270 mL/hr for the first 8 hours, then approximately 540 mL/hr for the next 16 hours
- E. Lactated Ringer's at approximately 540 mL/hr for the first 8 hours, then approximately 270 mL/hr for the next 16 hours

Question 42

Non-Neurological Trauma, Burns, and Thermal Injury · Abdominal compartment syndrome after burns · moderate

A 25-year-old man with 45% total body surface area burns, including circumferential burns of the torso, has received large-volume crystalloid resuscitation over the first 18 hours. He is intubated and sedated. Peak inspiratory pressures have risen from 26 to 42 cm H₂O, urine output has fallen to 8 mL/hr, and blood pressure is 84/50 mmHg. The abdomen is tense. Bladder pressure is 34 mmHg. Which of the following is the most appropriate next step?

- A. Increase the crystalloid infusion rate
- B. Begin a colloid infusion
- C. Perform escharotomy of the chest and abdomen
- D. Perform immediate decompressive laparotomy
- E. Initiate continuous renal replacement therapy

Question 43

Non-Neurological Trauma, Burns, and Thermal Injury · Accidental hypothermia and rewarming · moderate

A 57-year-old man is found unresponsive outdoors after a night of subfreezing temperatures. On arrival he is pulseless with ventricular fibrillation on the monitor. Esophageal core temperature is 27 degrees Celsius. Three defibrillation attempts have been unsuccessful. Which of the following is the most appropriate management?

- A. Withhold cardiopulmonary resuscitation until the core temperature reaches 32 degrees Celsius
- B. Passive external rewarming with warm dry blankets alone
- C. Aggressive diuresis to manage cold-induced volume shifts
- D. Continue cardiopulmonary resuscitation while initiating active core rewarming
- E. Administer empiric potassium to prevent hypokalemia during rewarming

Nutrition and Metabolism

Question 44

Nutrition and Metabolism · Energy and protein requirements in obesity · hard

A 58-year-old woman is admitted to the neurocritical care unit after aneurysmal subarachnoid hemorrhage and coiling. She is intubated and hemodynamically stable. Height is 157 cm and weight is 82 kg, giving a body mass index of 33 kg/m². Renal function is normal and she is not receiving dialysis. Indirect calorimetry is unavailable. Which of the following is the most appropriate nutrition prescription?

- A. 25 to 30 kcal/kg actual body weight with 1.2 to 2 g/kg actual body weight of protein
- B. 11 to 14 kcal/kg actual body weight with 2 g/kg ideal body weight of protein
- C. 22 to 25 kcal/kg ideal body weight with 2 to 2.5 g/kg ideal body weight of protein
- D. 25 to 30 kcal/kg ideal body weight with 1.2 g/kg ideal body weight of protein
- E. 11 to 14 kcal/kg ideal body weight with 1.2 to 2 g/kg actual body weight of protein

Question 45

Nutrition and Metabolism · Nutrition assessment and biochemical markers · easy

A 70-year-old man is in the neurocritical care unit on hospital day 5 after a large intracerebral hemorrhage. He is receiving continuous enteral nutrition through a nasojejunal tube. Serum albumin is 2.3 g/dL and prealbumin is low; C-reactive protein is elevated. Renal function is normal and urine collection is reliable. Which of the following best assesses the adequacy of his protein provision?

- A. A 24-hour urine urea nitrogen measurement to calculate nitrogen balance
- B. Serial serum albumin levels twice weekly
- C. Serial serum prealbumin levels twice weekly
- D. Serial C-reactive protein levels
- E. A serum transferrin level

Question 46

Nutrition and Metabolism · Timing and route of enteral nutrition · moderate

A 65-year-old woman is admitted after a motor vehicle collision with a Glasgow Coma Scale score of 8 and a right subdural hematoma with 12 mm of midline shift. She undergoes decompressive craniectomy and arrives in the intensive care unit sedated with propofol and fentanyl. She is hemodynamically stable with stable intracranial pressure and normal respiratory parameters. A large-bore nasogastric tube is on free drainage and a nasojejunal tube is in place. Computed tomography showed no significant intra-abdominal injury. Which of the following is the most appropriate nutrition plan?

- A. Start parenteral nutrition now through the central venous catheter
- B. Withhold nutrition until hospital day 7 or until sedation is discontinued
- C. Start a standard polymeric formula through the nasojejunal tube within 24 to 48 hours, counting the 1.1 calories per milliliter delivered by propofol
- D. Start enteral nutrition and check gastric residual volumes every 4 hours to guide advancement
- E. Start an elemental formula through the nasogastric tube and place it on free drainage between feeds

Question 47

Nutrition and Metabolism · Drug and enteral formula interactions · moderate

A 47-year-old man with a traumatic brain injury is receiving continuous enteral nutrition through a nasogastric tube and phenytoin for seizure prophylaxis. Despite appropriate weight-based dosing and good compliance with administration, his total phenytoin levels remain persistently subtherapeutic. Albumin and renal function are normal, and no interacting intravenous medications have been added. Which of the following is the most appropriate management?

- A. Switch the patient to parenteral nutrition
- B. Change to an elemental enteral formula
- C. Discontinue enteral nutrition for 24 hours and recheck the level
- D. Hold enteral nutrition 1 to 2 hours before and after each phenytoin dose and adjust the feeding rate over the rest of the day
- E. Crush an extended-release formulation and give it with the feeds

Question 48

Nutrition and Metabolism · Glycemic control in neurocritical care · moderate

A 71-year-old woman with type 2 diabetes mellitus is admitted 6 hours after an acute middle cerebral artery ischemic stroke. She is receiving enteral nutrition. Fingertick glucose values over the past 12 hours have ranged from 190 to 240 mg/dL. Which of the following glucose targets is most appropriate for this patient?

- A. 80 to 110 mg/dL
- B. 100 to 140 mg/dL
- C. 110 to 140 mg/dL
- D. 140 to 160 mg/dL
- E. Less than 180 mg/dL

Part II: Neurocritical Care

Acute Ischemic Stroke

Question 49

Acute Ischemic Stroke · Pathophysiology: core, penumbra, and ischemic thresholds · easy

A 68-year-old man presents 2 hours after sudden onset of left hemiparesis and neglect. CT angiography shows a right M1 occlusion. CT perfusion demonstrates a small region of severely reduced blood flow surrounded by a much larger region of moderately reduced flow. The stroke team explains that reperfusion therapy is directed at a specific tissue compartment. Which of the following best characterizes the ischemic penumbra?

- A.** Tissue below the infarct-necrosis threshold that is already irreversibly injured within minutes of occlusion
- B.** Tissue with normal cerebral blood flow that is transiently dysfunctional from remote deafferentation
- C.** Collaterally perfused tissue near the ischemic threshold of 18 to 20 mL/100 g/min that will infarct over 12 to 24 hours if flow is not restored
- D.** Tissue with mildly reduced flow that remains above the ischemic threshold and survives regardless of reperfusion
- E.** Tissue that restricts on diffusion-weighted imaging at the time of first presentation

Question 50

Acute Ischemic Stroke · IV alteplase dosing · easy

A 72-year-old woman weighing 80 kg presents 90 minutes after onset of aphasia and right hemiparesis, NIHSS 14. Noncontrast CT shows no hemorrhage, glucose is 118 mg/dL, INR is 1.0, and blood pressure is 168/92 mmHg. She is eligible for intravenous thrombolysis. Which of the following is the correct alteplase regimen?

- A.** 0.9 mg/kg total, maximum 90 mg, given entirely as a bolus over 5 minutes
- B.** 0.9 mg/kg total, maximum 90 mg, with 10% as a bolus over 1 minute and the remainder infused over 1 hour
- C.** 0.9 mg/kg total, maximum 100 mg, with 20% as a bolus over 2 minutes and the remainder infused over 2 hours
- D.** 0.6 mg/kg total, maximum 60 mg, with 15% as a bolus and the remainder infused over 1 hour
- E.** 0.25 mg/kg as a single intravenous bolus, maximum 25 mg

Question 51

Acute Ischemic Stroke · Contraindications to IV alteplase · moderate

A 66-year-old man arrives 2 hours after sudden onset of right hemiparesis and dysarthria, NIHSS 12. Noncontrast CT shows no hemorrhage and no significant edema. Which of the following additional findings, if present, would be an absolute contraindication to intravenous alteplase?

- A. Blood pressure of 178/104 mmHg after two doses of intravenous labetalol
- B. Serum glucose of 210 mg/dL
- C. International normalized ratio of 1.4 in a patient taking warfarin
- D. Elective total knee arthroplasty 10 days ago
- E. Age of 84 years

Question 52

Acute Ischemic Stroke · Blood pressure management after thrombolysis · easy

A 70-year-old man with a left MCA syndrome required two doses of intravenous labetalol to bring his blood pressure below 185/110 mmHg before alteplase was started. The infusion is now complete and his blood pressure has drifted up to 186/104 mmHg in the neurointensive care unit. Which of the following is the correct blood pressure goal for this patient?

- A. Below 220/120 mmHg for the next 24 hours
- B. Below 185/110 mmHg for the duration of the infusion only, with no goal thereafter
- C. Below 180/105 mmHg for 24 hours after the alteplase infusion
- D. Below 160/90 mmHg for 24 hours after the alteplase infusion
- E. Below 140/90 mmHg for 72 hours after the alteplase infusion

Question 53

Acute Ischemic Stroke · Blood pressure management without thrombolysis · moderate

A 74-year-old woman presents 14 hours after last known well with a right MCA syndrome. She is not a candidate for thrombolysis or thrombectomy. Blood pressure is 208/112 mmHg. ECG shows no ST elevation, troponin is normal, and there is no hematuria. Her home medications are metoprolol, clonidine, lisinopril, and amlodipine. Which of the following is the most appropriate blood pressure management?

- A. Withhold the lisinopril and amlodipine, continue metoprolol and clonidine at half the home dose, and treat only if blood pressure exceeds 220/120 mmHg
- B. Start a nicardipine infusion targeting a systolic blood pressure below 140 mmHg
- C. Resume all four home antihypertensives at full dose immediately
- D. Begin an intravenous sodium nitroprusside infusion now
- E. Lower the blood pressure below 185/110 mmHg with intravenous labetalol

Question 54

Acute Ischemic Stroke · Wake-up stroke and DWI-FLAIR mismatch · moderate

A 59-year-old woman awoke at 6:00 am with aphasia and right hemiparesis. She was last seen normal at 11:00 pm. She arrives at 7:30 am with an NIHSS of 9. Noncontrast CT shows no hemorrhage and no established infarct. CT angiography shows no large vessel occlusion. MRI shows a left MCA territory lesion that is hyperintense on diffusion-weighted imaging with no corresponding signal change on fluid attenuated inversion recovery. Which of the following is the most appropriate next step?

- A. Give aspirin 325 mg and admit for observation
- B. Repeat the MRI in 6 hours to look for FLAIR evolution
- C. Proceed to mechanical thrombectomy
- D. Start therapeutic intravenous heparin
- E. Administer intravenous alteplase at 0.9 mg/kg

Question 55

Acute Ischemic Stroke · Extended-window thrombolysis with perfusion selection · hard

A 68-year-old man presents 7 hours after last known well with mild right-sided weakness and dysarthria, NIHSS 8. Noncontrast CT shows no hemorrhage. CT angiography shows no large vessel occlusion. CT perfusion shows an ischemic core of 12 mL and a hypoperfused volume of 60 mL. Which of the following statements best supports the management decision here?

- A. Alteplase is contraindicated beyond 4.5 hours from onset regardless of the perfusion imaging findings
- B. Intravenous alteplase may be offered, since the EXTEND trial showed better functional outcome with perfusion-mismatch selection between 4.5 and 9 hours from onset
- C. Intra-arterial alteplase should be given as salvage therapy in place of the intravenous route
- D. He should proceed to thrombectomy on the basis of the DAWN trial criteria
- E. Tenecteplase is the only thrombolytic that has been studied beyond the 4.5-hour window

Question 56

Acute Ischemic Stroke · Tenecteplase · easy

A comprehensive stroke center uses tenecteplase rather than alteplase for patients with large vessel occlusion who are eligible for thrombectomy. Which of the following is the tenecteplase regimen studied in this population?

- A. 0.25 mg/kg as a single intravenous bolus, maximum 25 mg
- B. 0.9 mg/kg total, maximum 90 mg, with 10% as a bolus and the remainder over 1 hour
- C. 0.5 mg/kg as a single intravenous bolus, maximum 50 mg
- D. 0.4 mg/kg infused over 1 hour with no bolus
- E. 1.0 mg/kg as a single intravenous bolus, maximum 100 mg

Question 57

Acute Ischemic Stroke · Orolingual angioedema after alteplase · moderate

A 63-year-old woman taking lisinopril is 20 minutes into an intravenous alteplase infusion for an acute left MCA stroke when the nurse notes asymmetric swelling of her tongue and lower lip. She is phonating normally with no stridor and oxygen saturation is 98%. Which of the following is the most appropriate immediate management?

- A. Continue the alteplase infusion and give intravenous diphenhydramine alone
- B. Continue the alteplase infusion and observe with an airway cart at the bedside
- C. Stop the alteplase and proceed directly to surgical cricothyrotomy
- D. Stop the alteplase, hold the angiotensin-converting enzyme inhibitor, and give intravenous methylprednisolone, diphenhydramine, and famotidine
- E. Stop the alteplase and administer intravenous protamine sulfate and cryoprecipitate

Question 58

Acute Ischemic Stroke · Endovascular thrombectomy in the early window · moderate

A 70-year-old man presents 3 hours after onset of a left hemispheric syndrome with an NIHSS of 18. Noncontrast CT shows an ASPECTS of 8. CT angiography shows a left internal carotid artery terminus occlusion. Intravenous alteplase has been started and a neurointerventional team is immediately available. Which of the following statements best supports proceeding directly to thrombectomy?

- A. A mandatory observation period after alteplase should first be completed to see whether the deficit improves
- B. Thrombectomy benefit is limited to patients who are ineligible for or did not receive intravenous alteplase
- C. Thrombectomy benefit is restricted to patients younger than 80 years with an NIHSS below 20
- D. Thrombectomy improves recanalization rates but has not been shown to improve functional outcome
- E. Stent-retriever thrombectomy within 6 hours improves functional outcome with a number needed to treat of 2 to 3, across a broad range of ages and NIHSS scores

Question 59

Acute Ischemic Stroke · DAWN trial selection criteria · hard

An 83-year-old woman is brought in 20 hours after last known well with a right hemispheric syndrome and an NIHSS of 14. CT angiography shows a right internal carotid artery terminus occlusion, and automated perfusion software reports an infarct core of 18 mL. Which of the following best describes her candidacy for mechanical thrombectomy?

- A. She is ineligible because thrombectomy trials excluded patients older than 80 years
- B. She is eligible under DAWN group A criteria: age 80 years or older, NIHSS of 10 or more, and infarct core below 21 mL
- C. She is eligible only if the infarct core is below 31 mL and the NIHSS is greater than 20
- D. She is ineligible because the 6-to-24-hour window applies only to basilar artery occlusion
- E. She is eligible under DEFUSE 3, which enrolled patients presenting 6 to 16 hours after last known well

Question 60

Acute Ischemic Stroke · DEFUSE 3 trial selection criteria · moderate

A 61-year-old man presents 9 hours after last known well with an NIHSS of 12. CT angiography shows a left M1 occlusion. CT perfusion shows an ischemic core of 45 mL and a total hypoperfused volume of 110 mL. Which of the following sets of criteria establishes his eligibility for thrombectomy?

- A. Infarct core below 21 mL with an NIHSS of 10 or more
- B. Infarct core below 31 mL with age below 80 years
- C. Infarct core below 70 mL with a ratio of tissue at risk to core greater than 1.8, presenting 6 to 16 hours after last known well
- D. Infarct core of 31 to 50 mL with an NIHSS greater than 20
- E. Any infarct core volume, provided the patient presents within 24 hours

Question 61

Acute Ischemic Stroke · Antithrombotic timing after reperfusion therapy · moderate

A 66-year-old man presents 8 hours after last known well with an M1 occlusion and is taken directly for mechanical thrombectomy without intravenous thrombolysis, achieving complete recanalization. He returns to the neurointensive care unit 6 hours after the procedure, neurologically improved, with a stable examination. The 24-hour follow-up CT has not yet been obtained. Which of the following is the most appropriate antithrombotic management?

- A. Aspirin may be started within the first 24 hours, because antithrombotics are not contraindicated after endovascular therapy alone
- B. Withhold all antithrombotics for 24 hours, because the post-thrombolysis rule applies to any reperfusion therapy
- C. Start a therapeutic intravenous heparin infusion to prevent reocclusion
- D. Load with clopidogrel in addition to aspirin as routine post-stroke therapy
- E. Withhold all antithrombotics for 7 days given the size of the territory at risk

Question 62

Acute Ischemic Stroke · Malignant cerebral edema and decompressive hemicraniectomy · hard

A 48-year-old right-handed man had a left internal carotid artery terminus occlusion with failed recanalization. Thirty hours after onset he becomes progressively drowsy, opening his eyes only to noxious stimulation. Repeat CT shows hypodensity involving more than two-thirds of the left MCA territory including the basal ganglia, with 4 mm of midline shift. Which of the following is the most appropriate management?

- A. Intravenous dexamethasone 10 mg followed by 4 mg every 6 hours
- B. Induced hypothermia to 33 degrees Celsius
- C. Placement of an intracranial pressure monitor, treating only if the pressure exceeds 22 mmHg
- D. Start osmotherapy and proceed urgently to decompressive hemicraniectomy
- E. Defer surgery because the infarct involves the dominant hemisphere

Question 63

Acute Ischemic Stroke · Cerebellar infarction with mass effect and hydrocephalus · moderate

A 57-year-old woman with a right posterior inferior cerebellar artery territory infarct becomes increasingly somnolent on hospital day 2. CT shows cerebellar swelling with effacement of the fourth ventricle and enlarged lateral and third ventricles. Which of the following is the most appropriate management?

- A. Placement of an external ventricular drain as definitive therapy
- B. Mannitol alone with hourly neurological examinations
- C. Dexamethasone with continued observation
- D. Induction of barbiturate coma
- E. Suboccipital decompressive craniectomy, with ventriculostomy added as needed

Question 64

Acute Ischemic Stroke · Glucose management after acute ischemic stroke · easy

A 70-year-old man with type 2 diabetes received intravenous alteplase 6 hours ago for a right MCA stroke. His serum glucose is 265 mg/dL on two consecutive checks in the neurointensive care unit. Which of the following is the appropriate serum glucose target?

- A. 80 to 110 mg/dL using a continuous insulin infusion
- B. 110 to 140 mg/dL using a continuous insulin infusion
- C. 140 to 180 mg/dL
- D. 180 to 220 mg/dL
- E. No treatment unless the glucose exceeds 300 mg/dL

Acute Neuromuscular Disorders

Question 65

Acute Neuromuscular Disorders · Neuromuscular respiratory failure: PFT thresholds and airway decision · moderate

A 37-year-old woman presents with one week of ascending paresthesias and weakness, mild slurring of speech, and dyspnea that is worse when supine. Vital signs: temperature 37.0 C, heart rate 118/min, blood pressure 152/88 mm Hg, respiratory rate 28/min, oxygen saturation 96% on room air. She is anxious and diaphoretic and uses accessory muscles with each breath. Examination shows proximal greater than distal weakness in all four limbs, diffuse areflexia, and a stocking-glove sensory deficit. Chest radiograph is clear. Arterial blood gas: pH 7.42, PaCO₂ 44 mm Hg, PaO₂ 78 mm Hg. Bedside pulmonary function testing performed by a respiratory therapist with an adequate seal shows forced vital capacity 14 mL/kg, maximal inspiratory pressure -23 cm H₂O, and maximal expiratory pressure 32 cm H₂O. Which of the following is the most appropriate next step in management?

- A. Continue close observation with pulmonary function tests every 4 hours
- B. Initiate noninvasive ventilation with bilevel positive airway pressure
- C. Apply a nonrebreather mask at 15 L/min
- D. Perform elective endotracheal intubation and initiate mechanical ventilation
- E. Start high-flow nasal cannula oxygen with incentive spirometry

Question 66

Acute Neuromuscular Disorders · Bedside signs of diaphragmatic failure · easy

A 67-year-old man presents with progressive ascending paresthesias, weakness, and shortness of breath. He is anxious and pauses between words to catch his breath. Heart rate is 110/min and respiratory rate is 26/min with an oxygen saturation of 95%. Contraction of the sternocleidomastoid, pectoral, and latissimus dorsi muscles is noted with each inspiration. He can count only to 12 on a single breath. With one hand on the chest and one on the abdomen in the supine position, the abdomen is seen to move inward during inspiration. Examination also reveals facial diplegia, proximal greater than distal weakness, and areflexia. Which of the following findings is the most specific indicator of diaphragmatic failure?

- A. Facial diplegia
- B. Single-breath count of 12
- C. Sternocleidomastoid contraction with inspiration
- D. Heart rate of 110/min
- E. Inward movement of the abdomen during inspiration

Question 67

Acute Neuromuscular Disorders · GBS subtypes and pathophysiology (AMAN) · moderate

A 42-year-old man is admitted with generalized weakness and dyspnea. Two weeks earlier he returned from Mexico with a diarrheal illness that resolved with supportive care. He developed leg weakness one week later that ascended to involve the arms, followed by dysarthria and choking with meals. Examination shows facial diplegia, a weak cough and gag, proximal and distal weakness in all limbs, diffuse areflexia, and a normal sensory examination. Cerebrospinal fluid shows elevated protein with a normal white cell count. Serum GM1 antibodies are present and stool culture grows *Campylobacter jejuni*. Nerve conduction studies show diffuse reduction in the excitability of motor nerves. Which of the following best describes the mechanism of this patient's neuropathy?

- A. CD4 T cell-mediated attack on myelin proteins P2, P0, and PMP22 with macrophage-mediated demyelination
- B. Antibodies against gangliosides on the axolemma that activate macrophages attacking the nodes of Ranvier
- C. Autoantibodies against presynaptic voltage-gated calcium channels impairing acetylcholine vesicle release
- D. Autoantibody attack on postsynaptic acetylcholine receptors of the neuromuscular junction
- E. Thick-filament myosin loss with muscle fiber inexcitability from an acquired sodium channelopathy

Question 68

Acute Neuromuscular Disorders · GBS immunotherapy: IVIG regimen · easy

A patient with severe Guillain-Barre syndrome is to be treated with intravenous immunoglobulin. Which of the following is the standard dosing regimen for this indication?

- A. 0.4 g/kg per day for 5 days, for a total dose of 2 g/kg
- B. 0.4 g/kg as a single dose
- C. 1 g/kg per day for 5 days, for a total dose of 5 g/kg
- D. 2 g/kg per day for 2 days, for a total dose of 4 g/kg
- E. 0.1 g/kg per day for 10 days, for a total dose of 1 g/kg

Question 69

Acute Neuromuscular Disorders · GBS dysautonomia: adynamic ileus · hard

A 54-year-old man with severe Guillain-Barre syndrome is on day 8 of mechanical ventilation. His course has been complicated by dysautonomia, including two episodes of symptomatic bradycardia to the 30s requiring atropine. He receives fentanyl for severe neuropathic pain. Over 24 hours he develops abdominal distention, absent bowel sounds, and rising gastric residuals with intolerance of enteral feeds. Abdominal radiograph shows diffuse bowel dilation consistent with adynamic ileus, without evidence of mechanical obstruction or perforation. Which of the following is the most appropriate next step in management?

- A. Administer intravenous neostigmine
- B. Administer intravenous metoclopramide
- C. Place a nasogastric tube for intermittent suction and insert a rectal tube
- D. Administer oral pyridostigmine
- E. Begin parenteral nutrition without gastric or rectal decompression

Question 70

Acute Neuromuscular Disorders · GBS dysautonomia: bradyarrhythmia management · hard

A 61-year-old woman with severe Guillain-Barre syndrome is quadriplegic and mechanically ventilated. On hospital day 6, during tracheal suctioning, she develops asystole lasting 9 seconds with loss of pulse, treated with atropine and brief chest compressions. The following day she has a second episode of profound bradycardia with hypotension that occurs at rest, with no identifiable trigger. Telemetry shows markedly reduced R-R interval variability, and she has had recurrent severe hypertensive spells alternating with hypotension. Electrolytes are normal and she is on no nodal blocking agents. Which of the following is the most appropriate next step in management?

- A. Order scheduled intravenous atropine every 6 hours
- B. Start oral propranolol to blunt the autonomic swings
- C. Start a nicardipine infusion to control the hypertensive spells
- D. Administer a repeat course of intravenous immunoglobulin
- E. Arrange elective pacemaker placement

Question 71

Acute Neuromuscular Disorders · Myasthenic crisis: noninvasive ventilation · moderate

A 27-year-old woman with acetylcholine receptor antibody-positive myasthenia gravis presents with 3 days of worsening diplopia, nasal speech, dysphagia, proximal weakness, and dyspnea. Respiratory rate is 30/min, heart rate 113/min, oxygen saturation 94% on room air. She has prominent bulbar weakness with pooling of oral secretions and uses accessory respiratory muscles, but no paradoxical abdominal movement is seen and she is alert and protecting her airway with a weak but effective cough. Bedside pulmonary function testing is unreliable because facial weakness prevents an adequate mouthpiece seal and effort is poor. Arterial blood gas: pH 7.41, PaCO₂ 40 mm Hg, PaO₂ 82 mm Hg. Which of the following is the most appropriate next step in management?

- A. Perform rapid sequence intubation
- B. Initiate noninvasive ventilation with bilevel positive airway pressure
- C. Apply a nonrebreather mask at 15 L/min
- D. Observe with frequent pulmonary function tests alone
- E. Give high-dose intravenous methylprednisolone without ventilatory support

Question 72

Acute Neuromuscular Disorders · Myasthenia gravis: intubation and paralytic dosing · moderate

A 43-year-old man with known myasthenia gravis is in crisis. Despite 12 hours of bilevel positive airway pressure he becomes somnolent with paradoxical abdominal breathing, and arterial blood gas shows pH 7.28 with PaCO₂ 58 mm Hg. The decision is made to perform rapid sequence intubation. Which of the following neuromuscular blocking agents and doses is most appropriate?

- A. Succinylcholine 0.5 mg/kg
- B. Succinylcholine 1 mg/kg
- C. Rocuronium 0.5 mg/kg
- D. Rocuronium 1 mg/kg
- E. Rocuronium 1.5 mg/kg

Question 73

Acute Neuromuscular Disorders · Corticosteroids in myasthenic exacerbation · easy

A 58-year-old woman with generalized myasthenia gravis is admitted with moderate bulbar weakness and dyspnea. She is not intubated, has a PaCO₂ of 38 mm Hg, and is being managed with close monitoring and plasma exchange. The team plans to add high-dose intravenous methylprednisolone. Which of the following adverse effects of this specific therapy should be anticipated?

- A. Clinically meaningful worsening of weakness within days of starting treatment in one-third to one-half of patients
- B. Aseptic meningitis presenting as headache
- C. Acute renal failure
- D. Symptomatic hypocalcemia
- E. Line-related bacteremia

Question 74

Acute Neuromuscular Disorders · ALS in the ICU: secretion and cough management · moderate

A 63-year-old man with bulbar-onset amyotrophic lateral sclerosis diagnosed 2 years ago is admitted to the neurocritical care unit with aspiration pneumonia. He is on antibiotics and nocturnal bilevel positive airway pressure. He is alert, oriented, and states clearly that he does not want a tracheostomy. Examination shows a weak, ineffective cough with copious retained secretions; measured peak cough expiratory flow is 210 L/min. Which of the following interventions is most appropriate for his secretion clearance?

- A.** Proceed with tracheostomy and long-term mechanical ventilation
- B.** Start riluzole to improve cough strength
- C.** Start edaravone to improve respiratory muscle function
- D.** Provide a mechanical cough assist device with chest physiotherapy and nebulized mucolytics
- E.** Perform daily therapeutic bronchoscopy for secretion removal

Question 75

Acute Neuromuscular Disorders · Critical illness neuromyopathy: diagnosis · moderate

A 56-year-old man was admitted with septic shock and multiorgan failure requiring vasopressors, mechanical ventilation, and 5 days of continuous neuromuscular blockade, which was stopped 10 days ago. His infection has resolved, but he now repeatedly fails spontaneous breathing trials. Examination shows flaccid quadriparesis that is symmetric, depressed deep tendon reflexes, and preserved facial and extraocular movements. Serum creatine kinase is 180 U/L, electrolytes are normal, and brain and spine imaging and cerebrospinal fluid studies are unremarkable. Which of the following electrodiagnostic findings would most strongly support the diagnosis of critical illness neuromyopathy?

- A.** A greater than 10% decremental compound muscle action potential response with repetitive nerve stimulation at 2 to 5 Hz
- B.** Elevated cerebrospinal fluid protein with a normal cell count
- C.** Diffuse nerve root enhancement on magnetic resonance imaging of the spine
- D.** Markedly elevated creatine kinase with myoglobinuria
- E.** Severe reduction in compound muscle action potential amplitude with prolonged compound muscle action potential duration and muscle inexcitability on direct muscle stimulation

Clinical Evaluation of Coma and Brain Death

Question 76

Clinical Evaluation of Coma and Brain Death · Metabolic versus structural coma · easy

A 58-year-old woman with alcoholic cirrhosis is brought to the emergency department after two days of worsening confusion and is now unresponsive. Temperature is 37.1 C, blood pressure 118/70 mmHg, heart rate 96/min, and respirations are 26/min with a smooth crescendo-decrescendo pattern interrupted by brief apneic pauses. Pupils are 3 mm and briskly reactive bilaterally. Oculocephalic reflexes are intact. She withdraws symmetrically in all four limbs to nail bed pressure without posturing. Noncontrast head CT is unremarkable. Ammonia is elevated, and EEG shows diffuse slowing with triphasic waves. Which of the following examination findings most strongly supports a metabolic rather than a structural cause of her coma?

- A. The crescendo-decrescendo respiratory pattern with apneic pauses
- B. Symmetric withdrawal to noxious stimulation in all four limbs
- C. Preserved pupillary light reflexes
- D. The unremarkable noncontrast head CT
- E. Intact oculocephalic responses

Question 77

Clinical Evaluation of Coma and Brain Death · Intracranial pressure waveforms and the Monro-Kellie doctrine · moderate

A 24-year-old man with severe traumatic brain injury has an external ventricular drain in place. His baseline intracranial pressure is 15 to 18 mmHg. Over the course of the night, the nurse documents several discrete episodes in which the intracranial pressure abruptly rises to 55 to 60 mmHg and remains sustained at that level for roughly 20 minutes before falling back to baseline. During the episodes his blood pressure rises and he becomes less responsive, and between episodes his examination returns to baseline. The drain waveform is dampened only during the episodes and tracings are otherwise normal. Which of the following best explains these findings?

- A. Cushing reflex from established medullary compression
- B. Plateau waves, also known as Lundberg A waves
- C. Overdamped waveform artifact from a malpositioned external ventricular drain
- D. Nonconvulsive status epilepticus with autonomic accompaniments
- E. Progressive refractory intracranial hypertension from an expanding hematoma

Question 78

Clinical Evaluation of Coma and Brain Death · Locked-in syndrome · moderate

A 63-year-old man with atrial fibrillation who is not anticoagulated develops sudden quadriplegia and anarthria. On examination his eyes are open and he appears awake. He cannot move his face, limbs, or tongue and cannot phonate. He does not move his eyes horizontally, but he reliably blinks once for yes and twice for no and looks up and down on command, answering orientation questions correctly. Pupils are 2 mm and reactive. Which of the following is the most likely site of his lesion?

- A. Bilateral medial thalami
- B. Midbrain tectum and pretectum
- C. Bilateral frontal lobes
- D. Ventral pons
- E. Bilateral ventrolateral medulla

Question 79

Clinical Evaluation of Coma and Brain Death · Respiratory patterns in coma · moderate

A 70-year-old man with a large cerebellar hemorrhage and progressive brainstem compression is being observed off sedation before intubation. His breathing is quick and shallow, with irregular breaths of varying amplitude separated by random periods of apnea. There is no crescendo-decrescendo cycling, and no sustained inspiratory pauses are seen. Which of the following is the most likely level of the lesion producing this respiratory pattern?

- A. Pontomedullary junction
- B. Bilateral diencephalon
- C. Rostral pons and midbrain
- D. Bilateral ventrolateral medulla
- E. Bilateral cerebral hemispheres

Question 80

Clinical Evaluation of Coma and Brain Death · Oculovestibular reflex testing · easy

When performing cold caloric (oculovestibular) testing in a comatose patient with an intact tympanic membrane, to what angle should the head of the bed be elevated in order to obtain maximal stimulation?

- A. Flat, at 0 degrees
- B. 30 degrees
- C. 45 degrees
- D. 60 degrees
- E. 90 degrees

Question 81

Clinical Evaluation of Coma and Brain Death · Emergency management of raised intracranial pressure · moderate

A 55-year-old woman, 80 kg, is on day 3 after a large right middle cerebral artery infarction. She is intubated and sedated. She acutely develops a dilated nonreactive right pupil and extensor posturing on the left. Repeat noncontrast head CT shows 10 mm of midline shift with effacement of the right ambient cistern. Blood pressure is 160/85 mmHg, and the current arterial PaCO₂ is 42 mmHg. Neurosurgery has been called for consideration of decompressive craniectomy. Which of the following is the most appropriate immediate medical therapy?

- A. Dexamethasone 10 mg intravenously followed by 4 mg every 6 hours
- B. Sodium nitroprusside infusion titrated to a systolic pressure below 120 mmHg
- C. Place the bed flat to augment cerebral perfusion pressure
- D. Adjust the ventilator to a target PaCO₂ of 50 to 55 mmHg
- E. Mannitol 20 percent, 0.75 to 1.25 g/kg intravenously to a maximum of 100 g, with the head of bed at 30 degrees

Question 82

Clinical Evaluation of Coma and Brain Death · Glycemic target and the NICE-SUGAR trial · easy

Based on the NICE-SUGAR study, what is the recommended target blood glucose range for a critically ill comatose patient in the neurointensive care unit?

- A. 80 to 110 mg/dL
- B. 110 to 140 mg/dL
- C. 140 to 180 mg/dL
- D. 180 to 220 mg/dL
- E. 220 to 260 mg/dL

Question 83

Clinical Evaluation of Coma and Brain Death · Prerequisites for brain death determination · hard

A 44-year-old man had a witnessed cardiac arrest and underwent targeted temperature management. Fentanyl and midazolam infusions were discontinued 8 hours ago; his hepatic and renal function are normal. Core bladder temperature is now 34.8 C during passive rewarming. He is on norepinephrine with a mean arterial pressure of 55 mmHg. On examination he is comatose with no pupillary, corneal, oculocephalic, cough, or gag reflexes and no motor response to noxious stimulation. A urine drug screen is negative. The intensivist asks you to proceed with brain death determination. Which of the following is the most appropriate next step?

- A. Proceed directly to apnea testing, since the brainstem reflexes are already absent
- B. Obtain a cerebral angiogram and declare brain death if there is no intracranial flow
- C. Obtain an EEG to document electrocerebral silence before completing the examination
- D. Rewarm the patient to a core temperature of at least 36 C, correct the hypotension, and wait at least five elimination half-lives of the sedatives before performing the determination
- E. Arrange for two separate physicians to perform the clinical examination and then declare

Question 84

Clinical Evaluation of Coma and Brain Death · Movements compatible with brain death · moderate

A 31-year-old woman with a massive intracerebral hemorrhage and intraventricular extension is being evaluated for brain death. All prerequisites have been met. She is comatose with absent pupillary, corneal, oculocephalic, oculovestibular, cough, and gag reflexes. During nail bed pressure applied to the right foot, the examiner observes brief flexion at the hip, knee, and ankle on that side. Deep tendon reflexes are present in the legs and an extensor plantar response is noted bilaterally. Which of the following is the most appropriate interpretation?

- A.** These are spinally mediated responses that are compatible with brain death, and apnea testing should proceed
- B.** The motor response indicates preserved brainstem function, and brain death cannot be determined
- C.** The findings represent decorticate posturing and preclude a diagnosis of brain death
- D.** An ancillary blood flow study is now mandatory before the determination can continue
- E.** The examination should be repeated in 24 hours before any further testing

Question 85

Clinical Evaluation of Coma and Brain Death · Apnea testing · moderate

A 52-year-old man with a devastating brainstem hemorrhage is comatose with absent brainstem reflexes, and all prerequisites for brain death determination have been met. He is hemodynamically stable and is not a chronic carbon dioxide retainer. Observation of the ventilator confirms he is not triggering breaths above the set rate. Which of the following describes the appropriate performance of the apnea test and a result consistent with nonfunctional medullary chemoreceptors?

- A.** Preoxygenate with 100 percent FiO₂, adjust the ventilator to a baseline PaCO₂ of 25 to 30 mmHg, deliver oxygen at 10 L/min, observe for 3 minutes, and accept a post-test pCO₂ above 45 mmHg
- B.** Preoxygenate with 40 percent FiO₂, adjust the ventilator to a baseline PaCO₂ of 30 to 35 mmHg, deliver oxygen at 2 L/min, observe for 15 minutes, and accept a post-test pCO₂ above 50 mmHg
- C.** Preoxygenate with 100 percent FiO₂, leave the patient on the ventilator with the rate set to zero, observe for 5 minutes, and accept a post-test pCO₂ rise of 5 mmHg
- D.** Preoxygenate with 100 percent FiO₂, adjust the ventilator to a baseline PaCO₂ of 55 to 60 mmHg, deliver oxygen at 4 to 6 L/min, observe for 8 to 10 minutes, and accept any post-test pCO₂ above the baseline value
- E.** Preoxygenate with 100 percent FiO₂, adjust the ventilator to a baseline PaCO₂ of 40 to 45 mmHg, disconnect and deliver oxygen at 4 to 6 L/min through a catheter in the endotracheal tube, observe for 8 to 10 minutes, and accept a post-test pCO₂ above 60 mmHg or a rise of 20 mmHg

Question 86

Clinical Evaluation of Coma and Brain Death · Prognosis after cardiac arrest · hard

A 62-year-old man with hypertension and diabetes suffers a ventricular tachycardia cardiac arrest with return of spontaneous circulation after 25 minutes. He undergoes targeted temperature management at 32 to 34 C for 24 hours and is rewarmed to normothermia over 12 hours. Off all sedation after complete rewarming, he is comatose with no eye opening to noxious stimulation. Pupillary, corneal, and oculocephalic reflexes are present. He has extensor posturing to noxious stimuli and intermittent myoclonic jerking of the face, eyes, and upper extremities. The family asks about his prognosis. Which of the following findings, if present, would most strongly indicate a poor neurological outcome?

- A. Burst suppression on EEG
- B. Myoclonic jerking of the face and upper extremities
- C. Bilaterally absent N20 responses on median nerve somatosensory evoked potentials
- D. Diffuse slowing with triphasic waves on EEG
- E. Preserved pupillary and corneal reflexes off sedation

Hypoxic-Ischemic Brain Injury After Cardiac Arrest

Question 87

Hypoxic-Ischemic Brain Injury After Cardiac Arrest · Pathophysiology: primary vs secondary injury · easy

A 35-year-old woman suffers a witnessed pulseless electrical activity arrest in the emergency department from a saddle pulmonary embolism and achieves return of spontaneous circulation after alteplase, with an estimated total downtime of 5 minutes. Immediately after resuscitation she has preserved brainstem reflexes and localizes with both upper extremities. Over the next 4 days, despite hemodynamic stability off vasopressors, she becomes comatose with absent corneal reflexes and no motor response to noxious stimulation. A trainee asks what caused the brain injury at the moment circulation ceased. Which of the following best describes the mechanism of the primary ischemic insult?

- A. Loss of inhibitory interneurons with increased glutamatergic transmission
- B. Microvascular dysfunction producing the no-reflow phenomenon
- C. Adenosine triphosphate depletion with failure of transmembrane ionic gradients and anoxic depolarization
- D. Free radical formation and activation of nitric oxide synthase
- E. Systemic inflammatory response with microthrombotic events after reperfusion

Question 88

Hypoxic-Ischemic Brain Injury After Cardiac Arrest · Post-arrest coronary angiography and the COACT trial · moderate

A 58-year-old man has a witnessed out-of-hospital ventricular fibrillation arrest with bystander cardiopulmonary resuscitation and return of spontaneous circulation after two shocks. He is comatose and intubated. Blood pressure is 104/62 mmHg on low-dose norepinephrine, and the postresuscitation 12-lead electrocardiogram shows sinus rhythm with no ST-segment elevation. Troponin is mildly elevated and there is no shock or ongoing arrhythmia. Which of the following is the most appropriate next step?

- A. Admit to the intensive care unit for postarrest care and defer coronary angiography
- B. Proceed immediately to the catheterization laboratory for coronary angiography
- C. Administer systemic thrombolysis before intensive care unit admission
- D. Start a prophylactic amiodarone infusion to prevent recurrent ventricular arrhythmia
- E. Cannulate for extracorporeal cardiopulmonary resuscitation as a rescue strategy

Question 89

Hypoxic-Ischemic Brain Injury After Cardiac Arrest · TTM target selection with spontaneous hypothermia · hard

A 47-year-old man is resuscitated from an out-of-hospital ventricular fibrillation arrest after 12 minutes of downtime. On arrival to the emergency department he is intubated, does not follow commands, and withdraws to noxious stimulation. Esophageal core temperature is 32.4 degrees C, mean arterial pressure is 72 mmHg on low-dose norepinephrine, and there is no active bleeding or coagulopathy. Which of the following is the most appropriate temperature management strategy?

- A. Actively rewarm to 37.5 degrees C at 1 degree C per hour, then maintain controlled normothermia
- B. Rewarm to 36 degrees C over 2 hours and maintain that target for 24 hours
- C. Withhold temperature control because the patient is already below all recommended targets
- D. Maintain a target of 33 degrees C for 24 hours, then rewarm no faster than 0.25 to 0.5 degrees C per hour
- E. Deepen cooling to 30 degrees C for 24 hours to maximize metabolic suppression

Question 90

Hypoxic-Ischemic Brain Injury After Cardiac Arrest · HYPERION trial and nonshockable rhythms · moderate

A 68-year-old woman has an unwitnessed in-hospital cardiac arrest from asphyxia with an initial rhythm of asystole. Return of spontaneous circulation is achieved after 18 minutes. She remains comatose with a Glasgow Coma Scale score of 3 and requires norepinephrine for circulatory shock. Core temperature is 36.8 degrees C. Which temperature strategy is supported by the best available randomized evidence for this patient population?

- A. Controlled normothermia at 37.5 degrees C for 40 hours
- B. Targeted temperature management at 33 degrees C for 24 hours
- C. No active temperature control, with antipyretics as needed
- D. Targeted temperature management at 36 degrees C for 24 hours
- E. Targeted temperature management at 32 degrees C for 48 hours

Question 91

Hypoxic-Ischemic Brain Injury After Cardiac Arrest · TTM2 trial results · moderate

A cardiology colleague asks you to justify your unit's temperature protocol in light of the 2021 Targeted Hypothermia versus Targeted Normothermia after Out-of-Hospital Cardiac Arrest trial, known as TTM2. Which of the following statements most accurately reflects the findings of that trial?

- A. Hypothermia at 33 degrees C reduced 6-month mortality compared with targeted normothermia
- B. Hypothermia at 33 degrees C improved functional outcome but increased major bleeding
- C. The trial enrolled only nonshockable rhythms and demonstrated a benefit of 33 degrees C
- D. The normothermia group achieved its target without any active temperature control devices
- E. Mortality and functional outcome were similar, but arrhythmia causing hemodynamic compromise was more frequent with 33 degrees C, at 24 percent versus 16 percent

Question 92

Hypoxic-Ischemic Brain Injury After Cardiac Arrest · Shivering during targeted temperature management · moderate

A 52-year-old man is cooled to a target of 33 degrees C with external surface pads after an anterior ST-elevation myocardial infarction arrest and successful percutaneous coronary intervention. He is receiving scheduled acetaminophen, fentanyl at 200 mcg per hour, and propofol at 50 mcg per kg per minute. Four hours into the induction phase his esophageal temperature is 37 degrees C, the device water temperature reads 3 degrees C, and water flow through the pads is documented as good. Which of the following is the most appropriate next step?

- A. Exchange the surface pads for an endovascular cooling catheter because of device malfunction
- B. Add scheduled antipyretics for presumed central fever
- C. Administer a bolus of a neuromuscular blocking agent, such as vecuronium 0.1 mg per kg
- D. Lower the device set point to 32 degrees C to increase the temperature gradient
- E. Infuse 2 liters of 4 degrees C normal saline to accelerate induction

Question 93

Hypoxic-Ischemic Brain Injury After Cardiac Arrest · Ventilation and oxygenation targets after ROSC · easy

You are writing the ventilator order set for comatose patients admitted after return of spontaneous circulation. Which of the following sets of targets is most consistent with guideline recommendations and the brain-centered goals for postcardiac arrest syndrome?

- A. Oxygen saturation greater than 94 percent, partial pressure of carbon dioxide 35 to 45 mmHg, avoiding partial pressure of oxygen below 60 mmHg or above 300 mmHg
- B. Oxygen saturation greater than 88 percent, partial pressure of carbon dioxide 25 to 30 mmHg, partial pressure of oxygen above 300 mmHg
- C. Oxygen saturation of 100 percent on a fraction of inspired oxygen of 1.0, partial pressure of carbon dioxide 30 to 35 mmHg
- D. Oxygen saturation greater than 94 percent, partial pressure of carbon dioxide 20 to 30 mmHg, partial pressure of oxygen 60 to 80 mmHg
- E. Oxygen saturation greater than 90 percent, with routine targeted hypercapnia of 50 to 60 mmHg

Question 94

Hypoxic-Ischemic Brain Injury After Cardiac Arrest · Hemodynamic targets after cardiac arrest · easy

A 61-year-old woman is 8 hours out from an out-of-hospital cardiac arrest and is comatose on targeted temperature management. Her mean arterial pressure drifts between 55 and 60 mmHg with a systolic blood pressure in the 80s despite adequate volume resuscitation. Which of the following is the most appropriate blood pressure goal?

- A. Mean arterial pressure 85 to 100 mmHg with mixed venous oxygen saturation 65 to 75 percent
- B. Mean arterial pressure greater than or equal to 50 mmHg
- C. Systolic blood pressure greater than or equal to 140 mmHg
- D. Mean arterial pressure greater than or equal to 65 mmHg or systolic blood pressure greater than or equal to 100 mmHg
- E. Permissive hypotension with no specific target while sedated and cooled

Question 95

Hypoxic-Ischemic Brain Injury After Cardiac Arrest · Postanoxic myoclonus and EEG phenotypes · hard

A 44-year-old man is 36 hours out from a witnessed ventricular fibrillation arrest and remains comatose after rewarming. The bedside nurse reports generalized jerks of the limbs and trunk. Continuous electroencephalography shows a continuous background with narrow, midline vertex-predominant spike-and-wave discharges occurring in lockstep with the jerks. Somatosensory evoked potentials show preserved bilateral N20 peaks. The family asks whether this finding means recovery is impossible. Which of the following is the most accurate statement?

- A. This pattern is pathognomonic of a poor outcome and supports withdrawal of life-sustaining therapy
- B. Roughly half of patients with this electroclinical phenotype achieve functional independence, so aggressive treatment is warranted
- C. The finding is consistent with brain death and confirmatory testing should be arranged
- D. A burst-suppression background with high-amplitude polyspikes in lockstep with the jerks carries the same favorable prognosis
- E. Myoclonus occurring within the first 24 to 48 hours is always subcortical and therefore benign

Question 96

Hypoxic-Ischemic Brain Injury After Cardiac Arrest · Timing of multimodal neuroprognostication · moderate

A 66-year-old man treated with targeted temperature management at 33 degrees C completed rewarming and reached normothermia 12 hours ago. Sedation with midazolam and fentanyl was discontinued at the same time. He is now 48 hours from return of spontaneous circulation. His examination shows extensor posturing to noxious stimulation, giving a Glasgow Coma Scale motor subscore of 2, with preserved pupillary and corneal reflexes. The intensive care team asks whether they can tell the family that the prognosis is hopeless. Which of the following is the most appropriate next step?

- A.** Declare a poor prognosis on the basis of the Glasgow Coma Scale motor subscore of 2
- B.** Obtain somatosensory evoked potentials now and act on absent N20 responses
- C.** Send a neuron-specific enolase level and use a value above the cutoff to declare a poor outcome
- D.** Perform formal brain death determination
- E.** Defer a definitive prognostic statement until at least 72 hours after return to normothermia and discontinuation of sedatives, using multimodal testing

Question 97

Hypoxic-Ischemic Brain Injury After Cardiac Arrest · Neuron-specific enolase as a prognostic biomarker · moderate

In a substudy of the first Targeted Temperature Management trial, neuron-specific enolase measured at 48 hours after return of spontaneous circulation was assessed using the cutoff value of 33 ng per mL recommended by the 2006 American Academy of Neurology guidelines. Which of the following best describes the performance of that cutoff for predicting a poor neurologic outcome?

- A.** Sensitivity 65 percent and specificity 91 percent
- B.** Sensitivity 91 percent and specificity 65 percent
- C.** Sensitivity 100 percent and specificity 100 percent
- D.** Sensitivity 32 percent and specificity 100 percent
- E.** Sensitivity 61 percent and specificity 100 percent

Intracerebral Hemorrhage

Question 98

Intracerebral Hemorrhage · ICH volume by ABC/2 and the ICH Score · moderate

A 72-year-old man with long-standing hypertension is brought to the emergency department 1 hour after sudden left hemiparesis and vomiting. Blood pressure is 208/114 mmHg, heart rate 92 beats/min. His Glasgow Coma Scale score after resuscitation is 8 (E2, V2, M4). Noncontrast head CT shows a right putaminal hemorrhage with extension into the right lateral ventricle. On the axial slice containing the largest hematoma area, the longest diameter is 6 cm and the longest perpendicular diameter is 4 cm; hemorrhage is visible on six 0.5-cm slices, each containing more than 75% of the area of the largest slice. Which of the following is his ICH Score?

- A. 1
- B. 2
- C. 3
- D. 4
- E. 5

Question 99

Intracerebral Hemorrhage · Acute blood pressure control: choice of agent · easy

A 58-year-old man with untreated hypertension presents 90 minutes after sudden onset of right hemiparesis and aphasia. Blood pressure is 212/112 mmHg and heart rate is 88 beats/min. Noncontrast head CT shows a 20-mL left basal ganglia hemorrhage without intraventricular extension. He is awake and protecting his airway. Which of the following is the most appropriate agent for acute blood pressure lowering?

- A. Intravenous hydralazine boluses
- B. Sodium nitroprusside infusion
- C. Sublingual nifedipine
- D. Nicardipine infusion
- E. Intravenous enalaprilat boluses

Question 100

Intracerebral Hemorrhage · Vitamin K antagonist-related ICH reversal · moderate

A 66-year-old woman on warfarin for atrial fibrillation presents 2 hours after acute headache and left-sided weakness. Blood pressure is 186/98 mmHg, INR is 3.4, and platelet count is 220,000/microL. Noncontrast head CT shows a 25-mL right frontal hemorrhage. Which of the following is the most appropriate hemostatic management?

- A. Fresh frozen plasma 10-15 mL/kg intravenously alone
- B. Four-factor prothrombin complex concentrate plus vitamin K 10 mg intravenously
- C. Vitamin K 10 mg intravenously alone
- D. Andexanet alfa bolus followed by a 2-hour infusion
- E. Recombinant activated factor VII 80 mcg/kg intravenously

Question 101

Intracerebral Hemorrhage · Dabigatran-related ICH reversal and dosing · hard

A 74-year-old man taking dabigatran for atrial fibrillation is brought in 2 hours after his last dose with acute aphasia and right hemiparesis. Noncontrast head CT shows a 30-mL left temporal hemorrhage with intraventricular extension. Which of the following is the most appropriate reversal regimen?

- A. Andexanet alfa low-dose bolus followed by a 2-hour infusion
- B. Four-factor prothrombin complex concentrate 25 units/kg intravenously
- C. Protamine sulfate 50 mg intravenously
- D. Fresh frozen plasma 15 mL/kg intravenously
- E. Idarucizumab 5 g intravenously given as two 2.5-g doses no more than 15 minutes apart

Question 102

Intracerebral Hemorrhage · Antiplatelet-associated ICH and platelet transfusion · moderate

A 77-year-old woman taking aspirin 81 mg daily for primary prevention presents with sudden dysarthria and left arm weakness. Blood pressure is 178/92 mmHg and platelet count is 240,000/microL. Noncontrast head CT shows an 18-mL right thalamic hemorrhage without intraventricular extension. She is neurologically stable, and no surgical intervention is planned. Which of the following is the most appropriate management of her antiplatelet exposure?

- A. Transfuse one pool of platelets immediately
- B. Administer desmopressin 0.4 mcg/kg intravenously
- C. Continue standard medical care without platelet transfusion
- D. Transfuse platelets only after a platelet function assay confirms aspirin effect
- E. Administer recombinant activated factor VII

Question 103

Intracerebral Hemorrhage · Blood pressure targets and the INTERACT/ATACH trials · easy

A 62-year-old man presents 2 hours after onset of a 15-mL left thalamic hemorrhage. His initial systolic blood pressure is 190 mmHg and he is awake with a mild right hemiparesis. According to the 2015 American Heart Association and American Stroke Association guidelines for spontaneous ICH, which of the following is the most appropriate acute blood pressure target?

- A. Systolic below 220 mmHg
- B. Systolic below 140 mmHg
- C. Systolic below 180 mmHg
- D. Systolic below 200 mmHg
- E. Systolic below 110 mmHg

Question 104

Intracerebral Hemorrhage · Hemostatic agents: tranexamic acid and TICH-2 · hard

A 59-year-old man with hypertension presents 5 hours after onset of a 22-mL right frontal lobar hemorrhage. He takes no antithrombotic drugs, and his INR and platelet count are normal. A resident asks whether tranexamic acid should be given to limit hematoma expansion. Which of the following best describes the evidence?

- A. Tranexamic acid given as a 1-g bolus followed by 1 g over 8 hours improves 90-day modified Rankin Scale outcomes
- B. Tranexamic acid reduces 90-day mortality by approximately 10% in absolute terms
- C. Tranexamic acid is recommended specifically for patients with a positive spot sign
- D. Tranexamic acid modestly reduces hematoma expansion but does not improve clinical outcome and is not recommended
- E. Tranexamic acid is recommended only for ICH related to oral anticoagulants

Question 105

Intracerebral Hemorrhage · Cerebellar hemorrhage and surgical indications · moderate

A 68-year-old woman with hypertension develops sudden vertigo, vomiting, and truncal ataxia. Noncontrast head CT shows a right cerebellar hemorrhage measuring 3.5 cm in greatest diameter with effacement of the fourth ventricle and early hydrocephalus. Over 45 minutes her Glasgow Coma Scale score falls from 14 to 10. Which of the following is the most appropriate next step?

- A. Surgical evacuation of the cerebellar hematoma as soon as possible
- B. Placement of an external ventricular drain alone with continued observation
- C. Mannitol bolus and repeat CT in 6 hours
- D. Hyperventilation to a PaCO₂ of 25 mmHg maintained for 48 hours
- E. Medical management with a systolic blood pressure target below 140 mmHg alone

Question 106

Intracerebral Hemorrhage · Venous thromboembolism prophylaxis · moderate

A 64-year-old man is admitted with a 25-mL left putaminal hemorrhage. Intermittent pneumatic compression devices were placed on the day of admission. On hospital day 2, a repeat noncontrast head CT shows a stable hematoma, he has no coagulopathy, and his blood pressure is well controlled. Which of the following is the most appropriate next step for venous thromboembolism prevention?

- A. Add graduated compression stockings to the pneumatic compression devices
- B. Continue pneumatic compression alone until hospital day 10
- C. Begin therapeutic-dose low-molecular-weight heparin
- D. Obtain lower extremity duplex ultrasonography before any pharmacologic prophylaxis
- E. Begin subcutaneous unfractionated heparin 5,000 units every 8 hours

Question 107

Intracerebral Hemorrhage · Seizures and antiseizure prophylaxis · easy

A 61-year-old woman is admitted with a 20-mL right parietal lobar hemorrhage. She is alert with a mild left visual field deficit and has had no clinical seizure. Which of the following is the most appropriate approach to antiseizure medication?

- A. Start levetiracetam prophylactically for 7 days
- B. Start phenytoin prophylactically for 30 days
- C. Do not start an antiseizure medication and treat only if a seizure occurs
- D. Start prophylaxis because lobar location alone is an indication
- E. Start prophylaxis and continue indefinitely given the risk of late epilepsy

Question 108

Intracerebral Hemorrhage · Cerebral amyloid angiopathy · easy

An 83-year-old woman with 3 years of progressive memory decline and no history of hypertension presents with transient word-finding difficulty. She takes no antiplatelet or anticoagulant drugs. Blood pressure is 128/74 mmHg. Imaging shows an acute right parietal hemorrhage at the gray-white junction, and susceptibility-weighted MRI demonstrates numerous microbleeds with a lobar predominance. Which of the following is the most likely etiology?

- A. Chronic hypertensive small vessel disease
- B. Cerebral amyloid angiopathy
- C. Hemorrhagic metastasis
- D. Ruptured arteriovenous malformation
- E. Cerebral venous sinus thrombosis with hemorrhagic transformation

Question 109

Intracerebral Hemorrhage · CT angiography spot sign · hard

A 57-year-old man undergoes CT angiography 100 minutes after the onset of a right thalamic hemorrhage. The noncontrast study shows a 14-mL hematoma, and the CTA source images demonstrate a focus of contrast extravasation within the center of the clot. No aneurysm or arteriovenous malformation is identified. Which of the following does this finding best predict?

- A. An occult arteriovenous malformation requiring conventional angiography
- B. A high likelihood of benefit from recombinant activated factor VII
- C. An increased risk of nonconvulsive seizures on continuous EEG
- D. An increased risk of hematoma expansion and higher 3-month mortality
- E. An underlying hemorrhagic neoplasm

Question 110

Intracerebral Hemorrhage · Stepwise medical management of intracranial pressure · moderate

A 49-year-old woman with a large right hemispheric ICH and intraventricular hemorrhage has an external ventricular drain in place. Despite head of bed elevation to 30 degrees, a neutral neck position, adequate analgesia and sedation, and CSF drainage, her intracranial pressure remains 32 mmHg with a stable blood pressure. Which of the following is the most appropriate next step?

- A. Osmotic therapy with mannitol or hypertonic saline
- B. Sustained hyperventilation to a PaCO₂ of 25 mmHg for 48 hours
- C. Pentobarbital coma with continuous EEG monitoring
- D. Induced hypothermia to 33 degrees Celsius
- E. Neuromuscular blockade with continued sedation

Question 111

Intracerebral Hemorrhage · Secondary prevention and resumption of anticoagulation · moderate

A 71-year-old man with nonvalvular atrial fibrillation had a warfarin-associated 20-mL left basal ganglia hemorrhage that was reversed with 4-factor prothrombin complex concentrate and vitamin K. He has no mechanical heart valve. On hospital day 5 he is neurologically stable with a stable hematoma, and the family asks when anticoagulation for stroke prevention can be restarted. Which of the following is the most appropriate recommendation?

- A. Resume oral anticoagulation within 24 hours
- B. Resume oral anticoagulation after 7 to 10 days
- C. Resume oral anticoagulation after 72 hours
- D. Never resume oral anticoagulation after any ICH
- E. Resume oral anticoagulation after 4 weeks

Metabolic Encephalopathies and Delirium

Question 112

Metabolic Encephalopathies and Delirium · Pharmacologic management of ICU delirium · easy

A 68-year-old man is on hospital day 3 in the neurocritical care unit after clipping of a ruptured anterior communicating artery aneurysm. Overnight he becomes agitated, pulls at his lines, and is disoriented and inattentive, with fluctuating symptoms that worsen at night. He has no history of alcohol use. Temperature is 37.0 C, heart rate 96/min, blood pressure 138/78 mm Hg. Serum sodium, glucose, ammonia, and renal function are normal; urinalysis is unremarkable and head CT shows no new lesion. Electrocardiography shows a QTc of 430 msec and the serum magnesium is 2.0 mg/dL. Nonpharmacologic measures have failed to control the agitation. Which of the following is the most appropriate pharmacologic management?

- A. Haloperidol 1 to 5 mg intravenously every 4 to 6 hours with serial QTc monitoring
- B. Lorazepam 4 mg intravenously every 4 hours on a scheduled basis
- C. Olanzapine 20 mg orally daily
- D. Diphenhydramine 50 mg intravenously every 6 hours
- E. Bilateral wrist restraints without pharmacologic therapy

Question 113

Metabolic Encephalopathies and Delirium · Hepatic encephalopathy - first-line ammonia-lowering therapy · moderate

A 52-year-old man with alcohol-related cirrhosis is admitted with 2 days of increasing somnolence. He is lethargic but arousable to voice, disoriented to time, has asterixis, and performs subtraction poorly. He is jaundiced. Serum ammonia is 118 $\mu\text{mol/L}$, international normalized ratio is 1.6, and head CT shows no focal lesion or hydrocephalus. Electroencephalography shows triphasic waves. Which of the following is the most appropriate first-line therapy for his encephalopathy?

- A. Rifaximin 550 mg orally twice daily as monotherapy
- B. Intravenous flumazenil to reverse endozepine activity
- C. L-ornithine-L-aspartate infusion
- D. Lactulose 20 to 30 g orally two to four times daily
- E. Placement of an intraparenchymal intracranial pressure monitor

Question 114

Metabolic Encephalopathies and Delirium · Hyponatremic encephalopathy - correction limits · hard

A 47-year-old malnourished woman with a long history of heavy alcohol use is brought in with 4 days of progressive confusion. She is somnolent and disoriented but has had no seizures. Serum sodium is 104 mEq/L, potassium is 2.9 mEq/L, and a value of 106 mEq/L is documented from an outside clinic 5 days earlier. She is started on careful sodium correction. Which of the following is the maximum increase in serum sodium that should be permitted over the first 24 hours?

- A. 2 mEq/L
- B. 4 mEq/L
- C. 8 mEq/L
- D. 12 mEq/L
- E. 18 mEq/L

Question 115

Metabolic Encephalopathies and Delirium · Wernicke's encephalopathy - thiamine dosing · easy

A 44-year-old woman with metastatic gastric cancer has been receiving chemotherapy and has eaten almost nothing for 3 weeks. She develops inattention, indifference, and impaired memory. Examination shows horizontal nystagmus, bilateral lateral rectus palsies, and truncal ataxia. MRI FLAIR sequences show high signal in the medial thalami, mammillary bodies, periaqueductal gray, and midbrain tectum. Which of the following is the most appropriate initial treatment regimen?

- A. Thiamine 100 mg intravenously once daily
- B. Thiamine 200 to 500 mg intravenously every 8 hours
- C. Thiamine 50 mg orally once daily
- D. Intravenous dextrose infusion followed by oral multivitamins
- E. Methylprednisolone 1 g intravenously daily for 5 days

Question 116

Metabolic Encephalopathies and Delirium · Alcohol withdrawal - timing of syndromes · moderate

A 55-year-old man who drinks a liter of vodka daily was admitted 3 days ago after a fall. His last drink was approximately 72 hours before the current evaluation. He now has a temperature of 38.4 C, heart rate 132/min, blood pressure 176/98 mm Hg, profuse diaphoresis, coarse tremor, and vivid visual and tactile hallucinations, and he is disoriented to place and time. He has received no antipsychotics. Serum sodium, glucose, and ammonia are normal, and head CT is unremarkable. Which of the following is the most likely diagnosis?

- A. Alcohol withdrawal seizures
- B. Wernicke's encephalopathy
- C. Neuroleptic malignant syndrome
- D. Sepsis-associated encephalopathy
- E. Delirium tremens

Question 117

Metabolic Encephalopathies and Delirium · NMDA receptor encephalitis - initial management · moderate

A 22-year-old woman presents after a week of headache, fever, and vomiting, followed by paranoia, delusions, and echolalia. Over the next several days she develops orofacial dyskinesias, choreoathetoid limb movements, sialorrhea, tachycardia, and periods of hyperventilation. CSF shows lymphocytic pleocytosis with mildly elevated protein and oligoclonal bands, brain MRI is unremarkable, and continuous EEG shows rhythmic delta activity with superimposed bursts of 20 to 30 Hz beta activity riding on each delta wave. Serum and CSF N-methyl-D-aspartate receptor antibodies are positive. Which of the following is the most appropriate first step in her treatment?

- A. Imaging of the abdomen and pelvis to detect and remove an ovarian teratoma
- B. Immediate initiation of chronic azathioprine
- C. Rituximab combined with cyclophosphamide as the initial regimen
- D. Continuous propofol and sevoflurane sedation to suppress the dyskinesias
- E. Empiric high-dose acyclovir alone pending repeat CSF testing

Question 118

Metabolic Encephalopathies and Delirium · Posterior reversible leukoencephalopathy syndrome · moderate

A 34-year-old woman is 6 weeks post allogeneic stem cell transplantation and is maintained on tacrolimus. She develops severe headache, vomiting, visual disturbance, and a generalized tonic-clonic seizure. Blood pressure is 196/112 mm Hg. MRI shows bilateral parieto-occipital cortical and subcortical FLAIR hyperintensities without restricted diffusion, and additional patchy frontal lobe involvement. Which of the following is the most appropriate management?

- A. Intravenous thrombolysis for presumed posterior circulation ischemia
- B. Therapeutic anticoagulation for cerebral venous sinus thrombosis
- C. Induced hypertension to augment posterior circulation perfusion
- D. Emergent decompressive suboccipital craniectomy
- E. Blood pressure control and withdrawal of the tacrolimus

Multimodality Neuromonitoring

Question 119

Multimodality Neuromonitoring · Rationale and pathophysiology of secondary brain injury · moderate

A 32-year-old man is admitted to the neurocritical care unit after a high-speed motor vehicle collision. After intubation and adequate resuscitation his Glasgow Coma Scale score is 6 off sedation, with no eye opening and extensor posturing on the left. Blood pressure is 132/74 mmHg, pulse 88 beats per minute, and oxygen saturation 100%. A repeat noncontrast head CT at 6 hours shows stable traumatic subarachnoid hemorrhage and diffuse cerebral edema without herniation or mass effect requiring surgery. The attending proposes placing an intracranial multimodality monitoring bolt, and a trainee asks why serial neurological examinations and repeat CT imaging are not sufficient. Which of the following best explains the pathophysiologic rationale for invasive multimodality monitoring in this patient?

- A.** True ischemic burden after severe traumatic brain injury involves the majority of the brain volume, so global imaging reliably identifies tissue at risk
- B.** Secondary injury develops in patchy islands of metabolically vulnerable tissue in which blood flow is uncoupled from metabolism, and global imaging and the motor examination detect this only after injury is established
- C.** Neuroworsening after traumatic brain injury occurs in fewer than 5% of patients, so monitoring serves mainly to enroll patients in research protocols
- D.** Ischemia after traumatic brain injury is confined to the first 24 hours, after which normal flow-metabolism coupling is restored and monitoring adds little
- E.** Pupillary asymmetry and a declining motor Glasgow Coma Scale subscore are sensitive early markers of regional metabolic distress in comatose patients

Question 120

Multimodality Neuromonitoring · ICP monitoring: treatment threshold and device selection · moderate

A 45-year-old woman is admitted with severe traumatic brain injury and a post-resuscitation Glasgow Coma Scale score of 5. Head CT shows diffuse cerebral edema with slit-like lateral ventricles and no hydrocephalus. Her international normalized ratio is 1.0, platelet count is $210 \times 10^9/L$, and she takes no antiplatelet agents. The team plans intracranial pressure monitoring. Which of the following statements is most consistent with current guidelines and the comparative data on monitor type?

- A.** Treatment should be initiated once intracranial pressure exceeds 15 mmHg, and an external ventricular drain is preferred because it carries a lower infection risk than a parenchymal monitor
- B.** Intracranial pressure should be documented hourly alongside other vital signs, a practice that captures essentially all clinically important elevations
- C.** A ventriculostomy is required in this patient because parenchymal monitors cannot provide reliable measurements when the ventricles are small
- D.** Treatment should be initiated once intracranial pressure exceeds 22 mmHg, and a parenchymal monitor is reasonable here and carries lower hemorrhage and infection risk than an external ventricular drain
- E.** Intracranial pressure monitoring should be omitted because a randomized trial showed that monitored patients received more therapies and had worse functional outcomes

Question 121

Multimodality Neuromonitoring · Brain tissue oxygen: probe validation and the oxygen challenge · easy

A 28-year-old man is admitted after an unhelmeted motorcycle crash with a Glasgow Coma Scale score of 6 after intubation. A right frontal bolt with combined intracranial pressure and brain tissue oxygen catheters was placed 2 hours ago, and a confirmatory noncontrast head CT shows the catheters within subcortical white matter with no periprocedural hemorrhage. Current values are heart rate 80 beats per minute, mean arterial pressure 85 mmHg, intracranial pressure 18 mmHg, cerebral perfusion pressure 67 mmHg, and brain tissue oxygen 7 mmHg. Oxygen saturation is 99% on a fraction of inspired oxygen of 0.35, and hemoglobin is 10.2 g/dL. Which of the following is the most appropriate next step?

- A.** Increase the fraction of inspired oxygen to 1.0 for 10 to 15 minutes and observe the brain tissue oxygen value for a rise
- B.** Administer mannitol 1 g/kg intravenously over 15 minutes to lower the intracranial pressure
- C.** Start norepinephrine to raise the cerebral perfusion pressure above 70 mmHg
- D.** Transfuse packed red blood cells to a hemoglobin above 10 g/dL to increase oxygen delivery
- E.** Remove the bolt and replace the brain tissue oxygen catheter at a contralateral site

Question 122

Multimodality Neuromonitoring · Autoregulation, pressure reactivity index, and optimum CPP · moderate

A 34-year-old woman with Hunt-Hess grade 4 aneurysmal subarachnoid hemorrhage is on postbleed day 4 and postoperative day 3 after coiling. She remains intubated with a Glasgow Coma Scale score of 7. Multimodality monitoring shows intracranial pressure 24 mmHg, mean arterial pressure 80 mmHg, and cerebral perfusion pressure 56 mmHg. The bedside nurse reports that during the neurological examination the mean arterial pressure rose and the intracranial pressure fell precipitously, and that the intracranial pressure rose again when sedation was restarted and the mean arterial pressure fell. She has received a hypertonic saline infusion and her serum sodium is 158 mEq/L; she is euvolemic. Transcranial Doppler shows only very mild proximal vasospasm. Which of the following is the most appropriate next step in managing her intracranial pressure?

- A.** Administer mannitol 1 g/kg intravenously over 15 minutes
- B.** Start a neuromuscular blocking agent to reduce intrathoracic pressure and improve cerebral venous return
- C.** Start a vasopressor to raise the mean arterial pressure by approximately 10 mmHg and reassess the intracranial pressure
- D.** Take the patient to angiography for intra-arterial verapamil
- E.** Hyperventilate to a partial pressure of carbon dioxide of 25 to 30 mmHg

Question 123

Multimodality Neuromonitoring · Cerebral microdialysis: ischemia versus mitochondrial dysfunction · hard

A 40-year-old man is on hospital day 3 after severe traumatic brain injury with right frontal multimodality monitoring in place, including intracranial pressure, brain tissue oxygen, thermal diffusion flowmetry, and cerebral microdialysis catheters. Current values are intracranial pressure 14 mmHg, cerebral perfusion pressure 72 mmHg, brain tissue oxygen 28 mmHg, and regional cerebral blood flow 22 mL/100 g per minute. Serum glucose is 145 mg/dL, hemoglobin is 11.5 g/dL, and temperature is 37.2 degrees Celsius. Microdialysis shows a lactate-pyruvate ratio of 48, a pyruvate of 145 micromol/L, and a brain glucose of 1.6 mmol/L, or 29 mg/dL. Which of the following best explains this pattern?

- A.** Ischemia from inadequate delivery of substrate to the monitored tissue
- B.** Systemic hypoglycemia causing a brain substrate deficit
- C.** Large-vessel vasospasm within the anterior circulation
- D.** Artifact from failure to discard the first hour of collected dialysate
- E.** Mitochondrial dysfunction, also called energy failure, in the monitored tissue

Question 124

Multimodality Neuromonitoring · Brain glucose and glycemic targets · moderate

A 52-year-old woman with severe traumatic brain injury is on hospital day 4 with a right frontal multimodality monitoring bolt. She is on an insulin infusion per a unit protocol targeting a serum glucose of 90 to 110 mg/dL, and her current serum glucose is 100 mg/dL. Intracranial pressure is 14 mmHg, cerebral perfusion pressure is 70 mmHg, brain tissue oxygen is 26 mmHg, and temperature is 37.0 degrees Celsius. Cerebral microdialysis shows a brain glucose of 0.6 mmol/L, or 10.8 mg/dL, a lactate-pyruvate ratio of 32, and a pyruvate of 110 micromol/L. Which of the following is the most appropriate next step?

- A.** Intensify the insulin infusion to keep the serum glucose below 100 mg/dL
- B.** Administer hypertonic saline to lower the intracranial pressure
- C.** Start a vasopressor to raise the cerebral perfusion pressure above 80 mmHg
- D.** Liberalize the serum glucose target to 120 to 180 mg/dL
- E.** Replace the microdialysis catheter because the values are internally inconsistent

Question 125

Multimodality Neuromonitoring · Cortical EEG, invasive seizure detection, and spreading depolarizations · moderate

A 60-year-old man with severe traumatic brain injury and a post-intubation Glasgow Coma Scale score of 6 undergoes emergency craniotomy for evacuation of an acute subdural hematoma. At the end of the case, a six-contact subdural strip electrode is placed over the cortex and connected to a full-band amplifier as part of an integrated multimodality monitoring strategy, and continuous scalp electroencephalography is also running. Which of the following statements about cortical electroencephalography in this setting is correct?

- A.** Continuous scalp electroencephalography is more sensitive than a cortical strip electrode for detecting electrographic seizures
- B.** The strip electrode must be removed in the operating room at the conclusion of monitoring
- C.** Focal seizures are recorded from the cortex in roughly half of monitored patients, and approximately 40% of these are missed by scalp electroencephalography
- D.** Spreading depolarizations propagate at approximately 3 centimeters per second and appear as bursts of fast activity on the cortical recording
- E.** Spreading depolarizations are rare after traumatic brain injury, occurring in fewer than 10% of monitored patients

Question 126

Multimodality Neuromonitoring · BOOST-II and brain oxygen-directed therapy · easy

In the phase II Brain Oxygen Optimization in Severe Traumatic Brain Injury (BOOST-II) randomized trial, which compared management directed at intracranial pressure alone with management directed at both intracranial pressure and brain tissue oxygen, which of the following describes the brain tissue oxygen target used and the principal result?

- A.** A target above 15 mmHg, with a halving of mortality in the combined-management arm
- B.** A target above 20 mmHg, with the proportion of patients experiencing brain tissue hypoxia falling from 44% to 15% and a trend toward improved functional outcome
- C.** A target above 35 mmHg, with no reduction in the burden of brain tissue hypoxia in either arm
- D.** A jugular venous oxygen saturation target above 50%, with a significant improvement in 6-month functional outcome
- E.** A target above 20 mmHg, with a significant mortality reduction offset by a higher rate of acute lung injury

Question 127

Multimodality Neuromonitoring · Relative contraindications to invasive monitoring · easy

A 74-year-old man is admitted after falling down a flight of stairs. His Glasgow Coma Scale score after intubation and resuscitation is 6, and he does not follow commands. His alcohol screen is negative and he has received no long-acting sedative or paralytic agents. Head CT shows bifrontal contusions and diffuse cerebral edema with no surgical lesion. The team plans to place an intracranial multimodality monitoring bolt. Which of the following findings represents a relative contraindication to this procedure?

- A. An arterial lactate of 4.5 mmol/L
- B. A platelet count of $140 \times 10^9/L$
- C. An age of 74 years
- D. A shortened thromboelastography R time
- E. An international normalized ratio of 1.8

Question 128

Multimodality Neuromonitoring · Detecting delayed cerebral ischemia with regional monitoring · hard

A 45-year-old woman with Hunt-Hess grade 4 aneurysmal subarachnoid hemorrhage underwent clipping and remains intubated with a Glasgow Coma Scale score of 7. On postbleed day 3 a multimodality bolt was placed in the left frontal subcortical white matter, contralateral to the craniotomy, at the border zone between the anterior and middle cerebral artery territories, with intracranial pressure, brain tissue oxygen, thermal diffusion flowmetry, and microdialysis catheters. On postbleed day 7 she develops flaccid right leg weakness and abulia. Which of the following sets of monitoring values is most consistent with delayed cerebral ischemia within the monitored region?

- A. Intracranial pressure 17 mmHg, cerebral perfusion pressure 95 mmHg, brain tissue oxygen 7 mmHg, brain glucose 6.3 mg/dL, lactate-pyruvate ratio 42, regional cerebral blood flow 6 mL/100 g per minute
- B. Intracranial pressure 14 mmHg, cerebral perfusion pressure 90 mmHg, brain tissue oxygen 25 mmHg, brain glucose 25 mg/dL, lactate-pyruvate ratio 23, regional cerebral blood flow 25 mL/100 g per minute
- C. Intracranial pressure 18 mmHg, cerebral perfusion pressure 95 mmHg, brain tissue oxygen 18 mmHg, brain glucose 20 mg/dL, lactate-pyruvate ratio 28, regional cerebral blood flow 45 mL/100 g per minute
- D. Intracranial pressure 32 mmHg, cerebral perfusion pressure 70 mmHg, brain tissue oxygen 15 mmHg, brain glucose 23 mg/dL, lactate-pyruvate ratio 38, regional cerebral blood flow 15 mL/100 g per minute
- E. Intracranial pressure 8 mmHg, cerebral perfusion pressure 75 mmHg, brain tissue oxygen 30 mmHg, brain glucose 30 mg/dL, lactate-pyruvate ratio 15, regional cerebral blood flow 22 mL/100 g per minute

Neurocritical Care Nursing Special Considerations

Question 129

Neurocritical Care Nursing Special Considerations · Nursing interventions and intracranial pressure - endotracheal suctioning · moderate

A 58-year-old man with severe traumatic brain injury is intubated and has an intraparenchymal intracranial pressure monitor in place, with pressures running between 12 and 18 mm Hg. His oxygen saturation drifts to 91 percent and coarse secretions are audible in the endotracheal tube. The nurse orienting on the unit is reluctant to suction him for fear of raising the intracranial pressure. Which of the following is the most appropriate approach?

- A.** Withhold endotracheal suctioning entirely while the intracranial pressure monitor is in place
- B.** Hyperoxygenate with up to 100 percent oxygen for 30 to 60 seconds, then limit suctioning to one or two catheter passes of no more than 15 seconds each
- C.** Suction with four to five catheter passes of 30 seconds each to clear the secretions in a single episode
- D.** Suction only after a bolus of mannitol has been given to create intracranial pressure headroom
- E.** Suspend repositioning and chest physiotherapy for 24 hours because both reliably raise intracranial pressure

Question 130

Neurocritical Care Nursing Special Considerations · Enteral nutrition - gastric residual volumes · hard

A 66-year-old woman with a large left middle cerebral artery infarction is receiving continuous nasogastric tube feedings. On a routine check the gastric residual volume is 280 mL, and a repeat check returns 300 mL. Her abdomen is soft and nontender with normal bowel sounds, she has no nausea, vomiting, or distention, and the head of the bed is at 30 degrees. Which of the following is the most appropriate next step?

- A.** Stop the tube feedings and obtain an urgent abdominal radiograph
- B.** Stop the tube feedings permanently and begin parenteral nutrition
- C.** Hold the feedings for 12 hours and then restart at half the previous rate
- D.** Continue the feedings and notify the provider, who may order a promotility agent
- E.** Lower the head of the bed to flat and recheck the residual volume in 1 hour

Question 131

Neurocritical Care Nursing Special Considerations · Targeted temperature management - shivering · easy

A 61-year-old man undergoing targeted temperature management for controlled normothermia after a large hemispheric infarction begins to shiver, confirmed on the Bedside Shiver Assessment Scale. He has been extubated, is taking oral medications, and has a documented history of provoked seizures, so the team wishes to avoid agents that lower the seizure threshold. Counterwarming and adjusting the water-cooling device have not controlled the shivering. Which of the following antishivering medications is available in an oral formulation?

- A. Buspirone
- B. Meperidine
- C. Dexmedetomidine
- D. Magnesium sulfate
- E. Cisatracurium

Question 132

Neurocritical Care Nursing Special Considerations · Targeted temperature management - temperature measurement and devices · moderate

A 47-year-old woman with a subarachnoid hemorrhage is started on targeted temperature management for controlled normothermia after repeated fevers. The unit currently monitors her temperature with a tympanic thermometer and cools her with ice packs and an air-cooling blanket, but the recorded temperature swings widely and repeatedly overshoots below the target. Which of the following changes is most consistent with guideline recommendations?

- A. Continue tympanic monitoring but increase the frequency of readings to every 15 minutes
- B. Switch to axillary temperature monitoring and continue the air-cooling blanket
- C. Switch to oral temperature monitoring and add ice packs to the groin and axillae
- D. Change to an air-cooling blanket alone and monitor with a temporal artery scanner
- E. Switch to an esophageal or bladder temperature probe and use an intravascular catheter or gel-pad device

Neuroimmunology in the Neurocritical Care Unit

Question 133

Neuroimmunology in the Neurocritical Care Unit · Anti-NMDAR encephalitis: timing of tumor resection · moderate

A 22-year-old woman is admitted to the neurocritical care unit after 3 weeks of new psychiatric symptoms and short-term memory loss culminating in generalized seizures and orofacial dyskinesias. She is intubated for airway protection and has episodes of tachycardia to 150 beats/min alternating with sinus pauses. MRI of the brain is unremarkable. Continuous EEG shows an extreme delta brush pattern. CSF shows 28 white blood cells/microliter (88% lymphocytes), protein 48 mg/dL, glucose 62 mg/dL, and negative HSV PCR. Transvaginal ultrasound demonstrates a 4-cm complex right ovarian mass with calcification. Serum and CSF autoantibody panels have been sent, and the laboratory reports a 2- to 3-week turnaround. She has been started on high-dose methylprednisolone. Which of the following is the most appropriate next step in management?

- A. Withhold further immunotherapy and repeat the lumbar puncture in 1 week to document a rising cell count
- B. Administer tocilizumab to suppress the interleukin-6 driven inflammatory response
- C. Proceed with resection of the ovarian mass now, without waiting for the autoantibody results
- D. Defer any surgical intervention until CSF anti-NMDA receptor antibodies return positive
- E. Obtain a stereotactic brain biopsy of the mesial temporal lobe to confirm inflammation before escalating therapy

Question 134

Neuroimmunology in the Neurocritical Care Unit · Intracellular versus surface antigens: T cell-mediated paraneoplastic disease · hard

A 50-year-old woman with newly diagnosed ovarian carcinoma develops 6 weeks of progressive gait ataxia, dysarthria, and limb dysmetria, and is admitted after a fall with aspiration. MRI of the brain shows no cerebellar atrophy or enhancement. CSF shows 12 white blood cells/microliter, protein 62 mg/dL, and two oligoclonal bands not present in serum. Serum anti-Purkinje cell cytoplasmic antibody type 1 (anti-Yo) is positive. Which of the following is the most appropriate therapeutic strategy?

- A. Observation with serial examinations, since her tumor burden is small and the neurologic deficit is mild
- B. Physical therapy and referral to gynecologic oncology for tumor management alone
- C. Intravenous immunoglobulin alone, because antibody removal is the definitive therapy for this syndrome
- D. Aggressive immunotherapy directed at T cell-mediated pathology combined with aggressive treatment of the malignancy
- E. Plasma exchange alone, because circulating pathogenic autoantibodies drive the cerebellar injury

Question 135

Neuroimmunology in the Neurocritical Care Unit · CSF markers of CNS inflammation · moderate

A 45-year-old man is admitted with 8 weeks of progressive confusion, apathy, and two witnessed seizures. Toxicology, ammonia, thyroid studies, and vitamin B12 are normal. MRI of the brain shows subtle bilateral mesiotemporal T2/FLAIR hyperintensity without enhancement. Lumbar puncture shows 3 white blood cells/microliter, protein 38 mg/dL, and glucose 64 mg/dL with a serum glucose of 96 mg/dL. HSV-1 and HSV-2 PCR and metagenomic sequencing are negative, and cytology and flow cytometry are normal. Five oligoclonal bands are present in CSF and absent from a simultaneously drawn serum sample, and the IgG index is 0.85. Which of the following is the most appropriate interpretation of these results?

- A.** The oligoclonal bands and elevated IgG index indicate intrathecal immunoglobulin production and support starting empiric immunotherapy
- B.** The normal cell count and protein exclude an inflammatory process, so immunotherapy should be withheld
- C.** The oligoclonal band pattern is specific for multiple sclerosis and establishes that diagnosis
- D.** The IgG index is the more sensitive and specific test, so the oligoclonal band result can be disregarded
- E.** A CSF glucose below 45 mg/dL would be required before an inflammatory cause could be entertained

Question 136

Neuroimmunology in the Neurocritical Care Unit · Search for occult malignancy in paraneoplastic syndromes · easy

An 80-year-old woman with a 50-pack-year smoking history is admitted with 5 weeks of progressive truncal ataxia and opsoclonus. MRI of the brain with and without gadolinium is normal. CT of the chest, abdomen, and pelvis shows no mass. CSF shows 22 white blood cells/microliter with 90% lymphocytes and protein 68 mg/dL, with negative cytology and flow cytometry. Which of the following is the most appropriate next step in the workup?

- A.** Observe clinically and repeat CT of the chest, abdomen, and pelvis in 1 year
- B.** Obtain whole-body fluorodeoxyglucose PET/CT to search for an occult tumor
- C.** Obtain MR spectroscopy of the cerebellum to characterize the cerebellar dysfunction
- D.** Obtain continuous EEG to evaluate for nonconvulsive seizures as the cause of the ataxia
- E.** Obtain a stereotactic biopsy of the cerebellum to establish the inflammatory diagnosis

Question 137

Neuroimmunology in the Neurocritical Care Unit · Neurologic immune-related adverse events of checkpoint inhibitors · moderate

A 50-year-old man with metastatic melanoma who began pembrolizumab 6 weeks ago presents with 10 days of progressive ptosis, binocular diplopia, and fatigue that worsens through the day. He is now using accessory muscles and his negative inspiratory force is declining. Laboratory studies show creatine phosphokinase 2,400 units/liter, TSH 0.02 microunits/milliliter with a low free T4, and sodium 126 milliequivalents/liter. MRI of the brain shows no metastatic lesions but demonstrates an enlarged, enhancing pituitary gland. Which of the following is the most appropriate management?

- A. Continue pembrolizumab and add low-dose steroids for symptomatic relief of perilesional inflammation
- B. Discontinue pembrolizumab and switch to an alternative cytotoxic regimen for progressive melanoma
- C. Continue pembrolizumab and obtain a chest CT with plans for thymectomy
- D. Continue pembrolizumab and treat with tocilizumab for the cytokine-mediated syndrome
- E. Discontinue pembrolizumab and begin high-dose glucocorticoids

Question 138

Neuroimmunology in the Neurocritical Care Unit · CAR T cell-associated neurotoxicity: pharmacologic management · hard

A 61-year-old man with refractory diffuse large B-cell lymphoma received CD19-directed CAR T cell therapy 5 days ago. He was treated for cytokine release syndrome on day 2 with tocilizumab, and his fevers and hypotension resolved. On day 5 he becomes aphasic, then rapidly obtunded, and is intubated. Head CT shows diffuse sulcal effacement and loss of gray-white differentiation consistent with cerebral edema. He has received high-dose methylprednisolone. Which of the following is the most appropriate additional therapy for his neurotoxicity?

- A. Antithymocyte globulin
- B. Repeat dosing of tocilizumab
- C. Natalizumab
- D. Intravenous immunoglobulin
- E. Eculizumab

Question 139

Neuroimmunology in the Neurocritical Care Unit · Induction immunotherapy dosing · easy

Which of the following correctly states a standard induction regimen for acute immunomodulatory therapy in autoimmune neurologic disease?

- A. Methylprednisolone 125 mg intravenously daily for 14 days
- B. Plasma exchange of 0.25 plasma volumes daily for 10 consecutive days
- C. Rituximab 100 mg intravenously once monthly for 6 months
- D. Azathioprine 10 mg/kg/day started at full dose on day 1
- E. Intravenous immunoglobulin 2 g/kg divided over 3 to 5 days

Neuroinfectious Disease

Question 140

Neuroinfectious Disease · Bacterial meningitis - empiric antimicrobial therapy · moderate

A 68-year-old man with type 2 diabetes is brought to the emergency department with 12 hours of fever, headache, and progressive confusion. Temperature is 39.2 C, heart rate 112/min, blood pressure 104/62 mm Hg. He is oriented to person only and has marked nuchal rigidity without focal deficits. Noncontrast head CT is unremarkable. Lumbar puncture shows an opening pressure of 28 cm H₂O, CSF WBC 4,200/microliter with 92% neutrophils, glucose 18 mg/dL with a simultaneous serum glucose of 118 mg/dL, and protein 260 mg/dL. Gram stain and cultures are pending. Which of the following is the most appropriate empiric antimicrobial regimen?

- A. Ceftriaxone alone
- B. Vancomycin plus ceftriaxone
- C. Vancomycin plus ceftriaxone plus ampicillin
- D. Vancomycin plus ampicillin
- E. Ampicillin plus gentamicin

Question 141

Neuroinfectious Disease · Bacterial meningitis - adjunctive dexamethasone · easy

Adjunctive corticosteroid therapy is recommended in adults with suspected or confirmed pneumococcal meningitis. Which of the following describes the recommended dexamethasone regimen and its timing relative to antimicrobial therapy?

- A. 0.15 mg/kg every 6 hours for 2-4 days, with the first dose 10-20 minutes before or concomitant with the first antibiotic dose
- B. 0.15 mg/kg every 12 hours for 7 days, beginning once CSF cultures confirm the organism
- C. 0.30 mg/kg every 6 hours for 4-6 days, beginning 24 hours after the first antibiotic dose
- D. 0.50 mg/kg every 8 hours for 3-5 days, beginning with the second antibiotic dose
- E. 0.75 mg/kg every 8 hours for 1-3 days, given only if the CSF Gram stain shows gram-positive diplococci

Question 142

Neuroinfectious Disease · Bacterial meningitis - neuroimaging before lumbar puncture · moderate

Five patients present to the emergency department with fever and headache, and each is suspected of having bacterial meningitis. Empiric antibiotics are ordered for all of them. Which of the following findings most strongly indicates that head CT should be obtained before lumbar puncture?

- A. Severe nuchal rigidity with a positive Kernig sign
- B. A petechial rash over the trunk and extremities
- C. Photophobia with two episodes of emesis
- D. Temperature of 39.8 C with a peripheral WBC of 24,000/microliter
- E. A new left sixth cranial nerve palsy in a patient taking antiretroviral therapy for HIV

Question 143

Neuroinfectious Disease · Viral encephalitis - HSV treatment and dosing · moderate

A 44-year-old woman presents with 3 days of fever, headache, and malaise followed by confusion and an olfactory hallucination. Temperature is 38.9 C. Exam shows expressive aphasia and a right pronator drift. MRI shows T2 hyperintensity in the left medial temporal lobe and insula. EEG shows left temporal spike and slow wave activity. CSF shows 180 WBC/microliter with lymphocytic predominance, protein 92 mg/dL, glucose 62 mg/dL, and 90 RBC/microliter; HSV PCR is pending. Which of the following is the most appropriate therapy?

- A. Acyclovir 5 mg/kg IV every 8 hours for 7 days
- B. Acyclovir 10 mg/kg IV, dosed on ideal body weight, every 8 hours for 14-21 days
- C. Ganciclovir 5 mg/kg IV every 12 hours for 21 days
- D. Foscarnet 60 mg/kg IV every 8 hours for 14 days
- E. Ganciclovir plus foscarnet for 21 days

Question 144

Neuroinfectious Disease · Recurrent aseptic meningitis - HSV-2 · easy

A 55-year-old immunocompetent man presents with headache, photophobia, and nuchal rigidity. He reports this is the third such episode in 2 years; he received an antiviral after the first episode and has been on no therapy since, recovering completely each time. He recalls genital lesions about a week before the first episode. Temperature is 37.9 C and the neurologic exam is nonfocal. CSF shows 220 WBC/microliter with lymphocytic predominance, normal glucose, and mildly elevated protein. Which of the following is the most likely causative organism?

- A. Herpes simplex virus type 1
- B. Varicella-zoster virus
- C. Human herpesvirus 6
- D. Herpes simplex virus type 2
- E. Epstein-Barr virus

Question 145

Neuroinfectious Disease · Arboviral encephalitis - West Nile virus · easy

An 85-year-old woman is brought in confused and unable to walk. She was last seen at her baseline 5 days earlier on a camping trip. Temperature is 38.6 C. She is oriented to person only, has a coarse tremor and multifocal myoclonus, and has flaccid, areflexic paraplegia with preserved sensation. CSF shows 90 WBC/microliter with lymphocytic predominance, protein 88 mg/dL, and normal glucose. MRI shows T2 and FLAIR hyperintensity in the substantia nigra, basal ganglia, and thalami. Which of the following is the most likely pathogen?

- A. Herpes simplex virus type 1
- B. Cytomegalovirus
- C. Epstein-Barr virus
- D. Human herpesvirus 6
- E. West Nile virus

Question 146

Neuroinfectious Disease · Brain abscess - surgical versus medical management · moderate

A 47-year-old man with long-standing chronic otitis media presents with 10 days of progressive right-sided headache, low-grade fever, and 2 days of left arm weakness. He has had no prior neurosurgery and no head trauma. Contrast head CT shows a 3.5 cm ring-enhancing lesion in the right temporal lobe with surrounding hypodensity and 6 mm of midline shift; MRI shows a hyperintense DWI signal with low ADC within the lesion. Which of the following is the most appropriate management?

- A. Surgical aspiration for organism identification and mass effect reduction, plus empiric vancomycin, ceftriaxone, and metronidazole
- B. Empiric vancomycin, ceftriaxone, and metronidazole alone, with repeat imaging in 2 weeks
- C. Lumbar puncture for CSF culture, with antibiotics withheld until an organism is identified
- D. Empiric vancomycin, cefepime, and metronidazole alone for 6-8 weeks
- E. Dexamethasone and observation pending blood culture results

Question 147

Neuroinfectious Disease · Subdural empyema - diagnosis and imaging · hard

A 34-year-old man underwent a right frontal craniotomy for tumor resection 3 weeks ago. He returns with headache, fever to 38.7 C, meningismus, and a single generalized seizure. CT shows a crescentic hypodense extra-axial collection over the right convexity with peripheral enhancement. MRI shows the collection to be hypointense on T1 and hyperintense on T2, with reduced diffusion and an ADC value lower than that of normal cortical gray matter. Which of the following is the most likely diagnosis?

- A. Reactive subdural effusion
- B. Chronic subdural hematoma
- C. Subdural empyema
- D. Postoperative pseudomeningocele
- E. Epidural abscess with dural enhancement

Question 148

Neuroinfectious Disease · Healthcare-associated ventriculitis - empiric therapy · moderate

A 23-year-old man with a severe traumatic brain injury has had an external ventricular drain in place for 8 days. On ICU day 8 he becomes febrile to 38.8 C and his neurologic exam declines from following commands to localizing only. CSF from the drain shows 640 WBC/microliter with neutrophil predominance, glucose 32 mg/dL, and protein 180 mg/dL; Gram stain and culture have been sent. Which of the following is the most appropriate empiric antimicrobial regimen?

- A. Ceftriaxone plus vancomycin
- B. Vancomycin plus cefepime
- C. Meropenem plus linezolid
- D. Ceftazidime plus nafcillin
- E. Ampicillin plus gentamicin

Question 149

Neuroinfectious Disease · CSF lactate in the postoperative neurosurgical patient · hard

A 60-year-old woman is on postoperative day 4 from a posterior fossa tumor resection with an external ventricular drain in place. She develops a fever of 38.5 C and increased somnolence. CSF sampled from the drain shows 210 WBC/microliter, protein 96 mg/dL, glucose 46 mg/dL with a serum glucose of 128 mg/dL, and 3,400 RBC/microliter; Gram stain is negative and cultures are pending. The team notes that her blood-brain barrier injury and hemorrhagic CSF make the routine cell count, protein, and glucose difficult to interpret. Which of the following CSF findings would best support initiating empiric antimicrobial therapy while cultures are pending?

- A. A CSF red blood cell count above 1,000/microliter
- B. A CSF protein above 50 mg/dL
- C. A CSF opening pressure above 20 cm H₂O
- D. A CSF lactate concentration of 4.0 mmol/L or higher
- E. A single positive CSF culture with an otherwise normal CSF cell count, glucose, and protein

Question 150

Neuroinfectious Disease · Cryptococcal meningitis - hydrocephalus and CSF sampling · hard

A 45-year-old man with HIV who has been off antiretroviral therapy is found down by police and brought to the emergency department. Temperature is 38.9 C, blood pressure 128/74 mm Hg. He opens his eyes to voice, has a gait he cannot demonstrate because he will not stand, and has bilateral sixth nerve palsies without meningismus. CD4 count is 50 cells/microliter. Noncontrast head CT shows no ischemia or hemorrhage but demonstrates ventriculomegaly with transependymal flow. Which of the following is the most appropriate next step?

- A. Perform an immediate lumbar puncture for opening pressure and cryptococcal antigen
- B. Start fluconazole 800 mg IV daily as sole antifungal therapy
- C. Consult neurosurgery for placement of an external ventricular drain
- D. Send an HIV viral load and defer further intervention until it returns
- E. Begin high-dose dexamethasone for presumed inflammatory cerebral edema

Neurorehabilitation in the Neurocritical Care Unit

Question 151

Neurorehabilitation in the Neurocritical Care Unit · Timing of early mobilization after stroke (AVERT) · moderate

A 68-year-old woman is admitted to the neurocritical care unit 6 hours after onset of right hemiparesis and aphasia from a left middle cerebral artery infarct. She received intravenous thrombolysis. Her blood pressure is 148/82 mmHg, heart rate 78 and regular, and she is afebrile. NIHSS is 12. Repeat CT shows no hemorrhage. Eight hours after admission the nursing team asks whether the patient should be taken out of bed for frequent, high-dose mobilization sessions beginning now. Which of the following is the most appropriate approach to mobilization in this patient?

- A.** Begin frequent high-dose out-of-bed mobilization immediately, because very early mobilization improved 3-month outcomes in the AVERT trial
- B.** Use caution with out-of-bed mobilization during the first 24 hours while arranging a rehabilitation team assessment to determine candidacy for mobilization after the first day
- C.** Withhold any rehabilitation assessment or therapy involvement until at least 7 days after stroke onset
- D.** Mobilize aggressively only if her modified Rankin Scale score is greater than 2 at 24 hours
- E.** Defer all mobilization decisions until a videofluoroscopic swallow study has been completed

Question 152

Neurorehabilitation in the Neurocritical Care Unit · Posttraumatic agitation after TBI · hard

A 24-year-old man is in the neurocritical care unit on hospital day 6 after a severe traumatic brain injury with a right frontal contusion. He is arousable and follows simple commands intermittently but is restless, strikes at nursing staff, and pulls at his lines. Temperature is 37.1 C, white blood cell count is normal, urinalysis is negative, and there is no history of alcohol or benzodiazepine use. His agitation is tracked with the Agitated Behavior Scale. Scheduled acetaminophen has been started, opioids are timed before therapy sessions, the room has been made low-stimulus, and restraints have been minimized, but the agitated behavior persists. Which of the following is the most appropriate first-line medication?

- A.** Scheduled lorazepam
- B.** Haloperidol
- C.** Propranolol
- D.** Buspirone
- E.** Diphenhydramine

Question 153

Neurorehabilitation in the Neurocritical Care Unit · Autonomic dysreflexia in spinal cord injury · moderate

A 42-year-old woman with a T4 ASIA A paraplegia sustained six months ago returns to the ICU after uneventful revision spinal surgery. She arrives from the postanesthesia care unit on 2 liters per minute of supplemental oxygen with a peripheral intravenous line but no intravenous fluids running; her indwelling urinary catheter was removed just before transfer. Thirty minutes after arrival she reports a pounding headache, and she is flushed and sweating profusely above the level of her lesion with piloerection and nasal congestion. Blood pressure is 230/110 mmHg and heart rate is 54. Which of the following is the most appropriate immediate action?

- A. Administer intravenous morphine for presumed postoperative pain
- B. Administer intravenous naloxone for presumed opioid withdrawal
- C. Administer intravenous labetalol to lower the blood pressure urgently
- D. Lay the patient flat and place her in Trendelenburg position
- E. Sit the patient upright and immediately drain the bladder by replacing the urinary catheter

Question 154

Neurorehabilitation in the Neurocritical Care Unit · SSRIs and poststroke depression · easy

A 55-year-old woman with hypertension and diabetes is in the ICU after a large right middle cerebral artery infarct. She is awake and alert with a flat affect and dense left hemiparesis. Her family has read that fluoxetine helps stroke recovery and asks whether she should be started on it now. Which of the following is the most appropriate recommendation?

- A. Screen her with a validated depression instrument such as the 9-item Patient Health Questionnaire and start an antidepressant only if the screen is positive
- B. Start fluoxetine empirically in all stroke patients, based on the FLAME trial
- C. Avoid antidepressants entirely, since they are contraindicated in the immediate poststroke period
- D. Start fluoxetine for motor recovery based on the FOCUS trial, which demonstrated improved functional outcomes
- E. Defer any depression screening until at least 6 months after stroke, when poststroke depression typically appears

Pediatric Neurocritical Care

Question 155

Pediatric Neurocritical Care · Pediatric arterial ischemic stroke - acute recanalization · moderate

A 9-year-old previously healthy girl is brought to the emergency department 3 hours after the acute onset of right face and arm weakness with difficulty speaking. She had been entirely well the week before. Temperature is 37.0 C, heart rate 92/min, blood pressure 106/62 mmHg (persistently below the 95th percentile for age), and fingerstick glucose is 96 mg/dL. Her Pediatric NIH Stroke Scale score is 12. Non-contrast head CT shows no hemorrhage, and CT angiography demonstrates occlusion of the left middle cerebral artery M1 segment with a corresponding early infarct. Platelet count is 240,000/microL, PT is 13 seconds (INR 1.0), and PTT is normal. She has had no recent surgery, trauma, intracranial hemorrhage, or bleeding. Which of the following is the most appropriate next step in management?

- A.** Withhold thrombolysis because intravenous alteplase is contraindicated in all patients younger than 18 years
- B.** Consider intravenous alteplase, as she meets the typical selection criteria used for thrombolysis in children
- C.** Proceed directly to mechanical thrombectomy, which is the established standard of care for pediatric large vessel occlusion
- D.** Begin aspirin 3 to 5 mg/kg/day and admit for observation without recanalization therapy
- E.** Elevate the head of the bed to 45 degrees and induce hypertension to a systolic pressure 20 percent above the 95th percentile for age

Question 156

Pediatric Neurocritical Care · Post-cardiac arrest syndrome - targeted temperature management · moderate

A 4-year-old boy is resuscitated after an unwitnessed pool submersion. The first documented rhythm was pulseless electrical activity, and return of spontaneous circulation was achieved after 12 minutes of CPR. On arrival in the pediatric ICU he is intubated and comatose, with a core temperature of 38.1 C, blood pressure 88/50 mmHg on a low-dose epinephrine infusion, PaCO₂ 40 mmHg, and SpO₂ 97 percent. Continuous EEG has been started. Which of the following temperature strategies is most appropriate?

- A.** Targeted temperature management to prevent fever, maintaining core temperature between 36.0 C and 37.5 C for the first 48 hours
- B.** Therapeutic hypothermia to 33 C for 48 hours, which has been shown to reduce mortality compared with normothermia
- C.** No active temperature control, treating fever with acetaminophen only if the temperature exceeds 39 C
- D.** Deep hypothermia to 28 to 30 C for 24 hours followed by rewarming at 2 C per hour
- E.** Continuous neuromuscular blockade with sustained hypothermia to 32 C for 7 days

Question 157

Pediatric Neurocritical Care · Benzodiazepine-refractory convulsive status epilepticus · easy

A 5-year-old boy weighing 20 kg is brought to the emergency department with generalized tonic-clonic convulsions that began 20 minutes earlier. He receives two doses of intravenous lorazepam 0.1 mg/kg given 5 minutes apart. Ten minutes after the second dose he continues to convulse. His airway is patent, oxygen saturation is 96 percent on a non-rebreather mask, blood pressure is 96/58 mmHg, and glucose is 104 mg/dL. Which of the following is the most appropriate next step?

- A. Begin a propofol infusion titrated to seizure suppression
- B. Give a third dose of intravenous lorazepam 0.1 mg/kg
- C. Give pentobarbital 5 mg/kg by intravenous push
- D. Load with midazolam and begin a continuous midazolam infusion at 100 mcg/kg/hour
- E. Give levetiracetam 60 mg/kg intravenously, to a maximum of 4500 mg

Question 158

Pediatric Neurocritical Care · Severe TBI - tiered management of intracranial hypertension · hard

An 8-year-old girl with severe traumatic brain injury (post-resuscitation GCS 5) is in the pediatric ICU on day 2 with an external ventricular drain and an intraparenchymal ICP monitor. Her head is midline, the head of bed is at 30 degrees, the cervical collar has been loosened, and she is receiving fentanyl and midazolam infusions. Core temperature is 36.8 C, PaCO₂ is 38 mmHg, SpO₂ is 98 percent, and continuous EEG shows no seizures. Mean arterial pressure is 88 mmHg. Despite intermittent CSF drainage, her ICP has been 27 to 29 mmHg for the past 10 minutes. Serum sodium is 141 mEq/L and measured osmolality is 295 mOsm/L. Which of the following is the most appropriate next step?

- A. Begin a pentobarbital infusion titrated to burst suppression with an interburst interval of 3 to 5 seconds
- B. Institute sustained hyperventilation to a PaCO₂ of 25 to 30 mmHg
- C. Give a 3 percent hypertonic saline bolus of 2 to 5 mL/kg over 10 to 20 minutes, then start a continuous infusion at 0.1 to 1 mL/kg/hour
- D. Take her immediately for decompressive craniectomy
- E. Lower the head of bed to 0 degrees and give a 20 mL/kg normal saline bolus to raise cerebral perfusion pressure

Question 159

Pediatric Neurocritical Care · Postoperative diabetes insipidus · moderate

A 7-year-old boy is in the pediatric ICU on the evening of a transsphenoidal resection of a craniopharyngioma. Over the past 3 hours his urine output has been 6 mL/kg/hour. Urine osmolality is 120 mOsm/kg with a specific gravity of 1.003. Whole blood sodium is 149 mEq/L and calculated serum osmolality is 305 mOsm/kg. He is drowsy and cannot reliably drink. Heart rate is 128/min and blood pressure is 96/54 mmHg. No mannitol has been given. Which of the following is the most appropriate next step?

- A.** Restrict free water and begin a 3 percent saline infusion for the syndrome of inappropriate antidiuretic hormone secretion
- B.** Give a 20 mL/kg bolus of 0.9 percent saline and begin fludrocortisone for cerebral salt wasting
- C.** Observe with hourly sodium measurements and withhold treatment until serum osmolality exceeds 320 mOsm/kg
- D.** Start a vasopressin infusion at 0.2 milliunits/kg/hour and change maintenance fluid to D5 one-quarter normal saline with 20 mEq/L potassium chloride
- E.** Give oral desmopressin now and continue his current maintenance fluids unchanged

Question 160

Pediatric Neurocritical Care · Bacterial meningitis - adjunctive dexamethasone · easy

A previously healthy 3-year-old girl presents with 12 hours of fever to 39.4 C, vomiting, photophobia, and neck stiffness. She is irritable but arousable with no focal deficits and no papilledema. Lumbar puncture yields cloudy CSF with 4,200 white cells/microL (92 percent neutrophils), glucose 18 mg/dL, and protein 180 mg/dL; Gram stain shows gram-positive diplococci. No antimicrobial agent has yet been given, and vancomycin plus ceftriaxone is being drawn up. Which of the following best describes the appropriate use of adjunctive corticosteroid therapy?

- A.** Dexamethasone 0.15 mg/kg IV every 6 hours for 4 days, given 10 to 20 minutes before or together with the first dose of antimicrobial therapy
- B.** Dexamethasone 0.15 mg/kg IV every 6 hours for 4 days, started 24 hours after antibiotics have been established
- C.** Dexamethasone withheld unless *Haemophilus influenzae* is subsequently isolated from the CSF
- D.** No corticosteroid, because adjunctive steroids have shown no benefit for any organism in high-income countries
- E.** Methylprednisolone 30 mg/kg/day IV, to a maximum of 1 gram, for 3 to 5 days

Perioperative Neurosurgical Critical Care

Question 161

Perioperative Neurosurgical Critical Care · Posterior fossa craniotomy complications · moderate

A 61-year-old woman is 29 hours status post suboccipital craniectomy for resection of a cerebellar metastasis. She is awake and alert. Over the past hour the nurse has given two doses of intravenous labetalol and started a nicardipine infusion because the blood pressure keeps climbing; it is now 192/100 mmHg with a heart rate of 58. She appears uncomfortable and slightly more drowsy than earlier in the shift. Temperature is 37.2 C and oxygen saturation is 97 percent on 2 liters per minute by nasal cannula. Which of the following is the most appropriate next step?

- A. Give an additional as-needed dose of intravenous labetalol
- B. Titrate the nicardipine infusion upward until the blood pressure is controlled
- C. Give intravenous analgesia and reassess in one hour
- D. Perform a careful neurologic examination and obtain a stat head CT
- E. Place an external ventricular drain for presumed obstructive hydrocephalus

Question 162

Perioperative Neurosurgical Critical Care · Diabetes insipidus after trans-sphenoidal surgery · moderate

A 38-year-old woman is admitted to the ICU after endoscopic trans-sphenoidal resection of a large pituitary macroadenoma. Overnight her urine output rises to 400 mL per hour for three consecutive hours. Urine specific gravity is 1.003, serum sodium has risen from 140 to 149 mmol/L, and serum osmolality is elevated. She is intubated and sedated and cannot report thirst. Blood pressure is 104/62 mmHg and heart rate is 104. Which of the following is the most appropriate management?

- A. Restrict free water and repeat the serum sodium in 6 hours
- B. Replace urine output with hypotonic intravenous fluid and give desmopressin titrated to urine output, specific gravity, and serum sodium
- C. Give stress-dose hydrocortisone and observe, since the polyuria is most likely from cortisol insufficiency
- D. Give a normal saline bolus and attribute the polyuria to intraoperative fluid resuscitation
- E. Start a hypertonic saline infusion targeting a serum sodium of 155 mmol/L

Question 163

Perioperative Neurosurgical Critical Care · Neck hematoma after carotid endarterectomy · hard

A 68-year-old man with chronic obstructive pulmonary disease and obstructive sleep apnea is admitted to the neurocritical care unit after carotid endarterectomy under general anesthesia. Three hours after arrival he reports difficulty breathing. At the bedside his neck is visibly swollen and tense, he has inspiratory stridor, and oxygen saturation is 91 percent on 4 liters per minute by nasal cannula. He is anxious and using accessory muscles. Which of the following is the most appropriate next step?

- A.** Call the surgeon and anesthesia for emergent opening of the wound at the bedside followed by intubation
- B.** Call the surgeon and anesthesia for intubation first, followed by transfer to the operating room
- C.** Give supplemental oxygen and a racemic epinephrine nebulizer
- D.** Obtain a stat carotid duplex ultrasound
- E.** Obtain a stat neck CT and CT angiogram

Question 164

Perioperative Neurosurgical Critical Care · Hemodynamic management of acute spinal cord injury · easy

A 48-year-old man is admitted after a motorcycle crash with a C6 fracture-dislocation and an incomplete cervical spinal cord injury. He has undergone decompression and posterior instrumented fusion. In the ICU his blood pressure is 88/52 mmHg with a heart rate of 52, his extremities are warm, and there is no evidence of ongoing hemorrhage. According to current neurosurgical guidelines cited in this chapter, which of the following best describes the hemodynamic target and initial treatment?

- A.** Mean arterial pressure greater than 65 mmHg for 24 hours, treated with volume resuscitation alone
- B.** Systolic blood pressure greater than 140 mmHg for 72 hours, treated with high-dose methylprednisolone
- C.** Mean arterial pressure greater than 70 mmHg for 14 days, treated with albumin and hydrocortisone
- D.** Systolic blood pressure greater than 100 mmHg for 7 days, with permissive bradycardia and no vasopressor
- E.** Mean arterial pressure greater than 85 mmHg for 7 days, treated with volume resuscitation and, if needed, vasopressors such as dopamine, norepinephrine, or phenylephrine

Question 165

Perioperative Neurosurgical Critical Care · Postoperative seizures and seizure prophylaxis after craniotomy · moderate

A 72-year-old man is on postoperative day 1 after craniotomy for evacuation of an acute subdural hematoma. He is on levetiracetam started at the time of surgery. Overnight the nurse reports that his level of consciousness has been waxing and waning, without overt convulsive movements; he is intermittently unresponsive for several minutes and then follows commands again. Vital signs are stable and laboratory studies including sodium and glucose are normal. Which of the following is the most appropriate next step?

- A.** Increase the levetiracetam dose and continue observation
- B.** Switch from levetiracetam to phenytoin, which is superior for postoperative seizure prophylaxis
- C.** Obtain an urgent noncontrast head CT and arrange electroencephalography
- D.** Administer intravenous mannitol empirically for presumed cerebral edema
- E.** Discontinue antiseizure prophylaxis, since prophylaxis has no proven benefit after craniotomy

Question 166

Perioperative Neurosurgical Critical Care · Vasospasm and delayed cerebral ischemia after aneurysmal SAH · easy

A 49-year-old woman is on postoperative day 10 after craniotomy for clipping of a ruptured anterior communicating artery aneurysm. She has had an external ventricular drain since admission and has been neurologically stable with full strength in all four extremities. Today she develops left arm weakness, and CT angiography shows right middle cerebral artery vasospasm. She has been receiving nimodipine since admission. Which of the following best characterizes the role of nimodipine in this setting?

- A.** It reliably reverses established angiographic vasospasm and can be given as an intra-arterial rescue therapy
- B.** It improves outcomes and is a mainstay of treatment even though it has not been shown to directly reduce the incidence of vasospasm
- C.** It should be discontinued once angiographic vasospasm is documented so that induced hypertension can be achieved
- D.** It is effective only when administered intrathecally through an external ventricular drain
- E.** It prevents rebleeding of the secured aneurysm rather than affecting vasospasm

Pharmacology in the Neurocritical Care Unit

Question 167

Pharmacology in the Neurocritical Care Unit · Protein binding and therapeutic drug monitoring of phenytoin · easy

A 62-year-old woman is in the neurointensive care unit on hospital day 6 after a right basal ganglia intracerebral hemorrhage. She has been maintained on phenytoin for early post-hemorrhagic seizures. She is hemodynamically stable, afebrile, and has had no clinical or electrographic seizures, but the nurse reports she has become progressively more lethargic and has horizontal nystagmus on examination. Laboratory studies show sodium 138 mmol/L, creatinine 0.9 mg/dL, and albumin 2.1 g/dL. The total phenytoin concentration drawn at trough is 6.5 mcg/mL. Which of the following is the most appropriate next step?

- A. Increase the phenytoin maintenance dose until the total serum concentration is 10 to 20 mcg/mL
- B. Obtain a free (unbound) phenytoin concentration and target 1 to 2 mcg/mL
- C. Administer an additional phenytoin load of 20 mg/kg and recheck the total level in 2 hours
- D. Repeat the total phenytoin level immediately after the next intravenous infusion is completed
- E. Discontinue phenytoin and obtain a serum osmolar gap to assess for drug accumulation

Question 168

Pharmacology in the Neurocritical Care Unit · Augmented renal clearance and antiepileptic dosing · moderate

A 24-year-old man is admitted to the neurointensive care unit after a motorcycle collision with a severe traumatic brain injury. He is 4 days into his stay, is hemodynamically stable on no vasopressors, and his sequential organ failure assessment (SOFA) score is 3. Serum creatinine is 0.6 mg/dL and an 8-hour urine collection yields a measured creatinine clearance of 168 mL/min/1.73 m². He has been receiving levetiracetam 1,000 mg intravenously every 12 hours, and continuous electroencephalography now shows two brief electrographic seizures. Which of the following is the most appropriate adjustment to his levetiracetam regimen?

- A. Levetiracetam 500 mg intravenously every 12 hours
- B. Levetiracetam 1,000 mg intravenously every 24 hours
- C. Levetiracetam 1,500 mg intravenously every 12 hours
- D. Levetiracetam 1,000 mg intravenously every 8 hours
- E. Levetiracetam 1,000 mg intravenously every 12 hours with a free drug level in 48 hours

Question 169

Pharmacology in the Neurocritical Care Unit · Neuromuscular blocking agents after spinal cord injury · moderate

A 38-year-old man sustained a complete C5 spinal cord injury 3 weeks ago and was extubated 5 days ago. He now develops aspiration pneumonia with worsening hypoxemia and requires emergent reintubation. His blood pressure is 104/62 mm Hg, heart rate is 58 beats/min, and serum potassium is 4.6 mmol/L with a creatinine of 0.8 mg/dL. The team plans rapid sequence intubation. Which of the following is the most appropriate neuromuscular blocking agent for this intubation?

- A. Rocuronium 1.2 mg/kg intravenously
- B. Succinylcholine 1.5 mg/kg intravenously
- C. Pancuronium 0.1 mg/kg intravenously
- D. Atracurium 0.5 mg/kg intravenously
- E. Cisatracurium 0.1 mg/kg intravenously

Question 170

Pharmacology in the Neurocritical Care Unit · Monitoring scheduled mannitol therapy · moderate

A 58-year-old woman with a large right middle cerebral artery infarct and cerebral edema has been receiving scheduled mannitol 1 g/kg intravenously every 6 hours for 3 days for intracranial pressure control. Her intracranial pressure has been 12 to 16 mm Hg over the past 24 hours. This morning, serum sodium is 146 mmol/L, blood urea nitrogen is 28 mg/dL, glucose is 110 mg/dL, and creatinine has risen from 0.8 to 1.4 mg/dL. Measured serum osmolality is 332 mOsm/kg and calculated serum osmolality is 308 mOsm/L. Which of the following is the most appropriate next step?

- A. Continue the current mannitol regimen and repeat the osmolality in 24 hours
- B. Increase the mannitol dose to 1.5 g/kg every 6 hours to overcome tachyphylaxis
- C. Hold the next mannitol dose and extend the dosing interval before redosing
- D. Continue mannitol and add furosemide to augment the osmotic diuresis
- E. Continue mannitol but administer subsequent doses without an in-line filter to speed delivery

Question 171

Pharmacology in the Neurocritical Care Unit · Hyperchloremia and acute kidney injury with hypertonic saline · hard

A 30-year-old man with a severe traumatic brain injury has been receiving a continuous infusion of 3% sodium chloride at 50 mL/hr targeting a serum sodium of 145 to 155 mmol/L. Intracranial pressures have remained below 20 mm Hg for the past 24 hours. On day 3, laboratory studies show sodium 154 mmol/L, chloride 119 mmol/L, serum bicarbonate 15 mmol/L, potassium 3.1 mmol/L, and a creatinine that has risen from 0.7 to 2.1 mg/dL with a marked fall in urine output. Which of the following is the most appropriate next step?

- A. Continue the 3% sodium chloride infusion and increase the target sodium to 160 to 165 mmol/L
- B. Discontinue all hyperosmolar therapy and manage intracranial pressure with sedation alone
- C. Discontinue the 3% sodium chloride and begin scheduled mannitol 1 g/kg every 6 hours
- D. Discontinue the 3% sodium chloride and begin an equimolar hypertonic sodium acetate infusion
- E. Change the infusion to a compounded balanced hypertonic solution of sodium chloride plus sodium acetate

Question 172

Pharmacology in the Neurocritical Care Unit · Propofol-related infusion syndrome · moderate

A 19-year-old woman with refractory status epilepticus has been maintained on a propofol infusion at 90 mcg/kg per minute for the past 3 days along with intravenous fentanyl. Her body mass index is 18 kg/m² and she has required low-dose norepinephrine throughout. Overnight she becomes bradycardic to 42 beats/min with new hypotension. Laboratory studies show a lactate of 6.8 mmol/L, potassium 6.1 mmol/L, triglycerides 720 mg/dL, and creatine kinase 9,400 units/L. An echocardiogram shows a new reduction in ejection fraction. Which of the following best explains this presentation?

- A. Propylene glycol accumulation from the sedative vehicle
- B. Propofol-related infusion syndrome
- C. Fentanyl-induced histamine release with vasodilation
- D. Valproate-induced hyperammonemic encephalopathy
- E. Intensive care unit-acquired weakness from a steroid and aminosteroid combination

Question 173

Pharmacology in the Neurocritical Care Unit · Vasopressor selection in acute spinal cord injury · easy

A 45-year-old man is admitted to the neurointensive care unit after a motor vehicle collision with a complete C3-C4 spinal cord injury. He is intubated and adequately volume resuscitated. His mean arterial pressure is 50 mm Hg, heart rate is 41 beats/min, and temperature is 36.4 degrees Celsius. The team wishes to augment spinal cord perfusion pressure. Which of the following is the most appropriate agent?

- A. Phenylephrine infusion
- B. Dobutamine infusion
- C. Arginine vasopressin at a fixed dose of 0.03 units/min
- D. Norepinephrine infusion
- E. Oral midodrine

Pregnancy in Neurocritical Care

Question 174

Pregnancy in Neurocritical Care · Eclampsia: seizure management and antihypertensive choice · moderate

A 28-year-old woman at 34 weeks of gestation is brought to the emergency department after a witnessed generalized tonic-clonic seizure at home. For the past two days she has had a persistent occipital headache and blurred vision. Blood pressure is 172/112 mm Hg, heart rate 96/min, and urine dipstick shows 3+ protein. She is postictal but arousable, with no focal deficits. Noncontrast head CT is unremarkable; MRI shows symmetric parieto-occipital FLAIR hyperintensity. She has no prior history of epilepsy. Which of the following is the most appropriate treatment to prevent recurrent seizures in this patient?

- A. Intravenous lorazepam followed by a levetiracetam maintenance infusion
- B. Intravenous fosphenytoin load followed by oral phenytoin
- C. Intravenous magnesium infusion
- D. Intravenous enalaprilat with an oral valproate load
- E. Intravenous diazepam infusion titrated to burst suppression

Question 175

Pregnancy in Neurocritical Care · Acute ischemic stroke in pregnancy: thrombolysis and thrombectomy · moderate

A 37-year-old woman at 22 weeks of gestation develops sudden right hemiplegia and global aphasia while at work. She arrives 70 minutes after symptom onset. Blood pressure is 148/86 mm Hg, glucose is 104 mg/dL, and NIH Stroke Scale score is 18. Noncontrast head CT shows no hemorrhage and an early hyperdense left middle cerebral artery sign; CT angiography confirms a left M1 occlusion. Fetal heart tones are normal. Which of the following is the most appropriate management?

- A. Administer intravenous tissue plasminogen activator and proceed to mechanical thrombectomy
- B. Administer intravenous tissue plasminogen activator alone, because thrombectomy is unsafe in pregnancy
- C. Proceed directly to mechanical thrombectomy alone, because tissue plasminogen activator is teratogenic
- D. Give oral aspirin only, because thrombolysis and thrombectomy both endanger the fetus
- E. Obtain MRI and magnetic resonance angiography before any reperfusion therapy is considered

Question 176

Pregnancy in Neurocritical Care · Cerebral venous thrombosis in the puerperium · easy

A 31-year-old woman who delivered vaginally 10 days ago presents with a 4-day history of progressively worsening headache, nausea, and vomiting, followed this morning by a focal seizure with secondary generalization. Blood pressure is 126/78 mm Hg and she is afebrile. Noncontrast head CT shows a right parietal hemorrhagic lesion that does not conform to an arterial territory, and CT venography demonstrates filling defects in the superior sagittal and right transverse sinuses. Which of the following is the most appropriate treatment?

- A. Intravenous tissue plasminogen activator
- B. Oral warfarin titrated to an international normalized ratio of 2 to 3
- C. Oral apixaban
- D. Subcutaneous enoxaparin
- E. Oral aspirin plus clopidogrel

Question 177

Pregnancy in Neurocritical Care · Maternal cardiac arrest and perimortem cesarean section · moderate

A 30-year-old woman at 30 weeks of gestation collapses on the labor and delivery floor. She is pulseless and apneic, and the fundus is palpable well above the umbilicus. The team begins chest compressions and calls a maternal cardiac arrest code. Which of the following is the most appropriate approach to her resuscitation?

- A. Tilt the patient into a left lateral decubitus position and continue chest compressions in that position
- B. Manually displace the uterus upward and to the patient's left, and perform perimortem cesarean section within 4 minutes of starting compressions
- C. Establish femoral venous access, halve the epinephrine dose for fetal safety, and reassess at 10 minutes
- D. Continue standard supine compressions and defer perimortem cesarean section until 20 minutes of resuscitation have elapsed
- E. Perform perimortem cesarean section only after return of spontaneous circulation has been achieved

Question 178

Pregnancy in Neurocritical Care · Spontaneous spinal epidural hematoma: gestational age threshold · hard

A 29-year-old woman at 35 weeks of gestation develops sudden severe interscapular pain followed within an hour by dense bilateral leg weakness, a T6 sensory level, and urinary retention. Blood pressure is 132/80 mm Hg and she is afebrile; she takes no anticoagulants. Emergent MRI of the spine shows a spontaneous dorsal epidural hematoma extending from T4 to T7 with severe cord compression. Fetal heart rate tracing is reassuring. Which of the following is the most appropriate next step?

- A. Immediate surgical decompression with continuous fetal heart rate monitoring, deferring delivery
- B. Administer high-dose intravenous methylprednisolone and repeat MRI in 24 hours
- C. Administer intravenous magnesium for fetal neuroprotection and observe for spontaneous resolution
- D. Induce vaginal delivery with epidural anesthesia, then decompress the spine after delivery
- E. Perform cesarean section with intraoperative neurological monitoring, then proceed to spinal decompression

Prognostication in Neurocritical Care

Question 179

Prognostication in Neurocritical Care · Properties of prognostic models: discrimination versus calibration · easy

A newly published model predicts 6-month mortality after severe brain injury and reports an area under the receiver operating characteristic curve of 0.82. When the model is applied to a similar but separate cohort, it consistently predicts a 40% risk of death in patients whose observed mortality is only 15%, while performing accurately at the highest illness severities. Which property of this model is most clearly deficient?

- A. Discrimination
- B. Internal validity
- C. Calibration
- D. Sensitivity
- E. Interobserver reliability

Question 180

Prognostication in Neurocritical Care · Decompressive craniectomy in large hemispheric stroke: DESTINY II · moderate

A 66-year-old woman with a right middle cerebral artery infarct is admitted to the neurointensive care unit. On hospital day 2 she becomes progressively less arousable, and repeat CT shows a space-occupying infarct with 8 mm of midline shift. The neurosurgical team offers decompressive craniectomy, and the family asks what the operation would realistically accomplish. Based on the trial data specific to patients in this age group, which of the following statements is most accurate?

- A.** About half of patients who undergo craniectomy will walk unassisted at 12 months
- B.** Craniectomy reduces 12-month mortality from 71% to 22% and increases functional independence from 2% to 14%
- C.** Craniectomy does not reduce mortality in patients older than 60 years
- D.** Craniectomy reduces 12-month mortality from 76% to 43%, but functional independence is not achieved in either group
- E.** Craniectomy increases the likelihood of functional independence to about 14% without affecting mortality

Question 181

Prognostication in Neurocritical Care · Intracerebral hemorrhage: self-fulfilling prophecy and guideline-based counseling · hard

A 51-year-old man with hypertension presents with confusion, left hemiplegia, and dysarthria. He opens his eyes to pain, does not follow commands, and mumbles inappropriate words. Noncontrast head CT shows a right basal ganglia intracerebral hemorrhage of 47 mL by the ABC divided by 2 method, with 4 mm of midline shift and trace intraventricular blood without hydrocephalus. Six hours after admission, the distressed family states that he would never want to live in a vegetative state and asks about his chance of recovery and about a do-not-resuscitate order. Which of the following is the most appropriate approach?

- A.** Provide an individualized clinical estimate that explicitly conveys uncertainty and defer any new do-not-resuscitate order until after the second full day of hospitalization
- B.** Quote the 30-day mortality associated with his Original intracerebral hemorrhage score as his personal probability of death
- C.** Place a do-not-resuscitate order now and limit medical and surgical interventions accordingly
- D.** Recalculate his prognosis with a newer multivariable scale, which head-to-head studies have shown to be superior to the Original intracerebral hemorrhage score
- E.** Decline to discuss prognosis until his 90-day functional outcome can be observed

Question 182

Prognostication in Neurocritical Care · Aneurysmal subarachnoid hemorrhage radiographic grading · moderate

A 54-year-old woman presents with a thunderclap headache and vomiting. She is drowsy and mildly confused but has no focal deficit, corresponding to Hunt and Hess grade 3. Noncontrast head CT shows thick subarachnoid blood measuring more than 1 mm in the basal cisterns and sylvian fissures, with intraventricular blood in both lateral ventricles. Catheter angiography confirms a ruptured anterior communicating artery aneurysm. Which of the following best characterizes her radiographic risk stratification for symptomatic vasospasm?

- A. Modified Fisher grade 2, with an odds ratio of 1.6 compared with grade 1
- B. Modified Fisher grade 3, with an odds ratio of 1.6 compared with grade 1
- C. Fisher grade 2, with an odds ratio of 1.3 compared with grade 1
- D. Fisher grade 4, with an odds ratio of 1.7 compared with grade 1
- E. Modified Fisher grade 4, with an odds ratio of 2.2 compared with grade 1

Question 183

Prognostication in Neurocritical Care · Severe traumatic brain injury: withdrawal of life-sustaining therapy and detection of consciousness · moderate

A 24-year-old man is admitted after a motorcycle crash with a Glasgow Coma Scale score of 5 and diffuse traumatic axonal injury with traumatic subarachnoid hemorrhage on CT. On hospital day 2 he remains comatose on fentanyl and propofol infusions, and the family asks whether continued treatment is futile. Which of the following statements about prognosis in severe traumatic brain injury is most accurate?

- A. Coma on admission after severe traumatic brain injury invariably portends a poor functional outcome
- B. At least 30% of patients hospitalized with severe traumatic brain injury die in hospital, roughly 70% of those deaths are associated with withdrawal of life-sustaining therapy, and most withdrawal occurs within the first 3 days
- C. Most withdrawal of life-sustaining therapy occurs after the first week, once prognostic uncertainty has largely resolved
- D. Sedation, pain, and medical illness rarely confound bedside assessment of consciousness in this setting
- E. The IMPACT prognostic calculator discriminates 6-month outcome with an area under the curve greater than 0.90

Question 184

Prognostication in Neurocritical Care · Cognitive biases in surrogate decision-making · easy

During a family meeting for a 72-year-old man with a severe dominant hemisphere stroke, his daughter states that if he survives dependent on a wheelchair and unable to speak fluently, he would consider that no life at all. You are aware that patients living with comparable disability consistently rate their own quality of life higher than nondisabled surrogates predict they would. Which cognitive bias does this discrepancy specifically describe?

- A. Recall bias
- B. Focusing illusion
- C. Response shift
- D. Disability paradox
- E. Optimistic misinterpretation of quantitative prognosis

Sedation and Analgesia

Question 185

Sedation and Analgesia · Sedation depth targets before neuromuscular blockade · easy

A 53-year-old woman is in the neurocritical care unit on day 7 after a motorcycle collision with a traumatic subarachnoid hemorrhage. She is receiving fentanyl 50 mcg/hr and propofol 20 mcg/kg per minute and has been maintained at a Richmond Agitation-Sedation Scale (RASS) score of -1. She now develops bilateral infiltrates and severe hypoxemia with a ratio of arterial oxygen partial pressure to fraction of inspired oxygen of 130. The team plans to start a cisatracurium infusion for 48 hours as part of early management of acute respiratory distress syndrome. Which of the following is the most appropriate sedation target before the neuromuscular blocker is started?

- A. RASS 0 to -1
- B. RASS 0 to -2
- C. RASS -4 to -5
- D. RASS -2 to -3
- E. RASS -1 to -3

Question 186

Sedation and Analgesia · Propofol-related infusion syndrome · hard

A 17-year-old boy is in the intensive care unit after a motor vehicle collision with a severe traumatic brain injury. He has been receiving propofol 90 mcg/kg per minute and midazolam 5 mg/hr for elevated intracranial pressure. On day 4 his blood pressure falls to 68/40 mm Hg despite fluids. Laboratory studies show sodium 149 mmol/L, potassium 5.2 mmol/L, creatinine 3.0 mg/dL, creatine kinase 26,750 units/L, and triglycerides 550 mg/dL. The nurse reports that his urine has turned green. Which of the following is the most likely cause of this presentation?

- A. Accumulation of the active metabolite alphahydroxymidazolam
- B. Propofol-related infusion syndrome
- C. Propylene glycol toxicity from the midazolam infusion
- D. Contrast-induced acute tubular necrosis
- E. Nonconvulsive status epilepticus with autonomic instability

Question 187

Sedation and Analgesia · Propylene glycol toxicity from benzodiazepine infusions · moderate

A 60-year-old woman is intubated in the intensive care unit for severe alcohol withdrawal and has required high doses of intravenous lorazepam by continuous infusion for 4 days, along with as-needed acetaminophen and oxycodone. After several days of improvement she acutely becomes hypotensive and bradycardic. Laboratory studies show a lactate of 8.5 mmol/L, arterial pH 7.28, creatinine 2.3 mg/dL (baseline 0.8 mg/dL), and an elevated serum osmolar gap. Blood cultures are negative and there is no new imaging finding. Which of the following best explains her deterioration?

- A. Propylene glycol toxicity
- B. Opioid withdrawal from intermittent oxycodone
- C. Acetaminophen-induced hepatotoxicity
- D. Relative adrenal insufficiency
- E. Recurrent alcohol withdrawal with delirium tremens

Question 188

Sedation and Analgesia · Shivering control during targeted temperature management · moderate

A 58-year-old man is comatose after return of spontaneous circulation from an out-of-hospital cardiac arrest and is undergoing targeted temperature management. He is already receiving scheduled acetaminophen, buspirone, and skin counterwarming. The nurse palpates shivering localized to the neck and thorax only, giving a Bedside Shivering Assessment Scale score of 1, and the surface device is struggling to hold the target temperature. Which of the following is the most appropriate next intervention?

- A. Meperidine 100 mg intravenously every 6 hours
- B. Rocuronium 1 mg/kg intravenous bolus
- C. Propofol 50 mcg/kg per minute infusion
- D. Meperidine 12.5 mg intravenously every 6 hours
- E. Midazolam 5 mg intravenous bolus every 6 hours

Question 189

Sedation and Analgesia · Sedative choice in the patient approaching extubation · easy

A 45-year-old man was admitted to the intensive care unit 2 days ago for suspected seizure activity, which has since been ruled out on continuous electroencephalography. He is now meeting weaning criteria and the team plans extubation later today, but he has become increasingly agitated and is pulling at his lines, with a Richmond Agitation-Sedation Scale score of +2. His blood pressure is 138/78 mm Hg and heart rate is 88 beats/min. Which of the following is the most appropriate agent?

- A. Propofol infusion at 35 mcg/kg per minute
- B. Fentanyl infusion at 25 mcg/hr
- C. Lorazepam 2 mg intravenously every 4 hours
- D. Dexmedetomidine infusion at 1.5 mcg/kg per hour
- E. Dexmedetomidine infusion at 0.2 mcg/kg per hour

Question 190

Sedation and Analgesia · Neuroleptic selection and QT prolongation · moderate

A 71-year-old woman in the neurocritical care unit after evacuation of a subdural hematoma develops hyperactive delirium with visual hallucinations and is repeatedly climbing out of bed. Nonpharmacologic measures have failed and pain, hypoxemia, and hypoglycemia have been excluded. She has no history of Parkinson disease or other movement disorder. Her baseline electrocardiogram shows a QT interval corrected for heart rate of 446 msec, and she is receiving no other QT-prolonging medications. Which of the following is the most appropriate agent?

- A. Haloperidol 5 mg intravenously
- B. Olanzapine 5 mg intramuscularly
- C. Quetiapine 25 mg per tube twice daily
- D. Risperidone 1 mg per tube twice daily
- E. Midazolam 2 mg intravenously every hour as needed

Question 191

Sedation and Analgesia · Opioid selection in renal dysfunction · moderate

A 62-year-old man is intubated in the intensive care unit after extensive spinal surgery. His history includes diabetes, hypertension, stage 3 chronic kidney disease with a baseline creatinine of 2.3 mg/dL, and depression. His mean arterial pressure is 90 mm Hg and his heart rate is 120 beats/min, and the Critical Care Pain Observation Tool score is 6. The team believes pain is driving his vital sign changes. Which of the following is the most appropriate analgesic to initiate?

- A. Morphine 4 mg intravenously every 2 hours as needed
- B. Acetaminophen 1,000 mg by mouth every 4 hours as needed
- C. Hydromorphone 0.5 mg intravenously every 2 hours as needed
- D. Remifentanyl 0.5 mcg/kg intravenously every 4 hours as needed
- E. Meperidine 25 mg intravenously every 4 hours as needed

Question 192

Sedation and Analgesia · Sedation interruption in intracranial hypertension · hard

A 34-year-old man with a severe traumatic brain injury has an intracranial pressure monitor and a brain tissue oxygen probe in place. He is receiving propofol and fentanyl infusions, with intracranial pressures running 16 to 19 mm Hg and a brain tissue oxygen tension of 24 mm Hg. The covering team requests that a daily sedation interruption protocol be applied to him as it is for the medical intensive care unit patients. Which of the following is the most appropriate approach?

- A. Abruptly interrupt both sedation and analgesia daily until he is awake or appears uncomfortable
- B. Interrupt sedation only after starting a neuromuscular blocking agent to prevent agitation
- C. Substitute bispectral index monitoring for the clinical examination and continue uninterrupted sedation
- D. Wean sedation in a judicious, stepwise fashion with continuous intracranial pressure and brain tissue oxygen monitoring
- E. Abandon neurological examination during sedation and rely on serial computed tomography instead

Seizures and Status Epilepticus

Question 193

Seizures and Status Epilepticus · Definition of status epilepticus · easy

The International League Against Epilepsy defines status epilepticus along two time dimensions: t1, the time beyond which a seizure is unlikely to stop spontaneously and treatment should begin, and t2, the time beyond which long-term consequences are expected. For generalized convulsive status epilepticus, which of the following correctly pairs t1 with t2?

- A. t1 of 5 minutes and t2 of 30 minutes
- B. t1 of 5 minutes and t2 of 60 minutes
- C. t1 of 10 minutes and t2 of 30 minutes
- D. t1 of 30 minutes and t2 of 60 minutes
- E. t1 of 1 minute and t2 of 10 minutes

Question 194

Seizures and Status Epilepticus · First-line therapy for status epilepticus · easy

A 27-year-old woman with a history of epilepsy is brought to the emergency department after a witnessed generalized tonic-clonic seizure at home. In the department she is observed to have two additional generalized tonic-clonic seizures, each lasting about 2 minutes, without returning to her baseline mental status in between. Her temperature is 37.4 C, heart rate 112 beats per minute, blood pressure 148/86 mmHg, and oxygen saturation 94 percent on 4 liters by nasal cannula. Point-of-care glucose is 106 mg/dL and she has a functioning peripheral intravenous line. Which of the following is the most appropriate initial medication?

- A. Levetiracetam 60 mg/kg IV
- B. Lorazepam 4 mg IV
- C. Fosphenytoin 20 mg PE/kg IV
- D. Propofol infusion after endotracheal intubation
- E. Observation without medication, since each individual seizure is self-limited

Question 195

Seizures and Status Epilepticus · Prehospital management and the RAMPART trial · moderate

Paramedics are called to a 55-year-old man who has been having continuous generalized tonic-clonic activity for approximately 8 minutes. He is cyanotic with clenched jaw and copious oral secretions. Two attempts at peripheral intravenous access have failed because of ongoing convulsive movements. His fingerstick glucose is 128 mg/dL. Which of the following is the most appropriate next step?

- A. Withhold antiseizure medication until intravenous access is obtained in the emergency department
- B. Place an intraosseous line and give fosphenytoin 20 mg PE/kg
- C. Administer midazolam 10 mg intramuscularly
- D. Administer levetiracetam 60 mg/kg intramuscularly
- E. Administer phenobarbital 20 mg/kg intramuscularly

Question 196

Seizures and Status Epilepticus · Second-line therapy and the ESETT trial · moderate

A 44-year-old man with generalized convulsive status epilepticus has received two doses of lorazepam totaling 8 mg IV and continues to convulse. The team is choosing an urgent control (second-line) agent. Which of the following statements about the Established Status Epilepticus Treatment Trial (ESETT) best informs this decision?

- A. Levetiracetam was clearly superior, terminating seizures in about 70 percent of patients
- B. Fosphenytoin was clearly superior, with significantly less hypotension than the comparators
- C. Valproic acid was clearly superior and is now the guideline-preferred second-line agent
- D. Valproic acid, fosphenytoin, and levetiracetam were similarly effective, achieving the primary outcome in 46, 45, and 47 percent of patients respectively
- E. All three agents failed in more than 90 percent of patients and the trial was stopped for harm

Question 197

Seizures and Status Epilepticus · Benzodiazepine tachyphylaxis · moderate

A 62-year-old woman with herpes simplex encephalitis is on continuous EEG for altered mental status, which shows frequent nonconvulsive seizures. The first dose of lorazepam 2 mg IV produces transient electrographic seizure cessation and brief clinical improvement, but a second identical dose given 20 minutes later through the same visibly patent and well-flushed intravenous catheter produces no clinical or electrographic effect. Which of the following best explains the loss of efficacy?

- A. Reduced blood-brain barrier permeability to lorazepam
- B. Downregulation of N-methyl-D-aspartate receptors on postsynaptic neurons
- C. Increased surface expression of gamma-aminobutyric acid type A receptors
- D. Induction of hepatic cytochrome P450 enzymes accelerating lorazepam clearance
- E. Internalization of benzodiazepine-sensitive gamma-aminobutyric acid type A receptors with upregulation of N-methyl-D-aspartate receptors

Question 198

Seizures and Status Epilepticus · Nonconvulsive status epilepticus after treated convulsive status · moderate

A 27-year-old woman with no significant medical history presents after right-sided focal motor seizure activity that generalized. Clinical convulsions stop after a total of 6 mg of lorazepam IV, about 18 minutes after onset. Non-contrast head CT, complete blood count, basic metabolic panel, and hepatic function panel are unremarkable. Thirty-five minutes after the convulsions stopped, she remains unresponsive with a Glasgow Coma Scale of 7 (E1, V2, M4). Temperature is 37.6 C, pulse 95 beats per minute, blood pressure 138/57 mmHg, and oxygen saturation 99 percent. Which of the following is the most appropriate next diagnostic step?

- A. Continuous EEG monitoring
- B. Urine toxicology screen
- C. MRI of the brain
- D. CT venography
- E. Serum paraneoplastic antibody panel

Question 199

Seizures and Status Epilepticus · 2HELPS2B seizure risk score · hard

The 2HELPS2B score developed by Struck and colleagues stratifies the risk of seizures in hospitalized patients undergoing continuous EEG. Which of the following EEG findings is weighted at 2 points rather than 1 point in this score?

- A. Sporadic epileptiform discharges
- B. Brief ictal rhythmic discharges
- C. Lateralized rhythmic delta activity
- D. A periodic or rhythmic pattern with a frequency greater than 2 Hz
- E. Presence of plus features on a periodic or rhythmic pattern

Question 200

Seizures and Status Epilepticus · Drug-induced seizures in the ICU · easy

A 72-year-old man with chronic kidney disease is being treated in the medical ICU for hospital-acquired pneumonia with cefepime. On hospital day 3 he becomes progressively encephalopathic with intermittent eyelid twitching. Head CT is unremarkable, electrolytes and glucose are normal, and continuous EEG demonstrates nonconvulsive status epilepticus. Which of the following best explains the mechanism by which his antibiotic precipitated seizures?

- A. Direct agonism at N-methyl-D-aspartate receptors
- B. Blockade of voltage-gated sodium channels in cortical neurons
- C. Antagonism of gamma-aminobutyric acid receptors after crossing the blood-brain barrier
- D. Inhibition of astrocytic glutamate reuptake transporters
- E. Enhanced dopaminergic transmission in the mesolimbic system

Question 201

Seizures and Status Epilepticus · Second-line drug selection and drug interactions · hard

An 80-kg 40-year-old man is in the neuro ICU with an intracranial abscess being treated empirically with intravenous meropenem and vancomycin. While awaiting culture results he becomes difficult to arouse. Repeat head CT shows expected postoperative changes only. Continuous EEG shows evolving 3 Hz sharp wave discharges consistent with nonconvulsive status epilepticus. He receives a cumulative 8 mg of lorazepam IV with no change on EEG. His blood pressure is 132/74 mmHg and the QTc and PR intervals are normal. Which of the following is the most appropriate medication to give next?

- A. Valproic acid 3000 mg IV
- B. Levetiracetam 1000 mg IV
- C. Phenobarbital 500 mg IV
- D. Fosphenytoin 1600 mg PE IV
- E. Lacosamide 50 mg IV

Question 202

Seizures and Status Epilepticus · Refractory status epilepticus: choice of anesthetic · easy

A 34-year-old intubated woman with cryptogenic encephalitis remains in electrographic status epilepticus despite lorazepam 0.1 mg/kg IV and a full levetiracetam load of 60 mg/kg IV followed by maintenance dosing. Her blood pressure is 118/70 mmHg on no vasopressors. Which of the following is the most appropriate next step?

- A. Begin a pentobarbital infusion loaded at 5 mg/kg IV
- B. Initiate a ketogenic diet
- C. Give perampanel 12 mg by nasogastric tube
- D. Repeat lorazepam 0.1 mg/kg IV every 10 minutes until seizures stop
- E. Begin a continuous midazolam infusion with a 0.2 mg/kg bolus followed by 0.05 to 2 mg/kg/hr

Question 203

Seizures and Status Epilepticus · Titration and duration of anesthetic infusions · moderate

A patient with refractory status epilepticus has been started on a continuous anesthetic infusion. Which of the following best reflects current guideline recommendations for the endpoint and duration of that infusion?

- A.** Titrate to cessation of electrographic seizures or burst suppression, and maintain that endpoint for 24 to 48 hours before weaning
- B.** Titrate to cessation of electrographic seizures only, and wean after 6 to 12 hours
- C.** Titrate to a completely isoelectric EEG and maintain it for 72 hours
- D.** Titrate to burst suppression and maintain it for a minimum of 7 days
- E.** Titrate to resolution of clinical convulsions and wean immediately thereafter

Question 204

Seizures and Status Epilepticus · Breakthrough seizures during anesthetic weaning · hard

A 58-year-old man with refractory status epilepticus has been seizure free on a midazolam infusion for 36 hours and is on maintenance levetiracetam. During the wean, continuous EEG shows recurrent electrographic seizures. His serum levetiracetam level is at the low end of the therapeutic range and his blood pressure is stable. Which of the following is the most appropriate management?

- A.** Continue the wean at the current rate and reassess in 12 hours
- B.** Bolus midazolam and continue the infusion at a higher maintenance rate, and optimize the levetiracetam dose or add a non-anesthetic antiseizure medication before re-attempting the wean
- C.** Stop midazolam and transition directly to a pentobarbital infusion
- D.** Change the midazolam wean rate to 20 percent every 3 hours and make no other changes
- E.** Add therapeutic hypothermia to a target of 32 to 34 C for 24 hours

Question 205

Seizures and Status Epilepticus · Seizure prophylaxis in the critically ill · moderate

A 58-year-old man is admitted after a motor vehicle collision with a severe traumatic brain injury, a Glasgow Coma Scale of 6, and bifrontal contusions on CT. He has had no clinical or electrographic seizures. Which of the following best reflects guideline recommendations regarding antiseizure medication in this patient?

- A.** Levetiracetam is recommended over phenytoin on the basis of level 1 evidence in severe traumatic brain injury
- B.** Prophylaxis should be withheld, since guidelines recommend against antiseizure medication prophylaxis after all forms of acute brain injury
- C.** Phenytoin for 7 days is recommended to reduce the incidence of early post-traumatic seizures, although it does not change the long-term risk of epilepsy
- D.** Phenytoin should be continued for 6 months to prevent the development of post-traumatic epilepsy
- E.** Valproic acid should be started and continued until hospital discharge

Question 206

Seizures and Status Epilepticus · Hypothermia for status epilepticus (HIBERNATUS) · moderate

A 29-year-old woman is intubated and mechanically ventilated for convulsive status epilepticus. Her family read about cooling as a treatment for seizures and asks whether it should be added to her regimen. Which of the following best summarizes the findings of the HIBERNATUS multicenter randomized controlled trial of hypothermia as an add-on therapy in this population?

- A. Hypothermia improved 3-month functional outcome and is now recommended as adjunctive therapy
- B. Hypothermia significantly increased mortality and the trial was stopped early for harm
- C. Hypothermia significantly reduced progression to super-refractory status epilepticus
- D. Hypothermia improved electrographic seizure control, with nonconvulsive status epilepticus in 11 percent versus 22 percent, but did not improve functional outcome at 3 months
- E. Hypothermia worsened electrographic seizure control while improving functional outcome

Question 207

Seizures and Status Epilepticus · Treatment of nonconvulsive seizures (TRENdS) · moderate

A 70-year-old woman is on continuous EEG after cardiac surgery for persistent encephalopathy and is found to have frequent nonconvulsive seizures. She has a first-degree atrioventricular block on telemetry but is hemodynamically stable. The team discusses whether to load lacosamide or fosphenytoin. Which of the following statements is best supported by the only randomized, controlled, prospective trial to date that specifically evaluated treatment of nonconvulsive seizures?

- A. Fosphenytoin 20 mg PE/kg was significantly superior to lacosamide for terminating nonconvulsive seizures
- B. Lacosamide 400 mg IV was significantly superior to fosphenytoin, establishing it as the standard of care
- C. Both agents terminated nonconvulsive seizures in fewer than 20 percent of patients, so anesthetic infusion should be used first
- D. The trial was stopped early because of excess cardiac events in the lacosamide arm
- E. Lacosamide 400 mg IV terminated nonconvulsive seizures in 63.3 percent of patients versus 50 percent with fosphenytoin, a difference that did not reach statistical significance

Subarachnoid Hemorrhage

Question 208

Subarachnoid Hemorrhage · Diagnosis: negative CT and lumbar puncture · easy

A 45-year-old woman with a long history of migraine presents 14 hours after the abrupt onset of what she calls the most severe headache of her life, which she says is clearly different from her usual migraine attacks. She has an extensive smoking history and denies recreational drug use. Temperature 36.9 C, blood pressure 148/86 mm Hg, heart rate 88/min. Her neurologic examination, including cranial nerves and funduscopy, is normal. A noncontrast head CT is interpreted as showing no intracranial abnormality. Which of the following is the most appropriate next step?

- A. Administer subcutaneous sumatriptan and a migraine cocktail and discharge her from the emergency department
- B. Obtain brain MRI with gradient echo sequences
- C. Perform a lumbar puncture
- D. Perform catheter digital subtraction angiography
- E. Repeat the noncontrast head CT in 6 hours

Question 209

Subarachnoid Hemorrhage · Blood pressure management before aneurysm securing · moderate

A 58-year-old man with untreated hypertension presents with sudden severe headache and vomiting. He is awake with a stiff neck and no focal deficit, corresponding to World Federation of Neurological Surgeons grade II. Noncontrast CT shows diffuse blood in the basal cisterns, and CT angiography demonstrates a 7 mm anterior communicating artery aneurysm. Coiling is scheduled for approximately 10 hours from now. His blood pressure is 198/104 mm Hg with a mean arterial pressure of 135 mm Hg; his usual outpatient systolic pressures are in the 150s. Which of the following is the most appropriate blood pressure management?

- A. Start a nicardipine infusion at 5 mg/hr and titrate to a systolic blood pressure below 160 mm Hg
- B. Give intermittent intravenous hydralazine 10 mg boluses to a systolic blood pressure below 140 mm Hg
- C. Start a labetalol infusion titrated to a mean arterial pressure below 70 mm Hg
- D. Withhold antihypertensive therapy unless the systolic blood pressure exceeds 220 mm Hg
- E. Start a norepinephrine infusion targeting a mean arterial pressure 20 mm Hg above his baseline

Question 210

Subarachnoid Hemorrhage · Transcranial Doppler and the Lindegaard ratio · hard

A 47-year-old woman is on postbleed day 6 after coiling of a ruptured basilar apex aneurysm. She develops new right face, arm, and leg weakness with mild word-finding difficulty. Transcranial Doppler shows a left middle cerebral artery mean flow velocity of 201 cm/sec at a depth of 55 mm with a left extracranial internal carotid artery mean velocity of 40 cm/sec, and a right middle cerebral artery mean flow velocity of 122 cm/sec with a right extracranial internal carotid artery mean velocity of 52 cm/sec. Which of the following best characterizes these findings?

- A. Vasospasm of both middle cerebral arteries
- B. Vasospasm of the right middle cerebral artery only
- C. Hyperemia of both middle cerebral arteries
- D. A normal study, since velocities of this magnitude are expected on postbleed day 6
- E. Vasospasm of the left middle cerebral artery with hyperemia of the right middle cerebral artery

Question 211

Subarachnoid Hemorrhage · Delayed cerebral ischemia refractory to induced hypertension · moderate

The same 47-year-old woman with a left middle cerebral artery mean velocity of 201 cm/sec and a Lindegaard ratio of 5 is treated for delayed cerebral ischemia. She is maintained euvolemic on isotonic crystalloid and a norepinephrine infusion is titrated upward until her mean arterial pressure is 40 mm Hg above baseline, with a systolic pressure of 200 mm Hg. Her aphasia and right hemiparesis fail to improve after this trial. Which of the following is the most appropriate next step?

- A. Institute prophylactic hypervolemia and hemodilution to complete triple-H therapy
- B. Proceed to cerebral angioplasty and/or selective intra-arterial vasodilator therapy
- C. Perform prophylactic balloon angioplasty of the right middle cerebral artery because its velocity is also elevated
- D. Start an intravenous magnesium sulfate infusion targeting a supranormal serum magnesium level
- E. Add simvastatin 80 mg daily for the remainder of the vasospasm window

Question 212

Subarachnoid Hemorrhage · Prophylaxis against delayed cerebral ischemia · easy

A 52-year-old woman is admitted after aneurysmal subarachnoid hemorrhage and undergoes successful surgical clipping on hospital day 1. She is neurologically intact and hemodynamically stable. Which of the following interventions has the strongest evidence for reducing delayed cerebral ischemia and should be started as soon as possible?

- A. Intravenous magnesium sulfate infusion for 14 days
- B. Simvastatin 40 mg daily for 21 days
- C. Prophylactic hypervolemia with albumin and crystalloid
- D. Enteral nimodipine 60 mg every 4 hours for 21 days
- E. Prophylactic balloon angioplasty of the proximal intracranial vessels on day 3

Question 213

Subarachnoid Hemorrhage · Hyponatremia and cerebral salt wasting · moderate

A 60-year-old man is on postbleed day 6 after coiling of a ruptured anterior communicating artery aneurysm. He has become mildly lethargic. Serum sodium is 128 mEq/L, serum osmolality 265 mOsm/kg, urine sodium 60 mmol/L, and urine osmolality 450 mOsm/kg. His net fluid balance is negative 2.5 L over 48 hours, his weight is down 3 kg, and he is tachycardic with flat neck veins. Which of the following is the most appropriate management?

- A. Restrict total oral and intravenous fluid intake to 1 L per day
- B. Infuse 5% dextrose in water at 100 mL/hr and recheck sodium in 6 hours
- C. Infuse 0.9% sodium chloride to restore intravascular volume and add fludrocortisone 0.2 to 0.4 mg per day
- D. Start a 3% sodium chloride infusion together with free water restriction
- E. Discontinue nimodipine and observe

Question 214

Subarachnoid Hemorrhage · Antifibrinolytic therapy for rebleeding prevention · moderate

A 63-year-old woman presents to a rural emergency department 2 hours after sudden severe headache and is found to have Hunt and Hess grade 3 subarachnoid hemorrhage on noncontrast CT. Her systolic blood pressure has been brought below 160 mm Hg with a nicardipine infusion. Helicopter transfer to a comprehensive stroke center is arranged, and the earliest that her aneurysm can be secured is approximately 12 hours from now. She has no thrombotic history and no contraindication to antifibrinolytic therapy. Which of the following regimens is most appropriate?

- A. Tranexamic acid 1 g now, then 1 g every 6 hours until the aneurysm is secured, for a maximum of 72 hours
- B. Aminocaproic acid 4 to 10 g load then 1 to 2 g/hr, continued for 14 days after coiling
- C. Withhold antifibrinolytic therapy and begin it only if clinical rebleeding occurs
- D. Withhold antifibrinolytic therapy, since nimodipine 60 mg every 4 hours prevents rebleeding
- E. Aminocaproic acid begun 72 hours after ictus and continued until hospital discharge

Question 215

Subarachnoid Hemorrhage · Acute hydrocephalus after subarachnoid hemorrhage · moderate

A 47-year-old woman with Fisher grade IV subarachnoid hemorrhage underwent coil embolization of a basilar apex aneurysm on admission day 1. Two days after admission she becomes confused with a declining level of consciousness, impaired upward gaze, sluggish pupillary reactions, and a rising mean arterial pressure. Which of the following diagnostic tests is most appropriate to identify the likely cause of these changes?

- A. Continuous electroencephalography
- B. Serum sodium and osmolality
- C. Catheter digital subtraction angiography
- D. Transcranial Doppler ultrasonography
- E. Noncontrast head CT

Question 216

Subarachnoid Hemorrhage · ISAT and choice of aneurysm securing modality · easy

A 55-year-old woman presents with World Federation of Neurological Surgeons grade 2 subarachnoid hemorrhage from a ruptured 6 mm anterior communicating artery aneurysm. The multidisciplinary team judges the aneurysm to be equally amenable to surgical clipping and endovascular coiling. Which of the following statements about the International Subarachnoid Aneurysm Trial best informs the treatment decision?

- A.** Surgical clipping produced a lower rate of death or dependency at 1 year
- B.** Endovascular coiling produced a lower rate of death or dependency at 1 year, 23.7% versus 30.6%, and a lower risk of epilepsy
- C.** The two modalities produced identical rates of death or dependency at 1 year
- D.** Endovascular coiling produced both lower rebleeding and lower rates of incomplete aneurysm occlusion
- E.** The advantage of coiling seen at 1 year had disappeared by 10 years of follow-up

Question 217

Subarachnoid Hemorrhage · Seizure prophylaxis after subarachnoid hemorrhage · moderate

A 44-year-old man with World Federation of Neurological Surgeons grade 1 subarachnoid hemorrhage was started on levetiracetam in the emergency department and underwent coil embolization on hospital day 1. He has had no clinical seizures, and his neurologic examination is normal and reliable. Which of the following is the most appropriate management of his antiseizure medication?

- A.** Continue levetiracetam for 12 months
- B.** Change to phenytoin and continue for 6 months
- C.** Continue indefinitely until an electroencephalogram shows no epileptiform activity
- D.** Discontinue now that the aneurysm is secured; if prophylaxis is continued at all, limit it to a short course of 3 to 7 days
- E.** Continue through hospital day 14 to cover the entire period of delayed cerebral ischemia risk

Question 218

Subarachnoid Hemorrhage · Perimesencephalic subarachnoid hemorrhage · moderate

A 51-year-old man presents with sudden severe headache and neck stiffness. His neurologic examination is entirely normal. Noncontrast head CT shows blood centered in the prepontine and perimesencephalic cisterns immediately anterior to the midbrain and pons, without extension into the sylvian fissures or over the convexities. CT angiography is negative, and catheter digital subtraction angiography performed the same day shows no aneurysm or vascular malformation. Which of the following statements about this patient is most accurate?

- A.** Repeat catheter angiography in 7 to 14 days will identify an aneurysm in about one quarter of such patients
- B.** His risk of rebleeding, delayed cerebral ischemia, and hydrocephalus is comparable to that of aneurysmal subarachnoid hemorrhage
- C.** The source of bleeding is presumed venous, and delayed cerebral ischemia, rebleeding, and hydrocephalus are very rare in this population
- D.** This bleeding pattern accounts for approximately 80% of all cases of spontaneous subarachnoid hemorrhage
- E.** Nimodipine is contraindicated with this pattern of hemorrhage

Question 219

Subarachnoid Hemorrhage · Focal signs and aneurysm localization · easy

A 49-year-old woman presents with abrupt severe headache, vomiting, and neck stiffness. On examination she is drowsy but follows commands. The left pupil is 6 mm and nonreactive, there is left ptosis, and the left eye cannot adduct, elevate, or depress. Noncontrast head CT shows diffuse blood in the basal cisterns. Which of the following aneurysm locations is most likely responsible?

- A.** Left posterior communicating artery at its junction with the internal carotid artery
- B.** Anterior communicating artery
- C.** Left middle cerebral artery bifurcation
- D.** Left internal carotid artery within the cavernous sinus
- E.** Left vertebral artery at the origin of the posterior inferior cerebellar artery

Question 220

Subarachnoid Hemorrhage · Modified Fisher scale and delayed cerebral ischemia risk · hard

A 60-year-old woman presents with sudden severe headache and a Glasgow Coma Scale score of 14. Noncontrast head CT shows thick subarachnoid blood filling the basal cisterns and both sylvian fissures, together with layered blood in both lateral ventricles. Which of the following correctly gives her modified Fisher grade and the associated risk of delayed cerebral ischemia?

- A.** Grade 1, with a delayed cerebral ischemia risk of about 12%
- B.** Grade 2, with a delayed cerebral ischemia risk of about 21%
- C.** Grade 3, with a delayed cerebral ischemia risk of about 19%
- D.** Grade 4, with a delayed cerebral ischemia risk of about 12%
- E.** Grade 4, with a delayed cerebral ischemia risk of about 40%

Question 221

Subarachnoid Hemorrhage · Neurogenic cardiac and pulmonary complications · moderate

A 57-year-old woman is admitted with Hunt and Hess grade 4 subarachnoid hemorrhage. Within hours of admission she becomes hypoxemic with pink frothy endotracheal secretions and bilateral pulmonary infiltrates, and her blood pressure falls to 88/50 mm Hg. Troponin is 4.2 ng/mL and B-type natriuretic peptide is markedly elevated. The electrocardiogram shows deep symmetric T-wave inversions with a prolonged QT interval. Transthoracic echocardiography shows apical akinesis with a hyperkinetic base and an ejection fraction of 30%. Which of the following best explains her presentation?

- A. Acute coronary plaque rupture producing an anterior ST-elevation myocardial infarction
- B. Catecholamine-mediated neurogenic stunned myocardium with neurogenic pulmonary edema
- C. Nimodipine-induced hypotension with secondary demand ischemia
- D. Septic cardiomyopathy from early ventriculostomy-related infection
- E. Massive pulmonary embolism with acute right heart strain

Question 222

Subarachnoid Hemorrhage · Screening for unruptured intracranial aneurysms · easy

In which of the following groups do the American Heart Association and American Stroke Association guidelines recommend noninvasive screening for unruptured intracranial aneurysms with CT angiography or magnetic resonance angiography?

- A. Any patient with a single first-degree relative who has a known intracranial aneurysm
- B. Patients with autosomal recessive polycystic kidney disease
- C. All patients who both smoke and have hypertension
- D. Patients with two or more family members who have an intracranial aneurysm or subarachnoid hemorrhage
- E. All patients younger than 15 years of age

Question 223

Subarachnoid Hemorrhage · Angiogram-negative subarachnoid hemorrhage · hard

A 54-year-old man presents with sudden severe headache and photophobia. Noncontrast head CT shows diffuse thick subarachnoid blood in the basal cisterns extending into both sylvian fissures and the anterior interhemispheric fissure. CT angiography shows no aneurysm, and catheter digital subtraction angiography performed on the day of admission is also negative. He remains clinically stable. Which of the following is the most appropriate next step?

- A. Repeat catheter angiography 7 to 14 days after the initial presentation
- B. No further vascular imaging, since a negative digital subtraction angiogram excludes an aneurysm
- C. Repeat noncontrast head CT in 24 hours, with no further vascular imaging
- D. Perform a lumbar puncture to confirm the diagnosis of subarachnoid hemorrhage
- E. Substitute a CT perfusion study on day 4 for repeat angiography

Traumatic Brain Injury

Question 224

Traumatic Brain Injury · Cerebral hemodynamics and autoregulation · moderate

A 28-year-old man is admitted to the neurointensive care unit after a motorcycle crash with a severe traumatic brain injury (GCS 5). An intraparenchymal intracranial pressure (ICP) monitor is placed. Over the first 24 hours the team notes that every time mean arterial pressure (MAP) is raised with norepinephrine, ICP rises in parallel, and the moving correlation between MAP and ICP (pressure reactivity index) is strongly positive, indicating that pressure autoregulation is absent. Which of the following best describes the expected relationship between cerebral perfusion pressure (CPP) and cerebral blood flow (CBF) in this patient?

- A.** CBF will vary directly with CPP because cerebral vessel radius is effectively fixed
- B.** CBF will remain constant across CPPs of approximately 50 to 150 mmHg through changes in vessel diameter
- C.** CBF will be determined chiefly by vessel radius, with CPP having little impact
- D.** CBF will vary inversely with CPP because of compensatory vasoconstriction
- E.** CBF will become independent of blood viscosity once autoregulation is lost

Question 225

Traumatic Brain Injury · Surgical management of epidural hematoma · easy

A 22-year-old man is struck in the left temple during an assault. He was briefly unconscious, then awake and conversant with paramedics, and is now drowsy but follows commands, with a GCS of 14 and no focal deficit. Blood pressure is 138/78 mmHg and pulse is 72 beats per minute. Noncontrast head CT shows a left temporal lens-shaped hyperdense extra-axial collection that does not cross the coronal suture, measuring 45 cm³ in volume and 18 mm in maximal thickness, with 6 mm of midline shift. Which of the following is the most appropriate next step in management?

- A.** Admission for observation with a repeat head CT in 6 hours
- B.** Immediate surgical evacuation via craniotomy
- C.** Placement of an intracranial pressure monitor and medical management
- D.** Bolus mannitol with a repeat head CT in 24 hours
- E.** Bedside burr hole drainage in the intensive care unit

Question 226

Traumatic Brain Injury · Surgical management of subdural hematoma · hard

A 61-year-old woman falls down a flight of stairs. Her GCS at the scene was 10 (E3 V3 M4). On arrival in the emergency department 45 minutes later her GCS is 7 (E1 V2 M4). Pupils are 3 mm and briskly reactive bilaterally. Blood pressure is 142/80 mmHg and oxygen saturation is 98% after intubation. Noncontrast head CT shows a right convexity crescentic hyperdense extra-axial collection with a maximal thickness of 8 mm and 4 mm of midline shift, with patent basal cisterns. An intraparenchymal monitor is placed and reads an ICP of 14 mmHg. Which of the following is the most appropriate next step?

- A. Continued medical management with a repeat head CT in 6 hours
- B. Bolus hyperosmolar therapy and repeat imaging once the ICP exceeds 22 mmHg
- C. Placement of an external ventricular drain for cerebrospinal fluid diversion
- D. Surgical evacuation of the hematoma
- E. Observation with treatment triggered only if the midline shift exceeds 5 mm

Question 227

Traumatic Brain Injury · Basilar skull fracture and cerebrospinal fluid leak · moderate

A 19-year-old man is brought to the emergency department after a motor vehicle collision. He is alert with a GCS of 15. Examination shows retroauricular ecchymosis and hemotympanum on the right, and clear fluid dripping from the right nostril that tests positive for beta-2 transferrin. CT of the head demonstrates a nondisplaced right temporal bone fracture with a small amount of pneumocephalus and no intracranial mass lesion. The trauma team asks how to reduce his risk of meningitis. Which of the following is the most appropriate management?

- A. Begin intravenous ceftriaxone and continue until the leak resolves
- B. Begin intravenous vancomycin plus cefepime for 7 days
- C. Withhold prophylactic antibiotics, observe the leak, and pursue operative repair if it persists beyond 7 days
- D. Place a lumbar drain to divert cerebrospinal fluid and lower the leak pressure
- E. Proceed to immediate operative repair of the skull base defect

Question 228

Traumatic Brain Injury · Blood pressure targets · easy

According to Brain Trauma Foundation guidelines, what is the minimum systolic blood pressure that should be maintained in a 24-year-old patient with severe traumatic brain injury?

- A. 80 mmHg
- B. 90 mmHg
- C. 100 mmHg
- D. 105 mmHg
- E. 110 mmHg

Question 229

Traumatic Brain Injury · Tranexamic acid and CRASH-3 · moderate

A 47-year-old man falls from a ladder and arrives in the emergency department 90 minutes after injury. His GCS is 12, blood pressure is 128/72 mmHg, and pulse is 88 beats per minute. Noncontrast head CT shows a small right temporal contusion and a thin layer of traumatic subarachnoid hemorrhage. INR is 1.0, platelet count is 240,000 per microliter, and he takes no anticoagulant or antiplatelet drugs. Which of the following interventions has been shown in a large randomized controlled trial to reduce head injury-related mortality in patients such as this one?

- A. Recombinant activated factor VII
- B. Intravenous methylprednisolone
- C. Prophylactic platelet transfusion
- D. Intravenous tranexamic acid
- E. Prophylactic hypothermia to 33 degrees Celsius

Question 230

Traumatic Brain Injury · Intracranial pressure thresholds · easy

According to the current Brain Trauma Foundation guidelines, treatment of intracranial hypertension after severe traumatic brain injury should be initiated when the intracranial pressure exceeds which of the following values?

- A. 15 mmHg
- B. 20 mmHg
- C. 22 mmHg
- D. 25 mmHg
- E. 30 mmHg

Question 231

Traumatic Brain Injury · Osmotherapy and mannitol monitoring · moderate

A 34-year-old woman with severe traumatic brain injury and bifrontal contusions has received repeated 1 g/kg boluses of 20% mannitol over the past 24 hours for intracranial pressures in the high 20s. Current labs are sodium 152 mEq/L, BUN 28 mg/dL, glucose 126 mg/dL, creatinine 1.4 mg/dL (baseline 0.8), and a measured serum osmolality of 345 mOsm/kg. Blood pressure is 104/60 mmHg and urine output has been 400 mL/hr. Her intracranial pressure is again 26 mmHg. Which of the following is the most appropriate next step?

- A. Continue mannitol boluses because the serum sodium remains below 160 mEq/L
- B. Change to a continuous mannitol infusion to smooth the intracranial pressure profile
- C. Give furosemide to accelerate mannitol clearance and enhance the osmotic effect
- D. Withhold further mannitol and use hypertonic saline if additional osmotherapy is required
- E. Continue mannitol but reduce the dose to 0.25 g/kg per bolus

Question 232

Traumatic Brain Injury · Herniation and hyperventilation · moderate

A 40-year-old man with a severe traumatic brain injury and an external ventricular drain has an abrupt rise in intracranial pressure to 45 mmHg. His right pupil dilates to 6 mm and becomes nonreactive, and he develops extensor posturing. Blood pressure is 178/95 mmHg with a pulse of 52 beats per minute. Cerebrospinal fluid drainage produces no improvement. A stat head CT is ordered and the neurosurgical team is mobilizing for the operating room. In addition to a bolus of a hyperosmolar agent, which of the following is the most appropriate temporizing measure?

- A. Start a continuous infusion of mannitol
- B. Place a lumbar drain for cerebrospinal fluid drainage
- C. Administer intravenous dexamethasone
- D. Begin surface cooling to a core temperature of 33 degrees Celsius
- E. Transiently hyperventilate to an end-tidal carbon dioxide of 30 to 35 mmHg

Question 233

Traumatic Brain Injury · Refractory intracranial hypertension and tiered therapy · hard

A 26-year-old man with severe diffuse traumatic brain injury and no surgically evacuable mass lesion has intracranial pressures of 30 to 35 mmHg for the past 4 hours. He is already receiving deep sedation and analgesia, has the head of bed elevated with the neck in neutral position, is normocapnic with an arterial carbon dioxide of 38 mmHg, has had cerebrospinal fluid drained through an external ventricular drain, and has received repeated boluses of hypertonic saline with a serum sodium now of 158 mEq/L. A trial of neuromuscular blockade lowered the intracranial pressure only transiently. Cerebral perfusion pressure is 62 mmHg. Which of the following is the most appropriate next step?

- A. Induce therapeutic hypothermia to 33 degrees Celsius
- B. Administer intravenous methylprednisolone
- C. Give furosemide together with scheduled mannitol infusions
- D. Place a lumbar drain for additional cerebrospinal fluid drainage
- E. Proceed to decompressive craniectomy

Question 234

Traumatic Brain Injury · Decompressive craniectomy trials · hard

The family of a 30-year-old woman with severe traumatic brain injury and medically refractory intracranial hypertension asks what decompressive craniectomy is likely to accomplish. The neurosurgeon bases the discussion on the RESCUEicp trial, a multicenter randomized trial of 408 patients comparing last-tier decompressive craniectomy with continued medical management for refractory traumatic intracranial hypertension. Which of the following best describes the 6-month results of that trial?

- A. Both mortality and the rate of good functional outcome were significantly improved by surgery
- B. Mortality was unchanged but functional outcome was significantly better with surgery
- C. Mortality was significantly lower with surgery without a significant increase in good functional outcome
- D. Mortality was significantly higher with surgery
- E. Neither mortality nor functional outcome differed between the groups

Question 235

Traumatic Brain Injury · Cerebral perfusion pressure targets · moderate

A 52-year-old man with severe traumatic brain injury has an intracranial pressure of 16 mmHg and a mean arterial pressure of 95 mmHg with the arterial line transducer zeroed at the phlebostatic axis. He is euvolemic and off vasopressors. A trainee proposes starting norepinephrine and giving additional crystalloid to drive the cerebral perfusion pressure to 85 to 90 mmHg in order to improve cerebral oxygen delivery. Which of the following is the most appropriate management?

- A. Raise the cerebral perfusion pressure above 90 mmHg with norepinephrine
- B. Maintain the cerebral perfusion pressure between 60 and 70 mmHg and avoid pressor-driven elevation above 70 mmHg
- C. Maintain the cerebral perfusion pressure above 70 mmHg with aggressive fluid loading
- D. Lower the mean arterial pressure so that the cerebral perfusion pressure falls below 60 mmHg
- E. Titrate the cerebral perfusion pressure solely to keep the jugular venous oxygen saturation above 50%

Question 236

Traumatic Brain Injury · Posttraumatic seizure prophylaxis · easy

A 35-year-old man with a severe traumatic brain injury and bilateral frontal contusions was loaded with phenytoin in the emergency department and is now on hospital day 1 in the intensive care unit. He has had no clinical seizures, and continuous electroencephalography shows no electrographic seizures. The team asks how long to continue the prophylactic anticonvulsant. Which of the following is the most appropriate plan?

- A. Discontinue the phenytoin now, since prophylaxis is not recommended
- B. Continue prophylaxis for a total of 1 week after the injury
- C. Continue prophylaxis for 1 month after the injury
- D. Continue prophylaxis for 3 months after the injury
- E. Continue prophylaxis indefinitely because of the risk of posttraumatic epilepsy

Question 237

Traumatic Brain Injury · Paroxysmal sympathetic hyperactivity · moderate

A 24-year-old man is on hospital day 8 after a severe traumatic brain injury with diffuse axonal injury. As sedation is weaned he develops episodes lasting 20 to 40 minutes of heart rate 140 beats per minute, blood pressure 190/100 mmHg, respiratory rate 34 breaths per minute, temperature 38.9 degrees Celsius, profuse diaphoresis, and extensor posturing. The episodes occur several times a day and are reliably triggered by endotracheal suctioning and repositioning. Blood and urine cultures, chest radiograph, and a repeat head CT are unrevealing, and electroencephalography during an episode shows no seizure activity. Intravenous morphine aborts the episodes. Which of the following is the most appropriate maintenance medication to add?

- A. Propranolol
- B. Metoprolol
- C. Haloperidol
- D. Phenytoin
- E. Dantrolene

Traumatic Spinal Cord Injury

Question 238

Traumatic Spinal Cord Injury · Neurogenic shock and blood pressure targets · moderate

A 38-year-old man is admitted to the neurocritical care unit after a motorcycle collision. Imaging shows a C6 fracture-dislocation, and his examination is consistent with a motor-complete cervical spinal cord injury. He is intubated and normothermic. Heart rate is 48 beats/min, blood pressure 82/48 mmHg (mean arterial pressure 59 mmHg), and central venous pressure is 10 mmHg after 2 L of crystalloid. Hemoglobin is 12.9 g/dL, lactate 1.4 mmol/L, and focused ultrasound shows a hyperdynamic left ventricle with no free fluid. An arterial catheter is placed and a norepinephrine infusion is started. Which of the following is the most appropriate hemodynamic goal for the next 7 days?

- A. Systolic blood pressure greater than 90 mmHg
- B. Mean arterial pressure of 65 to 70 mmHg
- C. Mean arterial pressure of 85 to 90 mmHg
- D. Mean arterial pressure of 100 to 110 mmHg
- E. Central venous pressure of 12 to 15 mmHg with no specific pressure target

Question 239

Traumatic Spinal Cord Injury · Airway management in cervical spinal cord injury · hard

A 24-year-old man is brought to the emergency department after a diving accident. Computed tomography shows a C4 burst fracture, and examination reveals no motor or sensory function below C4, including absent sacral sensation and rectal tone. Over 20 minutes he becomes increasingly agitated with paradoxical abdominal movement, a respiratory rate of 34 breaths/min, and oxygen saturation falling to 86% on a nonrebreather mask. He is in a rigid cervical collar. Fiberoptic equipment is not immediately at the bedside. Which of the following is the most appropriate approach to securing the airway?

- A.** Noninvasive bilevel positive-pressure ventilation with reassessment in 1 hour
- B.** Awake nasotracheal intubation with the cervical collar left in place
- C.** Awake fiberoptic intubation once equipment is obtained from another unit
- D.** Rapid sequence intubation with the rigid cervical collar left in place
- E.** Rapid sequence intubation with the collar opened and manual in-line stabilization

Question 240

Traumatic Spinal Cord Injury · Pharmacologic therapy and steroids · easy

A 47-year-old woman is admitted 4 hours after a fall from a ladder with an ASIA B injury at T5. She is hemodynamically supported with norepinephrine and is scheduled for decompression and fusion. The referring surgeon asks whether high-dose intravenous methylprednisolone should be given because the patient is still within the early treatment window described in the National Acute Spinal Cord Injury Study (NASCIS) trials. Which of the following is the most appropriate response?

- A.** Withhold methylprednisolone, because steroids are not supported by evidence and are associated with harm
- B.** Give methylprednisolone as a bolus followed by a 24-hour infusion, since she is within 8 hours of injury
- C.** Give methylprednisolone as a bolus followed by a 48-hour infusion, since her injury is motor complete
- D.** Give dexamethasone instead, because it has fewer infectious complications than methylprednisolone
- E.** Give methylprednisolone only after surgical decompression is completed

Question 241

Traumatic Spinal Cord Injury · Timing of surgical decompression · moderate

A 55-year-old man arrives 6 hours after a motor vehicle collision with a C5-C6 fracture-dislocation and jumped facets. His examination shows 1/5 strength in the upper extremities, 0/5 in the lower extremities, and preserved perianal sensation. Blood pressure is 88/50 mmHg with a heart rate of 52 beats/min. Computed tomography angiography shows no cerebrovascular injury. He is hemodynamically stabilized with norepinephrine. Which of the following is the most appropriate next step in management?

- A. Continue closed traction only and plan definitive fixation at 5 to 7 days once edema resolves
- B. Proceed to decompression and stabilization within 24 hours of injury
- C. Maintain a rigid cervical collar and obtain flexion and extension radiographs before any operation
- D. Delay operative decompression until the mean arterial pressure has been maintained above 85 mmHg for 72 hours
- E. Place a prophylactic inferior vena cava filter before considering operative decompression

Question 242

Traumatic Spinal Cord Injury · ASIA impairment scale and prognosis · easy

A 44-year-old man is admitted with a C6 spinal cord injury after a fall. On examination he has partial loss of movement and sensation below C6. Voluntary motor function is present below the neurologic level, and most of the key muscle groups below the level of injury have a muscle grade of 3 or better, allowing movement against gravity. Sacral sensation and voluntary anal contraction are intact. How should this injury be classified?

- A. C6 ASIA A
- B. C6 ASIA B
- C. C6 ASIA C
- D. C6 ASIA D
- E. C6 ASIA E

Question 243

Traumatic Spinal Cord Injury · Autonomic dysreflexia · hard

A 29-year-old man with a motor-complete spinal cord injury at C7 sustained 3 weeks earlier develops a pounding headache, flushing and sweating above the clavicles, and nasal congestion during a routine turn. Blood pressure rises from a baseline of 104/62 mmHg to 212/118 mmHg, and heart rate falls from 78 to 46 beats/min. He is afebrile. His indwelling urinary catheter has drained no urine for several hours, and the bladder is percussed to the umbilicus. Which of the following is the most appropriate initial step?

- A. Administer atropine 0.3 mg intravenously for the bradycardia
- B. Start a nicardipine infusion and titrate to a systolic pressure below 140 mmHg
- C. Obtain computed tomography of the head to evaluate for intracranial hemorrhage
- D. Administer intravenous methylprednisolone for presumed cord edema
- E. Relieve the bladder outflow obstruction by flushing or replacing the urinary catheter

Question 244

Traumatic Spinal Cord Injury · Venous thromboembolism prophylaxis · moderate

A 61-year-old woman is 36 hours out from posterior decompression and fusion for a T4 fracture with an ASIA A injury. She has no intracranial hemorrhage, no ongoing surgical bleeding, hemoglobin is stable at 10.8 g/dL, and platelets are 210,000/microliter. The neurosurgical team has cleared her for pharmacologic prophylaxis. Which of the following is the most appropriate venous thromboembolism prophylaxis strategy?

- A.** Mechanical prophylaxis with pneumatic compression devices alone until she is mobilized
- B.** Placement of a prophylactic inferior vena cava filter, because complete injuries carry an especially high risk
- C.** Therapeutic anticoagulation with a direct oral anticoagulant started on postoperative day 7
- D.** Low molecular weight heparin started now, combined with pneumatic compression devices
- E.** Daily lower extremity duplex ultrasound surveillance with treatment only if thrombus is detected

Part III: Administration

Administrative and Managerial Principles

Question 245

Administrative and Managerial Principles · ICU staffing models and outcomes · moderate

A 68-year-old woman with a large left basal ganglia intracerebral hemorrhage is admitted to a community hospital ICU where any admitting practitioner may direct critical care and an intensivist consultation is optional. Over 3 days she requires intubation, an external ventricular drain, and blood pressure control, and her ICU stay is prolonged by a ventilator-associated pneumonia. Hospital leadership asks the newly hired neurointensivist to redesign the unit's physician coverage to improve outcomes. Which of the following changes is best supported by the evidence described for ICU staffing?

- A.** Convert to an open unit with 24-hour in-house hospitalist coverage and optional intensivist consultation
- B.** Retain the current model but add a daily telemedicine review by an off-site intensivist as the sole change
- C.** Convert to high-intensity staffing, in which patients are managed primarily by an intensivist or an intensivist consultation is mandatory
- D.** Retain low-intensity staffing but increase the nurse-to-patient ratio to two nurses for each patient
- E.** Convert to a mixed medical-surgical generalist ICU so that all critically ill patients are cared for in one unit

Question 246

Administrative and Managerial Principles · EMTALA and interhospital transfer · hard

A 57-year-old man is brought to a rural critical access hospital emergency department with a Glasgow Coma Scale score of 8, blood pressure 198/104 mmHg, and a noncontrast head computed tomography scan showing a 60 mL right putaminal hemorrhage with intraventricular extension. The hospital participates in Medicare, has no neurosurgical service, and has no ICU capable of external ventricular drain management. He is intubated and started on a nicardipine infusion, but he cannot be fully stabilized locally. A comprehensive stroke center 90 minutes away has an available neurocritical care bed. Under the Emergency Medical Treatment and Active Labor Act, which of the following is required of the transferring physician?

- A.** Obtain written consent from the patient's surrogate before the receiving facility may be contacted
- B.** Delay transfer until the emergency medical condition has been fully stabilized, regardless of local capability
- C.** Send the patient with any available transport, since qualified personnel and equipment are the receiving facility's responsibility
- D.** Withhold further treatment once transfer is arranged so that care is not duplicated at the receiving hospital
- E.** Certify that the benefits of transfer outweigh the risks and document the name of the accepting individual and facility

Question 247

Administrative and Managerial Principles · Neurocritical care unit design and bed planning · easy

A neurointensivist is recruited to build a new neurocritical care unit at a 450-bed public hospital that is both a Level 1 trauma center and a comprehensive stroke center. Neurotrauma patients will continue to be admitted to the existing trauma unit. As a first estimate for the initial business plan, how many neurocritical care beds should be proposed?

- A.** 2 to 4 beds
- B.** 8 to 12 beds
- C.** 18 to 24 beds
- D.** 30 to 36 beds
- E.** 45 to 55 beds

Quality Measurement in Neurocritical Care

Question 248

Quality Measurement in Neurocritical Care · Donabedian classification of quality measures · easy

A neurocritical care unit launches a quality improvement project to shorten ICU length of stay for patients with aneurysmal subarachnoid hemorrhage by standardizing transfer to the step-down unit once vasospasm surveillance is complete. To make sure the project does not simply shift harm downstream, the team also tracks the rate of unplanned readmission to the ICU within 48 hours of transfer. In the classification of quality measures, the readmission rate as used here is best described as which type of measure?

- A. Balance measure
- B. Structure measure
- C. Process measure
- D. Outcome measure
- E. Patient-reported experience measure

Question 249

Quality Measurement in Neurocritical Care · CMS value-based payment programs · moderate

A 550-bed academic hospital reviews its neurocritical care unit data and finds elevated rates of central line-associated bloodstream infection and catheter-associated urinary tract infection, placing the hospital in the lowest performing quartile nationally for hospital-acquired conditions. The quality officer is asked to quantify the financial exposure from the relevant Centers for Medicare and Medicaid Services program. Which of the following best describes the penalty this hospital faces?

- A. A 2% withholding of inpatient payments that may be partially or fully recouped for good performance
- B. A maximum 3% penalty on inpatient payments applied across six clinical cohorts
- C. A 1% reduction in inpatient payments under the Hospital-Acquired Condition Reduction Program
- D. Loss of eligibility for the American College of Surgeons Trauma Quality Improvement Program registry
- E. A 6% reduction in inpatient payments applied to all Medicare discharges for the reporting year

Question 250

Quality Measurement in Neurocritical Care · Risk adjustment and the mortality index · hard

Two comprehensive stroke centers care for similar populations of patients with high-grade aneurysmal subarachnoid hemorrhage, and chart review of matched cases shows comparable adherence to nimodipine administration, vasospasm screening, and external ventricular drain insertion bundles. Nevertheless, Hospital B reports a substantially higher mortality index than Hospital A on publicly reported rankings. Review shows that Hospital A consistently documents comorbidities and complications present on admission such as shock, coma, brain compression, and cerebral edema, whereas Hospital B does not. Which of the following best explains Hospital B's worse mortality index?

- A.** Hospital B's observed mortality is genuinely higher because of unmeasured differences in process of care
- B.** Hospital B's patients have a higher case mix complexity that the mortality index is designed to ignore
- C.** Hospital B is penalized because the mortality index counts only deaths occurring within 30 days of admission
- D.** Hospital B's index is inflated because publicly reported ranking systems weight reputation more heavily than mortality
- E.** Hospital B's expected mortality is underestimated because incomplete coding weakens risk adjustment

Answer Key

#	Ans	#	Ans	#	Ans	#	Ans	#	Ans
1	D	51	D	101	E	151	B	201	D
2	B	52	C	102	C	152	C	202	E
3	C	53	A	103	B	153	E	203	A
4	E	54	E	104	D	154	A	204	B
5	A	55	B	105	A	155	B	205	C
6	B	56	A	106	E	156	A	206	D
7	D	57	D	107	C	157	E	207	E
8	C	58	E	108	B	158	C	208	C
9	A	59	B	109	D	159	D	209	A
10	C	60	C	110	A	160	A	210	E
11	E	61	A	111	E	161	D	211	B
12	C	62	D	112	A	162	B	212	D
13	B	63	E	113	D	163	A	213	C
14	A	64	C	114	C	164	E	214	A
15	D	65	D	115	B	165	C	215	E
16	E	66	E	116	E	166	B	216	B
17	C	67	B	117	A	167	B	217	D
18	A	68	A	118	E	168	D	218	C
19	D	69	C	119	B	169	A	219	A
20	B	70	E	120	D	170	C	220	E
21	E	71	B	121	A	171	E	221	B
22	C	72	C	122	C	172	B	222	D
23	A	73	A	123	E	173	D	223	A
24	E	74	D	124	D	174	C	224	A
25	B	75	E	125	C	175	A	225	B
26	D	76	C	126	B	176	D	226	D
27	A	77	B	127	E	177	B	227	C
28	C	78	D	128	A	178	E	228	E
29	E	79	A	129	B	179	C	229	D
30	B	80	B	130	D	180	D	230	C
31	C	81	E	131	A	181	A	231	D
32	A	82	C	132	E	182	E	232	E

33	E	83	D	133	C	183	B	233	E
34	B	84	A	134	D	184	D	234	C
35	D	85	E	135	A	185	C	235	B
36	C	86	C	136	B	186	B	236	B
37	A	87	C	137	E	187	A	237	A
38	B	88	A	138	A	188	D	238	C
39	D	89	D	139	E	189	E	239	E
40	A	90	B	140	C	190	B	240	A
41	E	91	E	141	A	191	C	241	B
42	C	92	C	142	E	192	D	242	D
43	D	93	A	143	B	193	A	243	E
44	B	94	D	144	D	194	B	244	D
45	A	95	B	145	E	195	C	245	C
46	C	96	E	146	A	196	D	246	E
47	D	97	A	147	C	197	E	247	B
48	E	98	C	148	B	198	A	248	A
49	C	99	D	149	D	199	B	249	C
50	B	100	B	150	C	200	C	250	E